Grit And Self-Efficacy Beliefs on Classroom Management and Student Engagement: A Cross-Sectional Study of PreK-8 Teachers

By:

Malabika Chakraborty

A DISSERTATION

Submitted to the faculty of

Centenary University

in partial fulfillment of the requirements

for the degree of Ed.D. in Educational Leadership

Hackettstown,

New Jersey

December 10, 2024

Abstract

This quantitative, cross-sectional, correlational study explored the correlation between grit and self-efficacy beliefs in classroom management and student engagement among elementary and middle school teachers (Prek-8). Teachers in elementary and middle schools often have to teach multiple subject areas and ensure that classrooms are well-managed and students are engaged in the content being taught. The required skill sets extend beyond subject area expertise and involve personal traits that help teachers work through problems and succeed. The theoretical foundation for the study was Deci and Ryan's self-determination theory, Bandura's theory of self-efficacy and Duckworth's grit theory. The participants included 86 teachers from a convenience sample from elementary and middle schools across five public school districts in Northwestern NJ. A survey questionnaire consisting of demographic questions, the Short Grit Scale (Grit-S), and the Teacher Sense of Efficacy Scale (TSES) was used. Data collected from the Grit-S Scale and Teacher Self-Efficacy Scale was analyzed utilizing a product-moment correlation coefficient (Pearson's r) to answer the questions of the relationship between grit and self-efficacy beliefs of teachers in the area of classroom management and student engagement. The results of the study show a significant correlation between teacher grit scores and their effect on self-efficacy beliefs in the area of classroom management and student engagement. This study increased the body of knowledge in researching these constructs among elementary and middle school teachers.

Keywords: Grit, teacher self-efficacy, classroom management, student engagement, passion, quantitative

Dedication

This dissertation is dedicated to many people in my life who have stood by me and believed in me. I dedicate this dissertation to my mother who always believed that I could achieve anything I set out to and supported every endeavor that I undertook. Her mantra of "nothing is too far fetched when you put in the work and stay dedicated" has been my inspiration. I miss my mother's presence everyday but I know she is guiding me from above.

I also dedicate this dissertation to my daughters Shrijani and Danica. They have supported me during the whole process by being the best version of themselves and managing things at their end so that I could work without any worries. Their pride in having a mother who was working towards earning a "doctorate" kept me motivated despite feeling overwhelmed at times.

I would like to acknowledge my grandparents who always showered me with love and affection and took pride in everything I did since childhood. Their belief in me and their blessings gave me the courage to take on this endeavor.

Acknowledgments

This doctoral dissertation was completed with the support of several people. I would like to thank my committee chair Dr. Jenine Kastner who has always been generous with her time, guidance and encouragement throughout the process. It has been a privilege to learn from her. I would like to thank my committee members, Dr. Timothy Frederiks and Dr. Danielle Kovach. Both of them provided valuable feedback and guidance along the way. Dr. Frederiks is an amazing teacher and human being. He has helped and guided me from the very beginning. He was also always there to answer any question and clarify any doubts that I had. His knowledge of statistics and invaluable patience during data analysis is deeply appreciated.

I would also like to also acknowledge all the teachers who took the time to answer my survey. Without their enthusiastic participation, I would not have been able to complete my study. I would also like to acknowledge my colleagues who supported and encouraged me to stay strong and carry on.

Table of Contents

Abstract ii

Dedication iii

Acknowledgments iv

List of Tables viii

Chapter 1: Introduction to the Study 1

Background 2

Problem Statement 5

Purpose of the Study 7

Research Questions 8

Theoretical Foundation 8

Nature of the Study 9

Definitions 10

Limitations and Generalizability 12

Significance of the Study 13

Summary 15

Chapter 2: Literature Review 18

Background 19

Theoretical Framework 21

Self-Determination Theory 21

Self-Efficacy Theory 24

Grit Theory 26

Teacher Attrition 29

Teacher Burnout 31

Classroom Management 33

Growth Mindset....35
Conscientiousness....38
Grit and Teacher Efficacy....39
Grit Scale and TSES Scale....46
Instrumentation....48
Grit Short Scale (Grit S)....48
TSES Scale....49
Summary....50
Chapter 3: Research Methods....53
Overview....53
Research Design and Rationale....53
Research Questions....54
Participants and Sampling....55
Participant Recruitment and Data Collection....56
Archival Data....58
Validity and Reliability of Instrumentation....58
Ethical Considerations....59
Data Analysis Procedures....60
Variables....62
Limitations and Delimitations....62
Summary....64
Chapter 4: Findings....66
Descriptive Statistics....67
Quantitative Results....72
Linearity....80

Research Question 1 82

Research Question 2 84

Summary 86

Chapter 5: Discussions, Conclusions & Recommendations 87

Purpose of the Study 87

Discussion of the Research Results 90

Research Question 1 90

Research Question 2 92

Implications 94

Limitations 97

Recommendations for Future Research 98

Conclusion 99

List of Tables

Table 1: *Senses that Influence Teachers' Willingness to Stay in Their Job* 31

Table 2: Teacher Identities that are Interrelated 41

Table 3: Principle Sources that Frame Teacher Efficacy and Grit 46

Table 4: Summary of Steps Taken for Data Collection and Analysis 61

Table 5: Definitions and Measurement of Variables 62

Table 6: Frequency and Percentage Summaries for Teacher Work Experience 68

Table 7: Descriptive Statistics for Certification Pathway.....68

Table 8: Percentage of Teachers Teaching All Subjects and Special Area Subjects.....69

Table 9: Detailed Percentages of Subjects Taught 69

Table 10: Teacher Educational Qualification Levels.....71

Table 11: Descriptive Statistics of Study Variables-Grit, Self-efficacy Beliefs.....72

Table 12: Results of the Kolmogorov-Smirnov and Shapiro-Wilk Tests.....76

Table 13: Results of the Pearson's Correlation Test between Grit and Self-Efficacy Beliefs 84

List of Figures

Figure 1: Boxplot graph for Grit Score.....73

Figure 2: Boxplot Graph for Self-Efficacy Scores in Classroom Management 73

Figure 3: Boxplot Graph for Self-Efficacy Scores in Student Engagement 74

Figure 4: Bivariate Outliers Grit Score and Efficacy in Classroom Management 74

Figure 5: Bivariate Outliers Grit Score and Efficacy in Student Engagement 75

Figure 6: Histogram Showing Grit scores 76

Figure 7: Histogram Showing Self-Efficacy Scores for Classroom Management 77

Figure 8: Histogram Showing Self-Efficacy Scores for Student Engagement 77

Figure 9: Normal Q-Q Plot for Grit Scores 78

Figure 10: Detrended Q-Q Plot for Grit Scores 78

Figure 11: Normal Q-Q Plot for Self-Efficacy Beliefs in Classroom Management.....79

Figure 12: Detrended Q-Q Plot for Self-Efficacy Beliefs in Classroom Management 79

Figure 13: Normal Q-Q Plot for Self-Efficacy Beliefs in Student Engagement.....80

Figure 14: Detrended Q-Q Plot for Self-Efficacy Beliefs in Student Engagement 80

Figure 15: Scatterplot Grit Score and Self-Efficacy Beliefs in Classroom Management..81

Figure 16: Scatterplot Grit Score and Self-efficacy Beliefs in Student Engagement 82

Chapter 1: Introduction to the Study

Teacher retention is a dilemma in K-12 education across the United States. Research in the field of education shows that "teacher demand is on the rise, as a function of changes in student enrollment, shifts in pupil-teacher ratios, and most significantly, high levels of teacher attrition" (Sutcher et al., 2016, p. 2). School districts across the United States have been struggling to fill teacher positions. According to Sutcher et al. (2019) and Turner and Cohen (2023), some of the other factors that cause teacher attrition include insufficient compensation, lack of professional respect, and being overburdened with additional job responsibilities for which the teachers did not receive adequate training or institutional support to perform. The majority of the statistics highlight teacher attrition as the main issue but there is little focus on teacher retention, with an emphasis on why teachers stay in the field.

The purpose of this quantitative, cross-sectional, and correlational study was to explore if and to what extent a correlation existed between grit and self-efficacy beliefs of teachers in classroom management and student engagement in elementary and middle school (PreK-8). As illustrated in the literature review in Chapter 2, previous studies have focused on grit in relation to novice teachers (Robertson-Kraft & Duckworth, 2014), veteran teachers (Mott, 2020), alternately certified teachers (Rozell, 2019) special education teachers, and secondary teachers (Lee, 2020) and its relationship to self-efficacy beliefs of classroom management and student engagement while others were qualitative and phenomenological studies in relation to grit and growth mindset (Smith, 2023). There is, however, no significant literature about the relationship between grit and self-efficacy beliefs of teachers on classroom management and student engagement in elementary and middle schools.

Background

The question “what personal attributes make a teacher successful?” has become an area of interest in recent years as a result of high attrition rates among teachers across the United States. The number of teacher candidates is declining across the United States. Thirty-one percent fewer candidates completed a teacher prep program in the 2017–2018 academic year than in 2008–2009 (Weber, 2020). In the case of New Jersey, the percentage is significantly higher. There has been a 49% decline in candidates completing a teacher prep program in New Jersey (Weber, 2020).

The 2020–2021 school year saw approximately 3 million full-time teachers enter public schools across the United States (Irwin et al., 2023). About half that number, 1.5 of them, were elementary teachers. This number is lower than that of the fall of 2018 when 3.2 million teachers entered public schools across the country (McFarland et al., 2018).

According to the latest National Center for Education Statistics report, difficulty filling teaching positions was higher in 2020–2021 than in 2011–2012 across 12 reported subject areas (Irwin et al., 2023). During the same period, 2012–2013 to 2019–2020, the number of candidates completing traditional teacher preparation programs decreased by 28%, from 161,000 in 2012–2013 to 116,100 in 2019–2020 (Irwin et al., 2023).

While many initiatives have recently focused on increasing the percentage of new teachers entering the field, not many efforts have been directed toward retaining experienced teachers, many of whom have faced extensive challenges as a result of swift changes in instructional methods, new assessment systems, and declining healthcare benefits (Weber, 2020). Teachers who have been in the field for a long time have been facing burnout due to increased workloads (Ferguson et al., 2012).

Furthermore, the sudden and drastic changes brought on by the COVID-19 pandemic that led to changes in teaching methods, the use of new technology often on an experimental

basis and without the option of structured training, and constantly changing district policies had left teachers feeling overwhelmed and burdened. The burden of working from home without collegial support, learning on the go, and adapting to the new ways of student engagement and classroom management during a time when the world around them and they themselves in many cases were experiencing trauma that was beyond anything the modern era had ever imagined (Baird, 2020; Cullinane & Montacute, 2020; Viner et al., 2020). All these factors add to the major reasons for burnout and fatigue, which are the main causes of teacher attrition and effectiveness (Skaalvik & Skaalvik, 2017).

The NCES report on the recovery from the Coronavirus Pandemic in K–12 education indicates wide-ranging concerns from students and parents in the areas of academic, social, and health concerns (Irwin et al., 2023). Despite the implementation of strategies to support students' learning recovery, the effectiveness of these strategies varied (Irwin et al., 2023) and an average of 36% of students in public schools were reported to be behind grade level in at least one academic subject at the end of the 2021–22 school year in comparison to 50% at the beginning of the 2021–2022 school year (Irwin et al., 2023). Furthermore, data from the National Assessment of Educational Progress (NAEP) show declines in scale scores for 4th and 8th grade students from 2019 to 2022 in both mathematics and reading.

The cumulative effect of the reported statistics by NCES aligns with the demands teachers face in classrooms. Teachers are facing burnout with the added pressure of helping improve learning in subject areas in elementary and middle schools where students are already lagging behind a grade in critical academic subjects as well as reeling under the crippling social and emotional effects of social distancing and quarantines brought on by a pandemic. According to the New Jersey Public School factsheet for 2022–2023, the total number of full-time teachers in New Jersey was 116,698 (Douglas et al., 2024). Many of these teachers entered elementary and middle (PreK-8) school classrooms.

Teachers at the elementary and middle schools have the daunting task of not only making sure that students at varied levels in their classrooms are able to learn from the materials and plans created for them but also that students have opportunities to interact, work in groups, and be able to express learning in different formats. Taking all these factors into consideration, while a major part of a teacher's job will be focused on learning recovery another aspect of their focus will also fall on the social-emotional wellbeing of the students, addressing challenging behaviors in classrooms while ensuring efficient classroom management and meaningful student engagement.

Given the overwhelming demands of the job, it is critical to understand the attributes that help teachers be successful in classrooms. Grit is a personality trait that has emerged as a key predictor for success in many professions and also for those who have just entered the field of teaching (Robertson-Kraft & Duckworth, 2014; Ryan et al., 2017). Duckworth (2016) defined grit as including both perseverance and passion for long-term goals. Self-efficacy is another important attribute. It is based on Bandura's (1977) theory of self-efficacy and is based on an individual's belief in their ability to successfully cope with tasks, obligations, and challenges related to their professional role (Caprara et al., 2006; Tschannen-Moran & Hoy, 2007).

Studies show that teachers with high levels of self-efficacy report greater levels of job satisfaction, lower levels of job-related stress, and experience minimal difficulty in handling classroom-related challenging student behavior (Caprara et al., 2003). Therefore, more research must focus on understanding the precursors of self-efficacy (Barni et al., 2019). Furthermore, studies also show that "individuals flourish when they experience positive emotions (P), engagement (E), relationships (R), meaning (M), and accomplishments (A)" ([PERMA] Seligman, 2011). Several studies indicate the importance of personal attributes of teachers that contribute to their success in classrooms and have indicated reasons such as

unmotivating work environments, lack of autonomy in classrooms, and lack of administrative support as reasons for loss of teacher sense of self-efficacy. Other studies indicate the lack of grit among novice teachers as well as grit in relation to self-efficacy, as reasons for teachers leaving the profession. Despite research being available there has not been a change in attrition numbers of teachers in the United States.

School districts across the country still need to fulfill positions in crucial subject areas. One in five teachers leave the profession early on in their careers and teacher attrition rates are higher in school districts that struggle with funding (Henke et al., 2000). In other cases, those teachers who choose to remain in the profession suffer burnout, and as a result, do not put in much effort (Cherniss, 1980). Therefore, studies dedicated to identifying positive traits that help individuals remain committed and resilient even in the face of adversity might be the key to determining teacher effectiveness (Stanford, 2001).

Moreover, given the after effects of a recent pandemic and its devastating impact on teacher morale as a result of declining achievement scores in students and increasing workload on teachers, there are many veteran teachers who are leaving the profession or planning to retire, which will again cause disruption in student achievement scores due to changes in personnel, delays in hiring as a result of lack of qualified candidates applying for jobs and gaps in student learning in the process of transitioning between teachers and their varied teaching styles.

Problem Statement

The number of teacher candidates is declining across the United States. Available data indicates that novice teachers are leaving the field of education to pursue other careers within the first five years (Hise, 2022). Veteran teachers are leaving as a result of burnout and lack of motivation (Lee, 2020). Studies show that teachers who left attributed “low-to-no impact orientation, ineffective or non-existent mentorship, disconnected or absentee school

administration, workload, and ‘bait and switch’ situations during their tenure as teachers" (Muex, 2021, p. 2) to be some of the reasons.

Both novice teachers and veteran teachers are burdened with increased workloads due to constant changes in curriculum, implementing new teaching methods, new systems of student assessment, and use of advanced technology in teaching, while also having to manage classrooms with increased and challenging student behaviors (Lee, 2020). To be successful, teachers need to believe in their abilities to manage their classrooms and ensure learning through creative application of student engagement strategies to be successful (Bandura, 1977).

Teacher preparation programs are currently not designed to prepare teacher candidates for such a challenging role. The lack of success in the area of classroom management and student engagement leaves new teachers disheartened (Robertson-Kraft & Duckworth, 2014). Veteran teachers, on the other hand, despite having experience in managing classrooms and engaging students, suffer from burnout as a result of too many changes that require time to learn and implement (Tomkins, 1995). According to Jennings (2017), teaching is a highly stressful profession. Furthermore, the workload has been increasing (Ryan et al., 2017). As a result, teachers lack time and are overburdened, which affects their efforts to implement better student engagement strategies and classroom management techniques. Studies show that teachers who are resilient despite facing work-related stress are more confident, and generally have a better ability to handle setbacks (Stoffel & Cain, 2018).

Burnout has been found to be a major reason for qualified teachers to leave the profession and for qualified applicants to disregard teaching as a career option. In cases where teachers facing burnout stay back, they experience emotional exhaustion, depersonalization, and reduced personal accomplishment (Tomkins, 1995). Due to these factors often teachers facing burnout tend to be less empathetic towards students and

colleagues, show decreased tolerance for student issues, prepare inadequately for their teaching, and lose dedication and commitment to their work. (Tomkins, 1995). The overall impact affects teachers' beliefs of self-efficacy in classroom management, student engagement, and general success as a classroom teacher.

School districts across the United States are grappling to fulfill teacher positions in their districts. New Jersey is no different, and despite steps being taken by the state to reduce certification costs and complex steps to get teacher certification for new teacher candidates, the number of unfulfilled positions in school districts is increasing (Weber, 2020). Not many candidates are available and those who have been in the field for a long time are either leaving or reeling under the effects of lack of self-efficacy and contemplating leaving the profession despite being experienced and qualified in their field (Lee, 2020).

Purpose of the Study

The purpose of this quantitative correlational study was to explore the relationship between grit (measured by the level of tenacity, endeavor, and drive) and self-efficacy beliefs in the areas of classroom management and student engagement among elementary and middle (PreK-8) school teachers. The participants of this study were elementary and middle school (PreK-8) teachers in public schools in Northwestern New Jersey. The participants answered demographic questions and completed the short Grit-S scale measuring the independent variable of grit. Grit is defined as "perseverance and passion for long term goals" (Duckworth et al., 2007, p. 1087). The participants also completed the Teacher Sense of Self-Efficacy Scale (TSES) scale measuring the dependent variables of self-efficacy in classroom management as well as self-efficacy in student engagement. Self-efficacy is based on an individual's beliefs in their own capabilities that help the successful completion of tasks (Bandura, 1977). Self-efficacy in teachers is their belief in their abilities to provide instruction along with the ability to engage students and manage classroom environments

despite challenges related to student behaviors (Caprara et al., 2006). The inability to gain a certain outcome is defined as incompetence and it leads to individuals behaving less. On the other hand, the ability to attain an outcome promotes an intrinsic need within individuals which creates a shift that affects feelings, perceptions, and behaviors. Therefore, feeling competent is crucial for individuals to be able to cope with the demands of their surroundings (Deci & Ryan, 1980).

Research Questions

The research questions for the study were formulated based on the fact that for quantitative studies research questions are designed to find relationships among variables being investigated and are usually framed at the start of the study (Barroga et al., 2023). They were also formulated in a manner so that they are linked to the population being studied, the independent variable, the dependent variables, and the research design. This study sought to answer the following questions:

RQ1. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on classroom management?

RQ2. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on student engagement?

Theoretical Foundation

Based on available data, fewer candidates are entering the field of teaching (Weber, 2020). Teachers who are already in the field have been facing burnout due to increasing workloads for teachers each year (Ferguson et al., 2012). This is a result of the constant implementation of new and changing curriculums and teaching methods, changing student demographics in school districts, and challenging student behaviors in classrooms. Teachers feel overwhelmed, stressed, and less confident in their ability to manage classrooms and find

ways to engage students in lessons (Duckworth et al., 2009; Robertson-Kraft & Duckworth, 2014).

The ability of elementary and middle school teachers to persevere in the field of education while teaching multiple subject areas, dealing with constant challenges brought about by changes and demands in the work environment, feeling confident about one's skills related to classroom management and student engagement, and following a course of action to achieve them is grounded in Duckworth et al.'s (2007) theory of grit, Deci and Ryan's (1985) self-determination theory (SDT), and Bandura's (1977) self-efficacy theory.

Nature of the Study

A quantitative, correlational design was used to test if and to what extent there was relationship between the variables of teacher grit and self-efficacy beliefs in classroom management and student engagement in elementary and middle school teachers (PreK-8). Quantitative research involves the measurement of existing organizational data, physiological data, and self-prepared or published surveys that provide a quantified response set (Stichler, 2016). Quantitative research uses statistical processes to analyze the significance of the findings. A correlation is a "quantitative measure of the degree of correspondence" (Gay et al., 2014).

A correlational design is appropriate for this research as it involves the collection of data that helps determine whether a relation exists between two or more quantifiable variables or continuous scores. It also helps in studying the strength of the relationship between the variables (Gay et al., 2014). Continuous data is data that can be measured, it has values that are not fixed and have an infinite number of possible values. Furthermore, continuous data can be displayed in visualizations such as graphs because of its variable change over time. Correlational studies are often used in educational research (Gay et al., 2014). In this study, the Short Grit Scale (Grit-S) score was the independent variable and the

Teacher Self-Efficacy Scale (TSES) scores of classroom management and student engagement were the dependent variables.

Definitions

Attrition: Attrition is a term used to describe voluntary and involuntary terminations, deaths, and employee retirements that result in a reduction in the employer's physical workforce (SHRM).

Burnout: Burnout is defined as a syndrome that results from stress in the workplace and is not successfully managed. The characteristics of burnout have three dimensions. They are “feelings of energy depletion or exhaustion, increased mental distance from one’s job or feelings of negativism or cynicism related to one’s job, and reduced professional efficacy” (World Health Organization, 2019, para. 5).

Classroom Management: Classroom management is a teacher’s ability to build into their instruction, certain practices that engage and motivate students and have the ability to set up an organized environment while also completing mundane tasks (Aloe et al., 2014).

Conscientiousness: Conscientiousness is a continuum of constructs that outline individual differences in tendencies to be in control of themselves, responsible towards others, industrious, systematic, and obedient (Roberts et al., 2009).

Pandemic: A pandemic is defined as “an epidemic occurring worldwide, or over a very wide area, crossing international boundaries and usually affecting a large number of people” (Doshi, 2011, p. 540).

Grit: Grit is defined as “holding steadfast to that goal. Even when you fall down. Even when you screw up. Even when progress toward that goal is halting or slow” (Duckworth et al., 2009, p. 541).

Job Satisfaction: Job satisfaction is the normal attitude of an employee to their job. It is an overall conclusion that the employee develops by balancing satisfaction or dissatisfaction to different parts of their job (Kalleberg, 1977).

Mindset: Mindsets are a set of assumptions that help one filter complex worldviews into easily understandable information and then set expectations based on the input received (Dweck, 2016).

Motivation: Motivation involves energy, direction, persistence, and equifinality, encompassing all aspects of activation and intention (Ryan & Deci, 2000).

Passion: An interest or goal that an individual cares about so much that it marshals and provides meaning to everything they do (Duckworth et al., 2009, p. 541)

Perseverance: The ability to pursue one's goal, despite hurdles and setbacks and bring it to fruition/completion (Duckworth et al., 2009, p. 541).

Professional Development: It "is a systematic process that strengthens how professionals obtain and retain knowledge, skills, and attitudes. It consists of intentionally designed processes and activities developed to improve organizational approaches" (Centers for Disease Control and Prevention, n.d., p. 1).

Self-Determination Theory: Self-determination theory is the stimulus an individual has in order to make decisions and act on them (Deci & Ryan, 1985).

Self-Efficacy: Self-efficacy is an individual's belief in their capabilities to successfully organize and create actions that help attain goals (Bandura, 1997).

Student Engagement: It is a combination of behavioral, emotional, and cognitive involvement that creates a desire and ability within students to participate in activities through strategies implemented by a teacher that closely relate to motivation. This dimension is related to teachers' perceptions of their abilities to engage their students in class content (Tschannen-Moran & Hoy, 2001).

Teacher Self-Efficacy: Teacher self-efficacy is defined as teachers' perception that they can successfully navigate their classes and improve student achievement (Hoy, 2000).

Retention: The ability of a company or organization to keep long-term employees.

Veteran Reacher: Teachers who have spent at least a decade or more of consecutive years in the classroom actively teaching.

Limitations and Generalizability

Several assumptions were considered in the study. I assumed at least 86 teacher participants were willing and available to answer the survey's questionnaire. I assumed that the teacher participants voluntarily participated in the study after they received the email invitation through their school districts. I assumed that teacher participants had internet access to participate in the study. The survey was conducted using an online survey tool called Microsoft Forms. Therefore, I assumed that participants should have internet access to respond to the survey questionnaire. I also assumed that the study would have an accurate representation of teachers from different school districts who taught elementary and middle schools in Northwestern New Jersey. The following delimitations were present in this study:

1. The scope of this study was delimited to public schools in Northwestern New Jersey because I only had access to schools in this region and could only send the survey to those districts that had provided permission within this region.
2. The study was delimited to the period of 2 months of data collection to ensure that the data collected were applicable at the present time.

The following limitations were present in the study. According to Creswell (2013), "limitations are potential weaknesses or problems with the study identified by the researcher" (p. 199):

1. The study was limited to the constructs of grit and teacher self-efficacy measured using the survey instrument considered in the study. The survey used in this study

included constructs of teacher grit, self-efficacy beliefs of teachers in classroom management and student engagement

2. A limitation of this study was that participants might have different interpretations of survey questions, which might influence their responses.
3. Another limitation of this study was using a convenience sampling of teachers. Convenience sampling increased the likelihood that the participants were not a representative sample of the population of PreK-8 teachers in the United States.
4. Using a quantitative nonexperimental correlational design was a limitation. Quantitative nonexperimental correlational researchers could not infer cause-and effect; causality could only be determined in experimental studies.
5. Due to size limitations of the convenience sample, the study may have limited transferability and generalizability to a similar population. The research was analyzed with consideration to the implications of both transferability and generalizability of research.

Significance of the Study

Exploring teachers' attributes that make them successful in classrooms has been an area of interest for researchers in the field of teacher education. Current data indicate declining numbers in terms of candidates completing a teacher certification program in almost every state in the United States as well as many experienced teachers leaving the field of education due to burnout and fatigue (Weber, 2020). Previous studies conducted to study the phenomenon of burnout and attrition among teachers in the United States indicate that teachers in elementary schools are more likely to leave as compared to those teaching in secondary schools (Tomkins, 1995). This has reinforced the need to identify attributes among new teachers as well as, more importantly, among veteran teachers in elementary and middle schools (PreK-8) that help them persevere and succeed.

Understanding the relationship of grit and its impact on the self-efficacy beliefs of teachers in classroom management and student engagement would help several stakeholders. Administrators could benefit by finding the right candidates for positions, who can successfully navigate the challenges that come with teaching the very young learners in elementary and middle school (PreK-8) and grow professionally. It would also help school districts understand the need to provide support, training, and create work environments for experienced/veteran teachers to help them cope with the ever-increasing demands of the job.

Furthermore, creating more layers of support for veteran teachers so that they do not face burnout is essential to keep experienced teachers in districts. District human resource departments could devise better induction programs, provide opportunities for improved professional development, and introduce concepts of teacher self-care by providing resources that help identify burnout and provide ways to address it.

Grit as a personality trait has been identified as a factor in students in various fields of study that helps them persevere and complete their programs of study despite challenges. It has also been predicted as a trait that helps predict longevity in novice teachers (Robertson-Kraft & Duckworth, 2014), veteran teachers in secondary school (Lee, 2020), teachers in secondary schools who have used the alternate route for teacher certification teaching in Title 1 schools (Coronado, 2016).

This study has moved along the body of knowledge on grit and its relationship to self-efficacy beliefs on classroom management and student engagement in elementary and middle (PreK-8) school teachers. The findings of the study will provide policy-makers with insights into the importance of reforms in the area of teacher training, curriculum implementation, and teacher evaluations. School administrators will benefit from insights into the attributes that can help identify the right candidates, improve the levels of support for veteran teachers, and decrease attrition in their districts. Program directors in teacher certification programs would

benefit by understanding what to include in training programs in order to create better programs for new teacher candidates that would help train them to cope with the demands of the job and gain experience and confidence to become successful teachers in classrooms.

Summary

This study sought to investigate the relationship of grit and self-efficacy beliefs on classroom management and student engagement of elementary and middle school teachers (PreK-8) in public schools in Northwestern New Jersey. The study used a quantitative correlational approach to find the relationship between the variables. Literature related to the relationship between grit and self-efficacy beliefs of elementary and middle school teachers in the area of classroom management and student engagement specifically is lacking. Most of the previous studies focus on novice teachers, secondary teachers, and alternately certified teachers, and focus on reasons leading to attrition. The results of this study may serve administrators in elementary and middle schools (PreK-8), who could benefit by identifying attributes in candidates that are more demonstrative of individual grit and how it enables teachers to be effective in classroom management and student engagement. School administrators could also benefit from understanding the need to create more layers of support for veteran teachers so that they are able to cope with the demands of the job and not face burnout and prevent teacher attrition. Data obtained from the study could help human resources departments in school districts understand the importance of devising better teacher induction programs, provide opportunities for improved professional development, and introduce concepts of teacher self-care.

Chapter 1 introduced the topic, identification of the purpose, and problem to explain the need for this study. The chapter included background information, the problem statement, purpose of the study, the significance of this research, research questions, limitations, and delimitations of the study, and definitions of key terms used throughout the study. This study

sought to answer the following questions: Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on classroom management? Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on student engagement?

Four more chapters follow. Chapter 2 is a comprehensive review of the literature on attributes that are key to the retention of teachers, especially teachers who have continued to work, such as experienced and veteran teachers in school districts. In Chapter 2, the findings of previous researchers on teacher self-efficacy and teacher grit are discussed. The chapter also discusses the gaps in the literature related to a model for motivation for veteran and experienced teachers to stay in the teaching profession. The theoretical foundation of the study is based on Duckworth et al.'s (2007) theory of grit, Deci and Ryan's (1985) self-determination theory (SDT), and Bandura's (1977) self-efficacy theory and are discussed in the chapter in detail with relation to teacher abilities to work through problems and feel competent in their jobs. The implications of improving motivation in elementary and middle school teachers and its impact on their ability to succeed in the profession as well as the overall improvement in learning environments in the classroom is also discussed.

Chapter 3, discusses the study methodology. This chapter includes the purpose of the study, research questions, and rationale for using a quantitative correlational design for the study. The chapter also discusses the population of interest and sample selection method, instrumentation and data sources, and the instrument validity and reliability. Chapter 3 concludes with information regarding data collection and management, data analysis procedures, ethical considerations, assumptions, and delimitations.

Chapter 4 details the data analysis and results, including written and graphic summaries. It included information about the preparation of raw data gathered during the study. The chapter also provides descriptive findings and details of the data analysis procedures used. Chapter 4 then discusses the results of the study including any limitations.

Chapter 5 includes interpretations and discussion of the results. The study concludes with the interpretation of findings, implications, and recommendations for future research.

Chapter 2: Literature Review

The purpose of this correlational study was to extend knowledge about personal traits such as grit in teachers in elementary and middle (PreK-8) schools and its relationship to self-efficacy beliefs in the areas of classroom management and student engagement. The study addressed the research question, if and to what extent, was the relationship between individual teacher grit scores and self-efficacy beliefs in the area of classroom management and student engagement. The problem addressed in this quantitative non-experimental study was that more and more teachers are leaving the profession, both novice as well as veteran teachers. Research suggests that there is a need to identify, and look for interventions to manage teacher attrition by exploring the organizational and individual characteristics of teachers who leave and the teachers who stay in the profession despite challenges (Finster, 2015). Existing research on teacher self-efficacy (Korte & Simonsen, 2018), prestige and self-esteem related to teaching as a profession (Klimek, 2019), and attrition related to conditions at work (Harris et al., 2019) found similarities in the absence of certain attributes that lead to teachers leaving the profession. Therefore, understanding the relationship between personal attributes such as grit in teachers and its impact on self-efficacy beliefs is important to reach a better understanding on how to support teachers so that they can be successful in classrooms.

The organization of the literature review focused on examining the theoretical framework, which included the self-determination theory (SDT), the self-efficacy theory as well the grit theory, the variables that are being measured, and pertinent terminologies related to the topic. I examined whether there was a relationship between grit and self-efficacy beliefs in classroom management and student engagement among elementary and middle (PreK-8) teachers. Utilization of several databases with extensive research of scholarly articles gathered from Google Scholar, EBSCOhost Database, and ProQuest Database helped

retrieve the necessary information for this chapter. Additionally, the utilization of Centenary University's library databases, and public libraries provided access to books, peer-reviewed journal articles, and dissertations associated with the topic.

The literature review provides an introduction and background before focusing on the theoretical framework of self-determination theory (SDT; Ryan & Deci, 2000), Bandura's theory of self-efficacy (1986) is based on the Social Cognitive Theory (SCT) and Duckworth's (2016) theory of grit. Related topics and research related to them have been explored in the literature review. There was inconsistent research on whether grit had a positive or negative influence on teacher self-efficacy beliefs, especially among elementary and middle school teachers or whether grit was irrelevant to teacher self-efficacy (Saltman, 2014; Silvia et al., 2013).

More recent research however shows that teacher self-efficacy has a strong relationship to teaching quality, as well as learning (Künsting et al., 2016; Miller et al., 2017; Ryan et al., 2015). Through this literature review, the reader will gain a greater understanding about the correlation between grit and self-efficacy beliefs of teachers in the areas of classroom management and student engagement and review current research about the design, theories, and potential barriers that prevent teachers from continuing in the profession, and overall benefits of the importance of grit and its impact especially in field of educational practice.

Background

The number of teacher candidates entering the profession is declining across the United States. Several studies show high attrition rates among K-12 teachers (Carver-Thomas & Darling-Hammond, 2019; Keigher, 2010; Nyugen, 2020). Available data indicates that novice teachers are leaving the field of education to pursue other careers within the first 5 years (Hise, 2022). Studies also indicate that the main reason for novice teachers leaving their

profession is due to the lack of success in classroom management and student engagement, which leads to them getting disheartened (Robertson-Kraft & Duckworth, 2014).

Meanwhile, studies regarding veteran teachers show that burnout, and lack of motivation are some of the main reasons for them to be exiting classrooms (Lee, 2020; Tomkins, 1995). Although there are studies on the topic of teacher attrition, continued research into the reasons why teachers are leaving the profession will help current policy and practice in K-12 education (Billingsley & Bettini, 2019). Existing research on veteran teachers shows professional identity and commitment as being directly related to longevity in the field (Lowe et al., 2019).

However, the same research also points to a lack of the study of personal characteristics of veteran teachers that may help institutions nurture those traits in teachers to help them stay in the profession (Lowe et al., 2019). Similarly, attrition in teachers has not been studied in relation to personal characteristics of teachers nor has it been studied in relation to institutional dynamics leaving a gap in understanding the real reasons why some teachers leave and some stay in the profession (Kelchtermans, 2017).

An examination of literature on grit and its impact on long-term success revealed that it is an important personality trait that can be used as a predictor. Grit is defined as a voluntary continuation of goal-directed action, regardless of discouragements or difficulties (Park et al., 2004). Self-efficacy in teachers, involves their beliefs in their abilities. Self-efficacy helps teachers cope with the challenges related to their job without being weighed down by excessive stress (Caprara et al., 2006). Grit has been shown to have an impact on self-efficacy beliefs among secondary school teachers, alternatively certified teachers, preservice teachers, and novice teachers. The following sections focus on the theoretical framework of this study as well as available literature on grit and its relationship to self-efficacy beliefs.

Theoretical Framework

Self-Determination Theory

Interplay between the extrinsic forces that act on people and the intrinsic motives and needs essential to human nature is defined by Deci and Ryan's (1985) self-determination theory (SDT). The self-determination theory addresses life concerns, encompassing the fears, anger, and frustrations linked to unmet ongoing and core psychological needs (Deci & Ryan, 2020). According to Deci and Ryan (1980), the inability to gain a certain outcome is defined as directly related to a feeling of incompetence and it leads to individuals behaving less.

Contrary to that, the ability to attain an outcome promotes an intrinsic need within individuals, which creates a shift that affects feelings, perceptions, and behaviors. Therefore, feeling competent is crucial for individuals to be able to cope with the demands of their surroundings. It helps individuals have greater intrinsic motivation and self-determination to accomplish their desired outcomes that they set out to achieve. Feeling incompetent leads individuals to operate on automatic and less self-determined behaviors that are dependent on extrinsic motivation (Deci & Ryan, 1985). Based on SDT, the feeling of competence is important for every individual in order to feel motivated to continue to work.

Applying the same in context to teachers, the ability to manage classrooms while ensuring that students are engaged in content being taught, can make teachers more confident. Positive student response to teachers can provide the motivation for teachers to accomplish the desired outcomes that they have planned. However, this may not always be the case as teachers have to often deal with challenging behaviors in classrooms, other factors such as lack of time to prepare teaching materials, lack of support from administrators, can create hurdles to successful teaching. Some teachers work through these situations and learn from their experiences while others begin to believe that they are ineffective due to challenges with classroom management skills (Korpershoek et al., 2016).

Past studies show that when needs are satisfied, whether it is physiological or psychological, it promotes a feeling of well being (Hull, 1943). Dissatisfaction in terms of needs on the other hand causes individuals to have a feeling of “ill-being” (Hull, 1943). Studies also show that the need to feel competent (Harter, 1978; White, 1963), to have a sense of relatedness (Baumeister & Leary, 1995), and have autonomy in whatever an individual is involved in (DeCharms, 1968; Deci, 1975) is essential for facilitating natural tendencies towards growth and integration, while also effectively creating opportunities of social development and a feeling of personal well-being. Knowledge related to the elements that are essential for positive motivation and experience are therefore crucial as they have a direct influence on the well-being and performance of individuals especially in work places (Deci & Ryan, 2000).

The self-determination theory identifies motivation as the key to understanding individual responses to given situations and challenges in everyday environments whether it is a student who strives to excel in a subject or an employee who works hard to achieve results. According to SDT, there are two types of motivation, autonomous motivation, and controlled motivation (Deci & Ryan, 2008). Intrinsic or extrinsic factors can motivate action by an individual. Autonomous motivation is influenced by intrinsic or extrinsic factors. Actions as a result of autonomous motivation can result in a positive sense of self (Deci & Ryan, 2008). Extrinsic motivation is dependent on rewards and accolades to perform an action. Intrinsic motivation, which is the natural inclination to pursue new experiences and challenges, to expand and utilize one's abilities, to explore, and to acquire knowledge (Deci & Ryan, 2020), has been found to describe an individual’s natural tendency that is a source of “enjoyment and vitality” (Csikszentmihalyi & Rathunde, 1993; Ryan, 1995) throughout their lives. However, intrinsic motivation is not the same in all individuals. A subtheory within SDT is the Cognitive Evaluation Theory (CET) (Deci & Ryan, 1985). According to CET,

social and environmental factors can explain variability in intrinsic motivation among individuals. Based on the CRT, a core need among individuals is to feel competent when performing an action.

According to Deci and Ryan (1975), social and environmental contexts in work places have the ability to influence intrinsic motivation among employees such as positive or negative feedback. Furthermore, these factors also influence an individual's perception of their own competence (Vallerand & Reid, 1984). According to CET, when individuals undergo a behavior that is self-determined and accompanied with a feeling of competence or effectiveness it often is a result of inner resources within the individuals (Reeve & Deci, 1996). On the extreme opposite side of the self-determination continuum, based on Deci and Ryan's (1995) theory, is amotivation, a complete lack of motivation. Individuals who are in this state either lack the intention to act or they act without intention often just going through the motions. This is often the result of not valuing an activity (Ryan, 1995), due to a feeling of incompetence (Bandura, 1986), or performing an action often with the perception that it will not yield the expected or desired result (Seligman, 1975).

Controlled motivation is externally regulated. Behaviors are influenced by rewards or penalties (Deci & Ryan, 2008). Each type of motivation, autonomous and controlled, leads to different outcomes. However, autonomous motivation greatly influences an individual's successful pursuit of personal goals (Deci & Ryan, 2008). Motivation therefore is a critical variable that can bring about constant enhancement of knowledge and experience and promotes curiosity to learn and grow. Understanding the attributes within an individual that promote positive motivation, which in turn has the ability to help with their feeling of competence and well-being, has significant relevance in terms of creating conditions that can help not only with gaining knowledge but also behavior regulation (Sheldon et al., 1996).

Self-Efficacy Theory

Self-efficacy can be defined as an individual's confidence in his/her own ability to accomplish a goal when conditions are apparent (Holzberger et al., 2014; Meristo & Eisenschmidt, 2014; Miller et al., 2017; Ryan et al., 2015; Tschannen-Moran & Hoy, 2001). Studies show that self-efficacy is not an inherent trait. An individual can improve self-efficacy through learning and practice (Althauser, 2015; Knaggs & Sondergeld, 2015; Menon & Sadler, 2017). Self-efficacy is based on Bandura's (1977) theory.

Bandura's (1977) self-efficacy theory looks deeper into the aspect of an individual's need to feel competent. It is derived from the Social Cognitive Theory (SCT) and addresses the aspect of an individual's beliefs in his or her own capabilities (Bandura, 1977). The Social Cognitive Theory (Bandura, 1986) used three factors in its model. The first factor was personal attributes, such as internal thinking; the second factor was environmental factors, and the third factor was behavior (Lent et al., 1994). The concept of self-efficacy was built on SCT (Bandura, 1986).

Bandura (1977) defined self-efficacy as a set of "beliefs in one's capabilities to organize and execute the courses of action required to produce given attainments" (p. 2). According to Bandura's theory, the greater the feeling of self-efficacy, the more the chances of success in completing tasks. Self-efficacy is related to self-confidence, self-esteem, self-worth, and self-concept (Bandura, 1986). Furthermore, the perception of self-efficacy is greater than reality (Bandura, 1986). An individual's perception of their abilities is often a much better predictor of success than the actual measurement of their skills and ability (Bandura, 1986). The self-efficacy theory is based on the idea that individuals as part of their experience learn and adapt to the needs of the situation and the environment in which they work. This helps them develop skills, and confidence which then leads to a perception of expertise propelling them to take risks and continue to grow.

When applied in the context of teaching, studies have shown that teachers with higher self-efficacy have greater success in their field (Schunk & DiBenedetto, 2016). Self-efficacy helps teachers feel empowered, which affects their ability to engage students, manage the classroom, as well as effectively use instructional strategies (Schunk & DiBenedetto, 2016). Teacher self-efficacy has been shown to be a solid predictor of positive student outcomes, such as student behavior, academic achievement, and student-teacher relationship (Althauser, 2015; Holzberger et al., 2014). However, self-efficacy in teachers is not resistant to external factors. Studies show that teacher self-efficacy can be altered due to various causes; however, many of the factors still remain unidentified (Althauser, 2015; Menon & Sadler, 2017).

In the case of experienced/veteran teachers, since they have had time to gain experience and develop hands-on skills, self-efficacy is often thought to be higher. Individual perceived self-efficacy affects human behavior both directly and indirectly through individual perceptions, expectations, confidence, goals, and the personal ambitions and objectives they set for themselves (Bandura, 2006). This relates to some studies that indicate that despite the general perception of the many rewards of teaching, challenges and setbacks can be disheartening for teachers (Kane et al., 2006).

Teachers in their work environments often experience varying levels of stress as a result of having to manage student behaviors, student achievement levels, and decreasing parental involvement and responsibility (Herman et al., 2018; Lentfer & Franks, 2015; Skaalvik & Skaalvik, 2017). Stress has been shown to have a negative impact on teacher efficacy levels (Aloe et al., 2014). However, teachers who are able to navigate stress and develop a more positive outlook toward daily challenges have been shown to be able to improve their emotional well-being and commitment to their jobs (Zee & Koomen, 2016).

According to Caprara et al. (2006), self-efficacy in teachers involves their beliefs in their abilities and that helps them cope with the challenges related to their job without being

weighed down by excessive stress. According to Bandura (1977), individuals who avoid difficult tasks reduce their ability and opportunity to develop the required coping skills. As a result, it increases the fear of failure at tasks to develop within them (Bandura, 1977). Self-efficacy is therefore directly related to success or failure when performing a task (Bandura, 1977, 1994; Pfitzner-Eden, 2016). Individuals who gain proficiency through practice and experience increase their expectations. Based on studies on students who experience high levels of anxiety and stress with regards to a task, the results indicate that students often expect to be unsuccessful and as a result are less likely to attempt the task (Dale et al., 2018). Similarly, studies with regard to performance accomplishments show that individuals who upon experiencing failure, fear trying to reattempt, and tend to lower their expectations. Lowering of expectations leads to a decreased sense of self-efficacy (Pfitzner-Eden, 2016).

Other factors that also play an important role in an individual's sense of self-efficacy include personality traits such as conscientiousness. Studies show that individuals who are conscientious and open to new experiences have a higher sense of self-efficacy (Costa & McCrae, 1992; Djigić et al., 2014). According to Tschannen-Moran et al. (1998), self-efficacy varies substantially across different tasks and is dependent upon the environment of the task. This is indicative of the fact that despite being proficient and confident in managing classrooms as well as teaching content many veteran teachers seem to be overwhelmed and attrition statistics show increased numbers of veteran teachers leaving the profession. Further research into work environments and support systems in teaching environments may help improve attrition among veteran teachers.

Grit Theory

Grit is defined as “perseverance and passion for long term goals” (Duckworth et al., 2007, p. 1087). Grit has been found to be a valid predictor of long-term success (Duckworth et al., 2007). Duckworth et al.’s (2007) initial research focused on cadets at West Point. The

results indicated that cadets who had greater grit were more successful in completing the difficult cadet training program (Duckworth, et al., 2007). An important factor that was highlighted by Duckworth's study was that IQ did not play a role in cadets being able to complete the rigorous cadet training program, rather it was individual grit scores that reflected success. A similar study conducted on high-achieving students who competed in the National Spelling Bee competitions indicated that participants with higher grit levels were likely to make it to the finals (Duckworth & Quinn, 2009). There has been an increased interest in recent years about the effect of a personality trait called "grit" (Duckworth, 2016) and its relationship to pursuing long term goals.

According to Duckworth et al. (2007), grit indicates a tendency toward hard work with sustained effort and commitment toward a chosen area of pursuit. This personal attribute, which was identified as grit, helps individuals accomplish goals through "cumulative effort" (Duckworth et al., 2007). Individuals who demonstrate high grit levels are able to continue sustained long-term efforts toward achieving their goals (Park et al., 2004). During the process of working towards their goals, individuals who have higher grit can navigate hurdles and tolerate setbacks without giving up (Park et al., 2004). Furthermore, individuals with high grit set their own goals and pursue them (Park et al., 2004).

Grit is defined as a voluntary continuation of goal-directed action, regardless of discouragements or difficulties (Park et al., 2004). Hence, in the context of teachers, gritty teachers show the ability to persevere amidst heavy workloads and excessive demands on a long-term basis. Studies conducted on new teachers in the Teach for America program showed that higher grit scores were directly proportional to higher reports of job satisfaction among the new teachers (Duckworth et al., 2009).

Grit was found to have a direct correlation to achievement in the case of students (Duckworth et al., 2009). Grit has also proven to be an important attribute that is being

studied in the context of marriage, business, and the military (Duckworth, 2016; Eskreis-Winkler, et al., 2014). Therefore, studying grit in correlation to performance can be a great predictor in varied fields. Achievement has traditionally been reflected by IQ. However, individuals with more grit have been found to not necessarily have higher intellectual capacity or better talent. According to Fitzgerald and Laurian-Fitzgerald (2016), grit has an impact on an individual's intellect. When individuals intentionally persevere with passion, it leads to persistence which in turn leads to increased intellectual capacity. This is also supported by studies in the context of students which show that those students who persist become academically proficient (Duckworth, 2016). Grit has also been shown to have a direct correlation to short-term memory (Dale et al., 2018).

High grit scores using the Grit Short Scale (Duckworth & Quinn, 2009) among individuals have been shown to indicate higher levels of active working memory, and suggest that a positive relationship exists between grit and intellectual ability. Grit has been shown to be a strong predictor of success for pre-service teachers (Bashant, 2014). Grit can be shaped (Hochanadel & Finamore, 2015) and therefore learned (Duckworth, 2016; Hochanadel & Finamore, 2015). Since grit has been shown to increase an individual's intellectual ability over a period of time, fostering grit among teachers through professional development, by putting in place better supports to prevent teacher burnout should be an area that district administrators should focus on (Kim et al., 2019). Grit is "the tendency to pursue long-term goals with sustained zeal and hard work" and has been used, "to predict achievement in academic, vocational, and avocational domains" (Von Culin et al., 2014, p. 1). Therefore, identifying grit among teachers and its relationship to self-efficacy beliefs is important to understand what attributes help teachers persevere and follow their passion.

Teacher Attrition

Based on the report by the Pew Research Center, even though there has been an overall increase in college degrees, there has been “a profound drop in the numbers of education degree earners, and that decline has become more marked over the decades” (Douglas et al., 2024, p. 6). Based on available data the percentage of students earning degrees in education in 1970–1971 was 21% (Shaeffer, 2022). It was 8% in 2002–2001 and 4% in 2019–2020 (Douglas et al., 2024). In terms of teacher exits from the profession, the report indicates a steady trend. However, alarmingly “one-third of exits each year were by elementary school teachers” (Douglas et al., 2024, p. 16). Furthermore, 10% to 11% of exits were by middle school teachers” (Douglas et al., 2024, p. 16).

At least 10% of new teachers opt to leave the profession within their first 3 years (National Center for Education Statistics, 2023). Taking this factor into consideration as well as the statistics related to the rate of replacement of teachers, 2013–2014 through 2016–2017, the rate of replacement was three to one. This declined to two-to-one in 2020–2021 and is currently close to one-to-one (Douglas et al., 2024). The effects are visible in the face of school districts in New Jersey struggling to fill positions for teachers. It has definitely caused enough concern to have led to concerted efforts to shoring up routes into the profession in the past 2 years. Furthermore, a longitudinal study conducted by Sorensen and Ladd (2020) through the analysis of data over a 22-year period showed that a 3-year teacher turnover rate of 11% can show a drop of almost 8% in high stake testing scores in reading and 13% in mathematics.

Studies related to teacher attrition also show that factors that influenced teacher attrition were related to characteristics of the school, remuneration, school location, environmental conditions, teaching conditions, and teachers' satisfaction levels (Meyer et al., 2019). Furthermore, studies indicate that novice teachers with 1 to 5 years of experience are

often employed on a contractual basis. As a result, teachers face tensions and uncertainties surrounding contract renewal (Djonko-Moore, 2016). This contributes to teacher attrition within 5 years of their careers. Poverty also has an impact on attrition. Studies show that areas with high poverty margins had high attrition due to the teachers' hardships in such areas (Djonko-Moore, 2016).

Often teachers who are new and are employed on a contract basis in schools take on a high workload as they lack autonomy (Van Droogenbroeck et al., 2014). Novice teachers exhibit the highest burnout rates because they report the lowest autonomy and the highest workload than teachers occupying senior positions (Hise, 2022). Low tolerance for work-related stress and lack of skills, prevent novice teachers from performing effectively and efficiently (Bettini et al., 2017; Kucukoglu, 2014) According to Jang and Kim (2019), low grit and job stress were significant predictors of turnover intention. Studies show that grit has a mediating effect on the relationship between job stress and turnover intention in teachers and daycare providers (Jang & Kim, 2019; Shin et al., 2018).

Therefore, strategies geared toward teacher retention are critical to address the current teacher shortage. Teacher retention is the ability to reduce mobility (Hise, 2022). It involves strategies and interventions that are designed to promote teachers to stay in the profession (Shaw & Newton, 2014). Teacher retention would need to involve the creation of favorable conditions tailored to the prevailing needs of teachers (Hise, 2022). Based on teacher resilience studies by Bobek (2002) there are some senses that are important for teachers to feel in order to want to stay.

Table 1

Senses that Influence Teachers' Willingness to Stay in Their Job

Sense of belonging	When teachers, especially those new to the profession, find others who they can relate with and can empathize with what they are currently going through.
Sense of competence	A feeling of self-efficacy in the classroom through effective strategies and having the credentials to support their level of confidence.
Sense of ownership	Being able to set achievable targets and pursuing them with success that leads to rewards.
Sense of accomplishment	Recognition of accomplishments and efforts by administration reinforcers teachers positively and promotes perseverance in times when they are faced with challenges and roadblocks.
Sense of humor	Being able to handle difficult situations with a sense of humor essentially promotes collegial feeling among staff while helping with a better way to release tension.

Note. Adapted from "Teacher Resiliency: A Key to Career Longevity." by B. L. Bobek, 2002, *The Clearing House*, *75*(4), 202–205.

Educational stakeholders, such as school administrators, can ensure better rates of retention with the use of these interventions (Hise, 2022). Further, studies that delve into personal attributes that can enhance the ability to manage stress and persevere can help develop professional development for teachers that administrators can use to reduce attrition in schools and help more teachers stay in the profession.

Teacher Burnout

According to Maslach and Jackson (1981), burnout is a syndrome. It is characterized by a reduced sense of accomplishment, emotional incapacitation, and detachment. It often occurs among individuals who do "people work" of some kind. "People work" often involves individuals who work as human service professionals (Farber, 1991). Those working in educational institutions are human service professionals. Teachers are intensely involved with students' problems for a great deal of time (Tomkins, 1995). The emotional drain as a result

of repetitive stress for people involved in such work can lead to burnout (Maslach & Jackson, 1993).

Studies show that the combined effects of heavy workloads and burnouts cause anguish and eventually force employees to leave which is not uncommon among teachers (Torres, 2016). According to Johnson (1990) and Fennick (1992), oftentimes it is some of the most qualified teachers who choose to leave the profession as a result of burnout. The attrition that results from such situations is expensive for schools and affects students, and society as a whole (Pines & Aronson, 1988). In cases where teachers decide to stay despite facing burnout often there is very little enthusiasm or attachment on their part (Starnaman & Miller, 1992).

According to Maslach and Jackson (1981), burnout is internal. It involves attitudes, feelings, perceptions, and expectations and occurs at an individual level. A common symptom of burnout is a feeling of "inconsequentiality" (Farber, 1991, p. 25) on the part of the individual. According to West et al. (2009), burnout is a syndrome of emotional exhaustion, depersonalization, and low personal accomplishment that culminates in decreased effectiveness at work. However, even though burnout makes an individual feel inconsequential, most often burnout besets people who are highly motivated and believe their work provides meaning and substance to the world (Pines & Aronson, 1988). According to Qureshi et al. (2013), stress at work is positively correlated with turnover intention in the field of education mainly due to teachers having unique stressors (Hakanen et al., 2006; Pogere et al., 2019; Skaalvik & Skaalvik, 2017). Teacher job stress results from the inability to balance responsibilities and demands (McCarthy, 2019).

However, job-related stress cannot always be considered negative Gold (1984). Stress in some cases provides individuals an incentive to find ways to solve a problem. But when that same stress ceases to provide a stimulus to move forward it descends like a heavy weight

on the individual, causing damage (Cedoline, 1982). Stress is associated with all kinds of work hence burnout is a negative consequence of stress (Gold, 1984). According to Tomkins (1995), the role played by an individual teacher's overall attitude toward the teaching profession, along with administrative support and cooperation among the staff in a school and classroom autonomy are crucial in determining teachers' attrition. Job stress among teachers results from the inability to balance responsibilities and demands (McCarthy, 2019). Often this is mainly due to struggles related to adapting to the perceived roles and perceptions versus the reality of the actual job, which leads to stress and burnout.

Therefore, when individuals feel supported through positive interactions and actions around them, it provides them the incentive to persevere despite setbacks. According to a study by Wright et al. (2019), strategies that are designed to build resilience and perseverance within educators can increase retention. School leaders and administrators can benefit by understanding the importance of fostering greater cooperation among the entire school staff to minimize the negative perceptions held by teachers that most likely lead them to leave the profession.

Classroom Management

According to Marzano et al. (2003), one of the most critical challenges for teachers is classroom management. Studies show that most teachers who leave the profession primarily state classroom management as their main reason (Aloe et al., 2014; Korpershoek et al., 2016). Increasing teachers' sense of self-efficacy in the area of classroom management early in their careers is the key to helping teacher effectiveness and increasing their chances of succeeding long-term (Celep, 2000). Teaching content can only be successful when classrooms function in a cohesive and balanced manner. Teachers who lack training and skills in effective classroom management find it challenging to keep students engaged (Shaukat & Iqbal, 2012; Wolters & Daugherty, 2007). However, most formal teacher

preparation programs overlook effective classroom management as an essential component of their training program (Eisenman et al., 2015).

Previous studies show that districts often have been accused by The National Commission on Teaching America's Future (NCTAF) for employing unqualified teachers because of poorly enforced regulations (Fox & Peters, 2013). States are required to evaluate certification programs annually and report their findings to demonstrate accountability (U.S. Department of Education, 2014). The Department of Education evaluates the effectiveness of programs. "According to federal guidelines of the NCLB, a highly qualified teacher must have (a) a bachelor's degree, (b) full state certification or licensure, and (c) subject competency in the subject they teach" (Fox & Peters, 2013, p. 3). However, guidelines cannot replace mentoring and support in the workplace unless administrators are able to provide it in order for teachers to be successful once they enter classrooms (Lewis, 2008). Effective leadership has been linked to teacher effectiveness, and teacher effectiveness has also been linked to a teacher's sense of self-efficacy (Fox & Peters, 2013).

Some studies indicate that teachers who go through traditional certification programs are stronger in content and pedagogy when compared to those teachers who get certified through alternative certification programs (Corbell et al., 2010; Darling-Hammond, 2010). However, the same studies also recommend that having mentors who can provide the right guidance and influence make a huge difference in ensuring a novice teacher's sense of effectiveness in their practice. Other studies found no significant difference in teachers' self-efficacy and effectiveness due to certification type (Constantine et al., 2009; Tournaki & Podell, 2005).

Teachers with a higher sense of self-efficacy effectively manage student behavior in their classrooms (Weinstein, 1989). Since a teacher plays a vital role in the success of students in the classroom, having a strong sense of self-efficacy in teachers can directly

influence the effectiveness of teachers as well as students (Darling-Hammond et al., 2002). Often a highly qualified status is prioritized as the benchmark of teacher efficacy. However, studies show that it does not equate to increased teacher efficacy and effectiveness (Kane et al., 2008). It generally indicates that the teacher met all the required criteria prescribed by the federal and state departments of education such as having a degree, being certified, and demonstrating subject competency. It is not an indicator of effective classroom management skills.

The measure of effective classroom management is to reduce classroom disruptions, as a result of student aggression, lack of student engagement in content being taught and participation in classroom activities, vandalism, profanity, cell phone usage, and willful disobedience, to name a few. Effective classroom management strategies allow teachers to manage disciplinary issues without disrupting learning and keeping students engaged (Demirdag, 2015). Often the lack of time and resources, support from leadership, and relationships with colleagues influence the self-efficacy beliefs of teachers in the area of classroom management, which is a major cause of job dissatisfaction among teachers (Boyle et al., 1995). While recent changes in certification programs have attempted to incorporate more training for classroom management, teachers entering the profession still have trouble making the training applicable in the real world. Studies indicate however the type of certification has not been seen to have a significant relation to self-efficacy and classroom management.

Growth Mindset

Dweck (2006) theorized the growth mindset. Dweck reframed the mindset theory. According to the theory, when people are provided with a perspective that supports the growth mindset, struggle and effort follow that perspective. A growth mindset is the belief that one can improve and does not have a fixed amount of capability or intelligence and is

connected to brain plasticity (Dweck, 2006). Dweck (2007), theorized that mindsets are an individual's beliefs about basic qualities such as intelligence, talents, and personality. Studies show that mindsets play a significant role in motivation, self-regulation, achievement, and interpersonal processes (Dweck, 2017).

Based on Dweck's (2006) theory, two types of mindsets were identified, fixed (I think I can't) and growth (I think I can) situated at either end of a continuum. Individuals with a fixed mindset perceive their abilities as unchanging, apply less effort to succeed and dismiss challenging learning opportunities for fear of failure. Individuals with a growth mindset believe that their intelligence is moldable, they understand that mistakes are part of the learning process and persist when faced with challenges or setbacks (Dweck, 2007). According to Dweck (2016), an individual may not have one mindset or the other all the time but often may have a mixture of both depending on the task. Therefore, they may be placed along the mindset continuum for different tasks or abilities at different times of their life Dweck (2006).

The growth mindset is based on an incremental theory. It relates to individuals making positive judgments that they can develop and achieve (Dweck, 2006). According to the growth mindset, it is possible for an individual to develop to be happier, more successful, and build higher motivation (Dweck, 2006). On the other hand, when individuals engage in a fixed mindset that focuses on enduring traits, they are afraid of challenges and often devalue effort.

Studies provide extensive evidence that mindset is an essential component that influences an individual's belief that they can learn, have self-control, and develop skills over time (Dweck, 2007; 2008; Paunesku et al., 2015; Rattan et al., 2015; Yeager & Dweck, 2012). Studies in relation to academic success in students indicate that growth mindset strategies have a positive effect on individuals while shaping their efforts toward success.

According to Bandura (1997), individuals who have higher self-efficacy are more likely to persevere and therefore likely to be successful. Therefore, self-efficacy accounts for a major portion of performance and persistent behaviors in the context of academic performance (Multon, 1991). Individuals with high self-efficacy achieve success when they adopt a growth mindset as they are motivated to learn despite setbacks (Dweck, 1988). Self-efficacy acts as the hook within individuals that makes them believe in their capabilities (Bandura, 1977). Studies show that self-efficacy and a growth mindset can be strengthened through interventions (Wasylkiw et al., 2020). These interventions are mainly the sources of self-efficacy that promote an individual's confidence in their capacity (Bandura, 1977).

Mindset in the context of teachers, is their perceived ability as a classroom teacher to effectively engage their students in learning while being able to manage classroom environments. Teachers with a growth mindset are more likely to take instructional risks in their classrooms. Teachers with a growth mindset also believe that they can learn from others. They are not deterred when they make mistakes and take those opportunities to recognize areas that they need to work on (Dweck, 2014). Teachers with a growth mindset persevere even when faced with challenges in the classroom. Teachers with a fixed mindset can change and grow into a growth mindset (Bashant, 2014; Dweck, 2006; Hochanadel & Finamore, 2015).

Improving grittiness and mindset can help teachers improve their practice. Grit and a growth mindset affect teacher retention (McClendon, 2017). Creating avenues for professional development that enhance these non-cognitive skills, grit, and growth mindsets among teachers will help them develop perseverance, belief in abilities, and determination to keep moving forward despite challenges by making them more malleable.

Conscientiousness

Conscientiousness is a trait within the big five personality traits. Roberts et al. (2014) explained conscientiousness is a spectrum of constructs that describe individual differences in the propensity to be self-controlled, responsible to others, hardworking, orderly, and rule abiding. Conscientiousness is a trait that is used as a predictor of behaviors and has been shown to be highly correlated with academic achievement, employee performance, martial outcomes, income, success, and life satisfaction. Duckworth et al. (2007) recognized the overlap that conscientiousness and grit share. However, Duckworth et al. (2007) emphasized the distinct ability of individual grit that helps individuals maintain interest and effort over the short and long-term.

Grit and conscientiousness are associated but according to studies, grit demonstrated itself to be a stronger predictor (Duckworth et al., 2007). Similarly, Credé et al. (2017) found grit and conscientiousness to be very similar constructs. However, according to Tynan and Harms (2017), grit's distinction and strength lie within the perseverance of effort facet of the construct. Conscientiousness is traditionally identified as one of the big-five personality traits, however, it is also a key facet of the psychological construct of grit (Eskreis-Winkler et al., 2014; Fite et al., 2017).

Self-efficacy as a key internal psychological resource, has a strong relationship with the big five personality. According to the social cognitive theory, self-efficacy is an intermediary variable linking personality traits and social behavior (Xie, 2012). General self-efficacy refers to the degree of self-confidence individuals have about their ability to achieve specific accomplishments through their abilities (Bandura, 1977). Studies show that individual personality traits are part of core self-evaluation and self-concept, and general self-efficacy is one of its components (Judge et al., 1998).

The lack of belief in an individual's ability to solve problems may affect a person's neuroticism and will finally result in anxiety (Bandura et al. 1985). Therefore, there are reasons to believe that self-efficacy plays a part in the relationship between personality traits and anxiety. Based on a study conducted by Barańczuk (2021), individuals who demonstrated higher levels of extraversion, openness, agreeableness, and conscientiousness had a higher level of self-efficacy. According to Bruso et al. (2020), a positive personality can keep individuals motivated and improve their general self-efficacy by keeping their core self-assessment and acknowledgment of their abilities at a high level. Furthermore, self-efficacy is negatively associated with anxiety, and anxiety reduction could be achieved by improving an individual's general self-efficacy (Bruso et al., 2020). Therefore, districts can retain novice as well as veteran teachers through concerted efforts directed towards creating opportunities through training, professional development, support systems, and general work environments that promote autonomy, a sense of control, well-being, and a willingness to try.

Grit and Teacher Efficacy

According to Duckworth (2016), feeling connected to one's work and having a sense of purpose where an individual believes that their work makes a difference in the lives of others helps individuals work longer and better. Teachers focus on student growth and developing potential among them. Research shows that individuals who choose teaching as a profession believe that it is their way of helping society (Farkas et al., 2000). However, unlike other professions where efforts result in a tangible result, teachers oftentimes do not get to view results as it is eclipsed by many factors (Thompson, 1991). Therefore, in the case of teachers most often their strong calling to the profession and its larger goal despite the absence of visible immediate results is a driving factor that helps them continue in the profession (Haderer, 2020). To perceive that one's work in an area can affect change is a powerful fuel to one's purpose in life (Duckworth, 2016). This is one of the factors that helps

teachers continue in their field of work despite challenges (Robertson-Kraft & Duckworth, 2014).

Skill development in an area of interest usually leads individuals to practice more and develop mastery (Vallerand, 2012). Among teachers. a sense of purpose coupled with interest and practice helps them develop their abilities and confidence as they gain experience in the classroom. However, over the years, changing trends in demographics within classrooms, curricular changes, learning and using new technology efficiently for instructional needs, and managing behavioral issues in classrooms while ensuring consistent student engagement so that classrooms are managed efficiently bring about situations for teachers where they may start to doubt their beliefs of self-efficacy despite having adequate experience and mastery over subject areas. Moreover, there are no preordained responses to handling these situations. Teachers usually are responsible for countless and varied decisions on a regular basis along with their regular work load of teaching and managing classrooms. Research shows that reflection combined with a problem-solving approach, rather than engaging in assigning blame helped teachers develop persistence (Huisman et al., 2010).

Engaging in deliberate practice (Duckworth, 2016) by seeking help from experts through professional learning opportunities to develop mastery through experiential knowledge (Tschannen-Moran & Hoy, 2001) envelops within itself the aspect of grit as well as self-efficacy. Therefore, relationships whether informally with colleagues or formally through professional learning opportunities are key for teacher efficacy (Aldridge & Fraser, 2016). This includes school climate and administrator support that supports teachers by way of creating a community where teachers feel supported. In the case of novice teachers, having a positive relationship with mentors helps them feel supported (Yost, 2006).

Feeling supported reassures new teachers and helps them develop a sense of understanding of what they may need to adjust in their practice to do better as they continue

onto their next year. According to Carrillo and Flores (2018), understanding the factors that influence the identity of teachers is important for school leaders in order to successfully retain and lead them.

Table 2

Teacher Identities that are Interrelated

Personal	1. Teacher as a person 2. Personal circumstances 3. Family issues 4. Children related issues 5. Spiritual issues
Situated	1. Students 2. Colleagues 3. Administration 4. Families
Professional	1. Recurrent reforms 2. Culture of accountability 3. Professional development opportunities 4. Professional responsibilities 5. School context

Note. Adapted from “Issues influencing veteran teachers’ identity” by Carillo & Flores, 2018. p. 647.

Besides professional competence, identifying the internal and external issues that maintain teachers’ motivation and commitment to teaching and understanding the role of emotions in the transformation of veteran teachers' identities is important. Carillo and Flores (2018) categorized the factors into three separate sections that have permeable boundaries and are interrelated to each other in the form of personal, situated, and professional issues as shown in Table 2.

A characteristic of teacher efficacy is using reflection as a means to develop mastery (Tschannen-Moran & Hoy, 2001). When new teachers feel that they are successful they persist through challenges to gain experience which characterizes grit. Based on the social cognitive theory, teachers who believe they may not be successful in the classroom are less

likely to put in much effort in engagement, classroom management, and lesson preparation, despite having the knowledge of strategies to use (Tschannen-Moran & Woolfolk Hoy, 2007). A teacher's sense of self-efficacy is defined as their perception of the degree of success that they can achieve in their classrooms and improve student achievement (Bray-Clark & Bates, 2003; Fox & Peters, 2013; Hoy, 2000). When teachers are able to develop confidence in their ability to manage classrooms and teach their students it leads to a sense of increased self-efficacy. However, situations within the work environment such as collegial support, administrative support, and school climate may affect teachers' sense of self-efficacy and, therefore, can cause efficacy levels to vary (Tschannen-Moran & Woolfolk Hoy, 2007).

Studies on the relationship between teacher efficacy (Skaalvik & Skaalvik, 2010; Tschannen-Moran & Woolfolk Hoy, 2001; Tschannen-Moran et al., 1998; Zee & Koomen, 2016) and grit (Duckworth et al., 2009; Robertson-Kraft & Duckworth, 2014) indicated that these constructs have a direct impact on teacher beliefs in their self-efficacy and ability to stay in the profession. Furthermore, studies involving the relationship between the constructs of teacher efficacy and grit controlling covariates of years in teaching, placement (elementary, middle, or high school), and teaching as a second career indicate that teacher perception of their impact on student growth, relationship with colleagues, reflective practices that involve seeking informal help from peers as well formal learning through professional development (Haderer, 2020) help renew passion for teaching and staying in the profession.

According to Bandura (1997), mastery experiences are the most powerful because they show an individual's ability to take on a task. When an individual experiences success in doing a particular task repeatedly, their sense of self-efficacy increases when presented with a similar task in the future. However, if an individual has previously met with failure repeatedly with a particular task, then his or her sense of self- efficacy decreases when

presented with a similar task again. Vicarious experiences on the other hand occur by observing competent colleagues successfully model or perform a task. (Bandura, 1997). Similar to mastery experiences, the success of the individual providing the modeling directly impacts self-efficacy beliefs of the observer and therefore, it is vital that the person modeling the behavior is able to successfully do so. For teachers in their initial years of practice, this is usually their mentor. The right kind of model instills within the learner a sense of trust, someone they can identify with (Bandura, 2008; Tschannen-Moran & Hoy, 2007).

Similar to vicarious experiences, verbal persuasion affects self-efficacy beliefs. The trustworthiness and credibility of the person delivering verbal praise or feedback to the learner (Bandura, 1997) with regard to the completion of a task has a direct impact on self-efficacy beliefs (Bandura, 2008; Hamman et al., 2006). Verbal persuasion has a stronger effect on self-efficacy when it focuses on the feedback related learners improving. Physiological arousal on the other hand refers to reactions that are physiological or emotional with regard to completing a task (Bandura, 2008). An individual's self-efficacy level changes depending on their interpretation of stress, anxiety, and excitement. This may manifest as either fear or pleasure. In the case of teacher efficacy, these reactions are mostly established during the early years of their career. Therefore, it is essential to examine teacher self-efficacy early on because with time the beliefs get ingrained and the possibility of changing this response is reduced (Bandura, 1997).

Teacher achievement most often is viewed from the perspective of student achievement in classrooms. Past studies on professionals in the field of education have associated student performance with teacher effectiveness (Sanders & Rivers, 1996). Although there is definitely a relationship between the two, individual grit among teachers may be a better indicator of how teachers persevere and work passionately in their field to be successful. According to Duckworth (2016), when an individual's achievement is viewed

through the lens of grit there are two essential parts. The talent that an individual possesses and the amount of effort that the individual puts in. When these parts are combined in a ratio where effort is double that of talent it leads to skill in an area (Duckworth, 2016). Even an individual may have talent in a field without effort; talent alone cannot guarantee success. Similarly, individual skill along with sustained effort can help an individual hone their craft (Duckworth, 2016).

In the context of teachers, choosing teaching as a career has been indicated from studies to mostly be influenced as a result of their interest or calling for the profession (Farkas et al., 2000). When an individual enjoys what they do which can be associated with passion, they tend to persevere even though they meet obstacles based on Duckworth's (2016) theory of grit. Studies conducted on employee retention in different organizations indicate that those employees who are interested in their work, work better, they help fellow employees, and stay longer in the organization (Nye et al. 2012). Teachers, although driven by their interest and passion to bring about changes that influence others, have too often dealt with much more than just doing what they are interested in. Teachers, besides preparing lessons that are planned to interest students and engage them in meaningful activities that evoke varied methods of expression, have to deal with instances of disruptive behaviors, implement new curricula and methodologies, and use new and advanced technology while also worrying about changing teacher evaluation systems. This requires more than just passion to help sustain them in classrooms.

Dealing with constant changes and practicing them based on what they experienced can help teachers fine tune their skills. This may not just be associated with how many years a teacher may have spent in the classroom but by repeatedly doing the same thing many times to become effective in implementing it (Duckworth, 2016). Based on studies conducted on candidates in the Teach for America program, which places college graduates in school

districts that are mostly underfunded and therefore have more challenges to deal with in classrooms across the United States, indicated that although all of the candidates were novice teachers, grittier candidates persisted and persevered despite challenges (Duckworth, 2016). Therefore, developing expertise, setting challenges in terms of goals that one wants to achieve, along with the repeated practice of skills is a definition of continuous improvement (Duckworth, 2016) and in the context of classroom teachers it indicates ongoing practice of skills that defines the purpose that they set for themselves.

According to Duckworth (2016), “Interest without purpose is nearly impossible to sustain for a lifetime” (p. 91). The importance of the power to perceive that one’s efforts will result in something worthy (Duckworth, 2016) is an important way to keep one’s purpose alive. In the context of classroom teachers, their sense of purpose can increase their level of grit and help them persevere despite challenges. Studies by Von Culin et al. (2014) show that grittier individuals were more likely than less gritty individuals to see happiness through engagement” (p. 5). Thus, purpose along with effort over time has been shown to increase grit among individuals (Von Culin et al., 2014).

When individuals believe in a purpose, they are focused on working towards it by putting in the required effort and engaging in continuous improvement. Hope that one can achieve what they set out to, defines the kind of effort required. According to Duckworth (2016), “Hope is rising-to-the occasion kind of perseverance” (p. 91). Furthermore, “hope does not define the last stage of grit. It defines every stage” (p. 92). As individuals go through each stage of what they set out to achieve, the combined effects of the “four psychological assets of interest, practice, purpose and hope” helps individuals “learn to discover, develop, and deepen” (Duckworth, 2016, p. 92) their interests. According to Duckworth (2016), these four assets are not items that label an individual as “You have it or you don’t” (p. 92). One can “acquire the habit of discipline, cultivate a sense of purpose and meaning, and teach

oneself to hope" (Duckworth 2016, p. 92). Moreover, according to Duckworth et al. (2012), more conscientious individuals reported greater life satisfaction. Those who demonstrate higher grit and life satisfaction are able to maintain hope even though they may experience hurdles in their work. Studies found conscientiousness to be a key facet of the psychological construct of grit (Eskreis-Winkler et al., 2014; Fite et al., 2017). Therefore, both the constructs, teacher efficacy as well grit draw on several factors to be developed.

Table 3

Principle Sources that Frame Teacher Efficacy and Grit

Principle sources that frame Teacher Efficacy	1. Mastery Experience 2. Vicarious Experiences 3. Verbal Persuasion 4. Physiological Arousal
Principle sources that frame Grit	1. Interest 2. Practice 3. Purpose 4. Hope

Note. Adapted from "Exploring the relationship between teacher efficacy and grit" by Haderer, A. M. (2020). ProQuest Dissertations & Theses Global.

Grit Scale and TSES Scale

Duckworth et al. (2007) created the original instrument to measure the grit construct, the twelve-item Grit-O scale. However, Duckworth and Quinn (2009) conducted multiple studies and determined the shorter eight-item Grit-S was stronger psychometrically and more efficacious in measuring the perseverance of effort and consistency of interest dimensions of individuals. Both of these dimensions combine to give a total grit scale score. Two additional grit measures have been developed. Datu et al. (2017) developed the Triarchic Model of Grit Scale (TMGS), which examines three proposed dimensions of grit; perseverance of effort, consistency of interests, and adaptability to situations. Though this instrument was found to be psychometrically sound, to date it has only been used in two studies, both conducted by

the developers of the TMGS. As such, it has not been widely used in research and the added dimension of adaptability to situations has not been examined in the literature. This is an area that future research may wish to examine. A final grit measure has been developed, the Clinical Nurses Grit Scale (CNGS) by Park et al. (2020). However, this scale has thus far only been used in the original study and is specific for clinical nurses. With these understandings, the Grit-O or the Grit-S would be the most appropriate instruments of measure to utilize for the measurement of grit in elementary level teachers. Yet, within the review of literature, the vast majority of research with educators examining job stress and / or turnover intention have utilized the Grit-S scale (Jang & Kim, 2019; Shin et al., 2018). Therefore, the propensity of researchers utilizing the Grit-S in quantitative correlational studies provided support for its use in the current study. Additionally, the Grit-S has been shown to be a more valid measure of grit (Duckworth & Quinn, 2009) and is the most appropriate choice for this research to examine the extent to which job stress, teacher self-efficacy, and grit, directly and interactively, predict the turnover intention of elementary level public-school teachers in the United States of America when controlling for duration of service.

The Grit-S and TSES instruments have been used in quantitative correlational educational research successfully (Al-Ghamdi, 2017; Coronado, 2016; Duckworth & Quinn, 2009; Jang & Kim, 2019; Riddle, 2018; Shin et al., 2018). The studies verify the use of the Grit-S and TSES instruments to examine correlations between grit, and teacher self-efficacy beliefs. This study is focused on expanding quantitative correlational research to examine the extent to which teacher self-efficacy beliefs are related to grit, among elementary and middle school level (PreK-8) public-school teachers in Northwestern New Jersey. The propositions of this research may lead to more effective and intuitive hiring and professional development practices amongst educational leaders and human resource partners to reduce high turnover

rates of public school teachers and to lower teacher shortages. The repercussions of having qualified teachers and structured administrative support would result in increased student engagement and well managed classrooms thus effectively increasing teachers' sense of self-efficacy and willingness to stay in the profession.

Instrumentation

Grit Short Scale (Grit S)

The Grit-S was developed at the University of Pennsylvania (Duckworth, 2016; Duckworth & Quinn, 2009). Grit, as a personality trait, is defined as the combination of perseverance and passion (Duckworth & Quinn, 2009). The scale contains the two subscales of perseverance of effort and consistency of interest. There are eight items included in the scale with four in each subscale. The Grit-S contains eight items and was developed based on the Original Grit Scale (Grit-O) in response to feedback that raised questions regarding the reliability and validity of the initial scale, which consisted of 12 items (Duckworth & Quinn, 2009). Scoring is determined based on a 5-point Likert scale ranging from 5 = *very much like me* to 1 = *not like me at all*. Possible total scores range from 5 (*extremely gritty*) to 1 (*not at all gritty*). For purposes of the instrument, gritty refers to the spectrum of lower or higher scores received on the Grit-S scale (Duckworth et al., 2007). Total scores are divided by eight using questions such as "Setbacks don't discourage me, I am a hard worker, I finish whatever I begin, and I am diligent" (Duckworth & Quinn, 2009). Reliability for the scale and subscales was established through confirmatory factor analysis from four of the six samples. Reliability for the scales was reported using Cronbach's alpha across the four samples that ranged from 0.73 to 0.83, 0.73 to 0.79 for the consistency of interest subscale, and 0.60 to 0.78 for the perseverance of effort subscale (Duckworth & Quinn, 2009). Predictive and consensual validity was confirmed by collecting data twice over a 1-year period for three of

the six samples ($r = 0.68$, $p < 0.001$). This instrument has been used in numerous studies including Bowman et al. (2015) and Strayhorn (2014).

TSES Scale

A variety of instruments are available that measure self-efficacy of an individual. However, there are limited choices in instrumentation when looking specifically at measuring teacher self-efficacy. This study utilized the Short Form of the Teachers' Sense of Efficacy Scale (TSES). The TSES Short Form is a 12-question version of the longer 24-question TSES scale developed by Tschannen-Moran and Hoy (2001). The TSES Short Form uses a Likert type rating system for all 12 questions ranging from (1) *Not At All* to (9) *A Great Deal*. The TSES Short Form encompasses three subscales for Efficacy in Student Engagement, Efficacy in Instructional Strategies, and Efficacy in Classroom Management. The composite score of the instrument is determined by calculating the sum of the responses from participants and finding the mean score. The score ranges from 1 to 9. Scores of 5 and below shows low teacher self-efficacy with score 6 and above showing higher levels of teacher self-efficacy (Tschannen-Moran & Hoy, 2001). Tschannen-Moran and Hoy (2001) found the TSES Short Form to be a valid measure of teacher self-efficacy with a reliability of 0.90. Duffin et al. (2012) performed a confirmatory factor analysis of the TSES with practicing and preservice teachers and found the total score scale model had CFI values greater than 0.95, confirming its validity and reliability. The TSES has been used in multiple quantitative correlational studies (Duffin et al., 2012). Additionally, Riddle (2018) used the TSES and the GritS in preservice teachers to examine correlations between grit and teacher self-efficacy. Coronado (2016) used the TSES as well as the Grit-S to determine correlations between teacher grit, teacher self-efficacy, and student achievement. A review of the literature found few studies that use the TSES to examine correlations between grit and teacher self-efficacy among elementary and middle school teachers (PreK-8). Therefore, this study is designed to find the

correlation between these variables. Examination of the variables and the instruments together provide empirical evidence with regards the aspects relating to educator's grit scale score and teacher self-efficacy beliefs.

Summary

There is an acute shortage of qualified teachers across the United States. Not only has there been a decline in candidates choosing it as career path (Sutcher et al., 2019) but also attrition of veteran teachers (Weber, 2020) as result of burnout, and new teachers leaving within the first 5 years of their service (Hise, 2022) due to challenges in classroom management, stress brought on by work overload, and lack of support from colleagues and administration (Aloe et al., 2014; Zhu et al., 2018). Districts have employed a variety of strategies to moderate teacher shortages, including options such as offering professional development, lowering qualification standards, and examining culture and climate to fill positions (McHenry-Sorber & Campbell, 2019; Sutcher et al., 2019). Reducing educator qualifications is counterproductive in an age of education accountability and studies that investigate personal traits among teachers who have remained in the profession despite challenges and a multitude of changes in education need to be conducted to identify ways to retain teachers. Studies in the area of the relationship between grit and self-efficacy beliefs (Dale et al., 2018; Robertson-Kraft & Duckworth. 2014; Zee & Koomen, 2016) among teachers indicate that there is a positive correlation between the two.

Teacher self-efficacy has been consistently shown to be negatively correlated with teacher turnover intention (Afzal et al., 2019; Shin & Kim, 2017; Skaalvik & Skaalvik, 2016; Swanson & Huff, 2010). Grit is defined by consistency of interest and perseverance (Duckworth, 2018; Duckworth et al., 2007). Grit indicates an individual's resilience, as well the capacity to persevere, despite setbacks or challenges. It is a distinct construct (Duckworth

et al., 2007; Perkins-Gough, 2013). Grit has been shown to be positively correlated with employee retention (Duckworth et al., 2007; Eskreis-Winkler et al., 2014).

Most of the literature on grit relates to grit as a significant predictor of academic achievement. The studies focus on the correlation between higher levels of grit in individuals and increased academic achievement as measured by grade point averages, course completion, and individual course grades (Clark & Malecki, 2019; Credé et al., 2017). Based on studies by Duckworth (2018) and Dugan et al. (2018), grit can be cultivated through repetition, practice, and reflection. Grit is a two-dimensional construct consisting of consistency of interest and perseverance of effort that has predictive value in education and organizational settings as demonstrated through various studies (Fong & Kim, 2019; Jachimowicz et al., 2018; Lai Lam & Zhou, 2020). Although research on grit is relatively new, the evidence from different studies point to its reliability and predictive power.

Robertson-Kraft and Duckworth's (2014) study on novice teachers to find if correlations existed between higher levels of grit, retention through the academic year, and teacher effectiveness as measured by academic outcomes indicate that grit was positively and significantly correlated with novice teacher retention as well as with higher performance outcomes of students. Studies on grit and life satisfaction (Khan & Khan, 2017) found that teachers with higher levels of grit reported higher levels of happiness and life satisfaction.

Within the workplace, the level of grit displayed by supervisors and employees was significant in minimizing the negative effects of the climate in the workplace (Jordan et al., 2019). School districts need to create an environment where teachers have the opportunity to be mentored adequately in their early years. Positive school climate promotes professional learning communities where knowledge can be shared. Especially through vicarious experiences that can help novice teachers learn and build their repertoire in terms of strategies that can be used for student engagement and classroom management. Developing

professional training for teachers that are geared towards developing grit and consequently helping teachers increase their self-efficacy beliefs in classroom management and student engagement can build success among classroom teachers, help reduce burnout and stress, and reduce attrition among teachers. Studies show that employees with higher levels of grit believe their work is meaningful and therefore they demonstrate higher work engagement (Singh & Chopra, 2018). Hiring candidates with higher grit could be useful for district administrators when hiring new teachers for positions. Although as seen in the literature review, several studies have been conducted to find correlations between grit and self-efficacy beliefs of teachers there is a gap in literature with regards to studies specific to grit and self-efficacy beliefs of teachers in elementary and middle schools (PreK-8). This quantitative correlational study was conducted primarily for the purpose of finding the correlation between grit and self-efficacy beliefs of elementary and middle school teachers (PreK-8) teachers in the area of classroom management and student engagement.

Chapter 3: Research Methods

Overview

The research methods utilized for the study are explained in this chapter. Specifically, the research methodology overview includes the following components: (a) an overview of the research design, (b) research design and rationale, (c) a description of the survey instruments used to collect data, (d) a description of the sample used for analysis, (e) an overview of the statistical procedures conducted, (f) ethical considerations, and (g) discussion on the limitations of the study.

The purpose of this quantitative correlational study was to explore the relationship between teacher grit (measured by the level of tenacity, endeavor, and drive) and teachers' beliefs in self-efficacy in the areas of classroom management and student engagement. Teachers in elementary schools (PreK-8) often teach multiple subject areas while also managing classroom environments. Teachers are expected to keep all students actively engaged in learning while maintaining a learning space that allows students freedom to express, move and interact. The skill sets needed to achieve these are more than subject area expertise. The overall goal of this quantitative correlational study is to expand on the knowledge of the impact of personal traits such as passion and perseverance to pursue a long-term goal among teachers especially in elementary schools (PreK-8) and its overall effect on teachers' self-efficacy beliefs in classroom management and student engagement.

Research Design and Rationale

This study used a quantitative, non-experimental correlational design to test the relationship between teacher grit and teacher self-efficacy beliefs in classroom management and student engagement. To investigate the research questions, data was collected by utilizing a convenience sampling that used an internet-based survey to collect teacher grit scores using the Grit-S scale, and the TSES scale to collect teacher self-efficacy scores. Besides the two

scales four demographic questions related to teacher experience in years, certification pathway, subjects taught and educational qualifications were also collected.

Quantitative research involves measuring existing organizational data, physiological data, and self-prepared or published surveys that provide a quantified response set (Stichler, 2016). It uses statistical processes to analyze the significance of the findings. A correlation is a quantitative measure of the correspondence degree (Gay et al., 2014). One of the most commonly recognized means to develop new knowledge is through establishing relationships between variables or phenomena (Rumrill, 2004).

A correlational design is appropriate for this research as it involves collection of data that helps determine whether a relation exists between two or more quantifiable variables or continuous scores. Correlational research aims to uncover relationships between variables, describing their direction (positive or negative) and strength, without implementing any intervention to alter the outcome variable (Cook & Cook, 2008; Gay et al., 2014).

Continuous data is data that can be measured, it has values that are not fixed and have an infinite number of possible values. Also, continuous data can be displayed in visualizations such as graphs because of its variable change over time. Correlational studies are often used in educational research (Gay et al., 2014). In this study, the Short Grit Scale (Grit-S) score was the independent variable and the Teacher Self-Efficacy Scale (TSES) scores in classroom management and student engagement were the dependent variables.

Research Questions

Research questions for quantitative studies are designed to find relationships among variables being investigated and are usually framed at the start of the study (Barroga et al., 2023). The questions are formulated in a manner so that they are linked to the population being studied, the independent variable, the dependent variables, and the research design.

Quantitative comparative research questions are designed to investigate differences between groups within the context of an outcome variable (Ratan & Ratan, 2019).

This study sought to answer the following questions:

RQ1. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on classroom management?

RQ2. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on student engagement?

Participants and Sampling

This section describes and justifies the selection of the research setting, participants, and sampling, providing the background and issues related to the problem. The shortage of teachers in public schools has been a growing concern. Retention of current teachers (Zhang & Zeller, 2016) as well as the inability of schools to find candidates to fill positions has been attributed to dissatisfaction with the profession due to conditions at work, lack of support from administration, remuneration, lack of respect to name a few. Studies indicate that personal traits and qualities in teachers such as grit and years of teaching impact career success and retention of teachers (Burton, 2021).

Personal traits, such as grit, have been enhanced in students with growth mindset practices and interventions, and this could be true for teachers, too. Articulated theories and principles aligned to improving teachers' personal traits such as passion and perseverance, also known as grit, and their beliefs in self-efficacy in the areas of classroom management and student engagement will need further exploration.

The participants for this study were therefore elementary and middle school teachers (PreK-8) in public schools in Northwestern New Jersey. The participants completed a short four question demographic survey, the short Grit-S scale measuring the independent variable of grit where grit is defined as "perseverance and passion for long term goals" (Duckworth et

al., 2007, p. 1087) and the Teacher Sense of Self-Efficacy Scale (TSES) short scale measuring the dependent variables of self-efficacy in classroom management as well as self-efficacy in student engagement. Self-efficacy is based on an individual's beliefs in their own capabilities that help the successful completion of tasks (Bandura, 1977). self-efficacy in teachers is their belief in their abilities to provide instruction along with the ability to engage students and manage classroom environments despite challenges related to student behaviors (Caprara et al., 2006).

Participant Recruitment and Data Collection

Participants were recruited by first obtaining permission from school districts. Emails to district Superintendents were sent with a detailed description of the purpose of the study, the survey and invitation to participate. Emails were sent to 22 elementary public school superintendents and principals in Northwestern New Jersey in the first week of August seeking permission to send out surveys to teachers in their district. Out of the 22 school districts, five districts provided permission letters to send out the survey to teachers in their district.

After obtaining approval from the IRB committee, I sent out the email with the invitation to participate and the survey link to the superintendents and principals who then sent it to teachers in their districts using district email addresses during the week of October 16, 2023. This method was chosen so that I did not have access to district email addresses directly. Consent to participate was obtained from participants prior to entering the survey.

The survey was opened on October 16, 2023. A reminder to participate in the voluntary survey was sent out on December 8, 2023 to principals and superintendents to remind staff to complete the survey. The survey responses were closed on December 15, 2023. Microsoft Forms was the tool utilized for data collection in this study. This tool allowed me to collect strictly anonymous responses. By default, Microsoft Forms records the

respondent's email address in the survey results. To maintain anonymity, I turned on "Anonymous Responses" to prevent Microsoft Forms from tracking emails. Additionally, by default, most collectors record the IP addresses of respondents in survey results. I turned on "Anonymous Responses" to prevent IP tracking. The projected number of participants expected was 40–100. There were 86 (N = 86) eligible responses to the survey.

The survey consisted of three parts. The first part of the survey consisted of four demographic questions. Demographic data related to teacher age, sex, and ethnicity were not taken. The four questions in the demographic portion of the survey collected data on teacher experience in relation to number of years they worked, pathway to teacher certification, subjects taught, and educational qualifications. The second part of the survey had the first instrument, the Grit Short Scale (Duckworth, 2016).

The Grit-S scale consisted of eight straightforward statements written in first person. The participant needed to respond, out of five options, if that statement is "Very much like me" to "Not like me at all." Each answer was given an automatically generated grit score upon completion. The approach of the Grit Scale allowed for there to be something against which to measure their answers from the next portion. Additionally, this Grit Scale survey was given via the online platform called Microsoft Forms that automatically aggregates the data while ensuring confidentiality.

The third part of the survey was the Teachers' Sense of Efficacy Scale (TSES) short form. The TSES Short Form is a 12 question version of the longer 24-question TSES scale developed by Tschannen-Moran and Hoy (2001). The TSES Short Form uses a Likert type rating system for all 12 questions ranging from (1) *Not At All* to (9) *A Great Deal*. The TSES Short Form encompasses three subscales for Efficacy in Student Engagement, Efficacy in Instructional Strategies, and Efficacy in Classroom Management. The composite score of the instrument is determined by calculating the sum of the responses from participants and

finding the mean score. The TSES Scale survey was given via the online platform called Microsoft Forms that automatically aggregate the data while ensuring confidentiality.

Participation in the survey was voluntary and consent was obtained via a consent form attached to the survey. The eligibility requirement for participation was teachers who taught grades Prek-8. All responses were anonymous. All responses on the survey were self-reported.

Archival Data

Data will be stored electronically for 5 years on a password-protected computer drive, which is stored at my home address. During the 5 years, this data will not be shared with anyone and will only be accessible by myself. Ultimately, data will be deleted from the drive and then into 'trash', which will be emptied to remove it from the hard drive.

Validity and Reliability of Instrumentation

Short Grit Scale: This instrument began as a 27-item response document to assess grit and the researchers eliminated 10 of these items due to redundancy and vocabulary. Once data analysis was completed to assess for validity and reliability of the tool, the researchers reduced the scale from 17 to 12 items through a 2-factor structure and signaled the completion of the measurement tool (Duckworth et al., 2007). The reliability of the Grit-S was tested and had acceptable reliability with Cronbach's alpha ranging from 0.73 to 0.83 (Duckworth & Quinn, 2012)

Teacher Sense of Efficacy Scale: Tschannen-Moran and Hoy (2001) developed the TSES to assess teachers' sense of efficacy regarding the teaching tasks involved in student engagement, classroom management, and instructional practices. They tested the TSES in three separate studies where the original 52 items were reduced to two forms of the measure: a 24-item long form and a 12-item short form. I utilized the 12-item short Likert item to measure teacher self-efficacy. The composite score of the instrument is determined by

calculating the sum of the responses from participants and finding the mean score. The score ranges from 1 to 9. Scores of 5 and below shows low teacher self-efficacy with score 6 and above showing higher levels of teacher self-efficacy (Tschannen-Moran & Hoy, 2001). Tschannen-Moran and Hoy (2001) found the TSES Short Form to be a valid measure of teacher self-efficacy with a reliability of 0.90. The TSES has been used in multiple quantitative correlational studies (Duffin et al., 2012). Cronbach's alphas for the TSES were 0.94 overall, 0.87 student engagement component, 0.91 instructional strategies component, and 0.90 classroom management component (Tschannen-Moran & Hoy, 2001).

Ethical Considerations

I upheld the ethical standards of research from the Belmont Report (Office of Human Subjects Research, 1979), which summarized key principles to protect human participants. The key ethical principles in research included (a) the role of assessment of risk-benefit criteria in determining the appropriateness of research involving human subjects, (b) appropriate guidelines for the selection of human subjects for participation in such research, and (c) the nature and definition of informed consent in various research settings. The participants and the topic of the study were not considered vulnerable populations, and they were assessed to have minimal risk. To protect the participants while identifying the sample, I did not collect any personal information such as emails, names, age, sex, school district etc. from the participants.

A unique identifier was used for each participant. To obtain informed consent, prospective participants received an email invitation with a link to Microsoft Forms to access the informed consent form. Microsoft Forms, the means of data collection, saved the data on a secure computer. There were no anticipated risks to any participant participating in the online survey. After the data was downloaded, every participant remained anonymous by coordinating and identifying letter/number combinations for each participant. These were no

longer identifiable once the data were changed. I analyzed the data according to the respective methodology. No unauthorized persons could access the data because these were stored in a strong password-protected file on my computer. After 5 years, the hard copies of the data will be destroyed via shredding and permanently deleting the electronic files.

Additionally, a debriefing page was available immediately after the last question on the survey was completed. Contact information for myself and my dissertation chairperson was provided to the participants. Survey data was populated into an Excel spreadsheet from Microsoft Forms. All the IRB procedures for ethical considerations were followed. Additionally, all necessary approvals were granted proper to the start of the research process.

Data Analysis Procedures

Descriptive statistics were used in reporting and analyzing the results of the anonymous survey sent out to elementary teachers in grades PreK-8 in public schools in Northwestern New Jersey. Data collected from the survey were downloaded to the IBM Statistical Package for the Social Sciences (SPSS) version 29.02 for analysis of the resulting quantitative data. Data was analyzed utilizing the product moment correlation (Pearson *r*). Table 4 shows the summary of data analysis and the collection process. Table 5 shows the definitions of the variables and their measurement. The statistical software suite Statistical Package for the Social Sciences (SPSS) version 29.02 was used to conduct the analysis

I reported descriptive statistics of the data for the predictor and criterion variables. A frequency and percentages summary was obtained for categorical variables, while the measure of central tendencies of means, standard deviations, and minimum and maximum values were conducted for continuous variables. Parametric assumptions were conducted, which included linearity, the normality of the standardized residuals, homoscedasticity, and outlier detection. To assess linearity and homoscedasticity, plots of the standardized residuals and the standardized predicted values were examined. If the plots were not curvilinear, then

there was no violation of the assumption of linearity (Field, 2013; Tabachnick & Fidell, 2013).

Additionally, if the plots formed a rectangular pattern or did not flare out on either end of the distribution, then there was no violation of the assumption of homoscedasticity (Field, 2013; Tabachnick & Fidell, 2013). ShapiroWilk's was performed to determine if the data were normally distributed (Field, 2013; Tabachnick & Fidell, 2013). Additionally, kurtosis and skewness statistics were generated to assess normality further. Testing for a linear relationship between the criterion variable and each predictor variable was performed by visually inspecting scatterplots. The next step involved testing homoscedasticity using a scatterplot to check if residuals were equally spread over the predicted values of the criterion variable. Residuals were tested to check if they were approximately normally distributed using the casewise diagnostics option in SPSS. Bivariate outliers were also tested for. The data was analyzed for correlations and directionality of the data.

Table 4

Summary of Steps Taken for Data Collection and Analysis

Process Step	Task
Step 1	Clean data to remove incomplete data and estimate missing values where appropriate
Step 2	Calculate descriptive statistics, including demographic statistics
Step 3	Perform bivariate analysis and assess scatter plots for linear relationships between variables
Step 4	Test data for: Normal distribution of the criterion variables (Kolmogorov-Smirnov test, kurtosis/skewness statistics, histograms, and Q-Q plots) Linearity of the relationships between variables, Outlier detection
Step 5	Assess significance at the 5% level.

Variables

Table 5

Definitions and Measurement of Variables

Variables	Conceptual definition	Operational definition	Variable Measurement
Teacher Self-Efficacy	Teachers' sense of efficacy is their confidence in their ability to have an impact on student learning and outcomes. It is the sense from an educator that s/he can make a difference to students even if they appear to be unmotivated or are challenging (Tschannen-Moran & Hoy, 2001).	Teacher self-efficacy was operationalized as and measured by the Short Form of the Teachers' Sense of Efficacy Scale (TSES) (Tschannen-Moran & Hoy, 2001).	Technically ordinal, approximated to interval
Self-reported Grit	The intrinsic desire or want to accomplish or succeed at something (Duckworth, 2018; Duckworth et al., 2007). Grit is a quality that consists of perseverance and consistency of interest in the face of impediments and hurdles to still succeed and achieve (Duckworth, 2018; Duckworth et al., 2007).	Grit was operationalized as and measured with the Grit Short Scale (Grit-S) (Duckworth & Quinn, 2009).	Technically ordinal, approximated to interval.

Limitations and Delimitations

This section identifies the assumptions and specifies the limitations, as well as the delimitations, of the study. Several assumptions were considered in the study. I assumed at least 86 teacher participants were willing and available to answer the survey questionnaire in the study. Teacher participants voluntarily participated in the study after receiving the email invitation from their school districts. I assumed that teacher participants had internet access to participate in the study. The survey was conducted using an online survey tool called

Microsoft Forms. Therefore, participants should have Internet access to respond to the survey questionnaire. I also assumed that the study would accurately represent teachers from different school districts who taught elementary and middle schools in Northwestern New Jersey. The following delimitations were present in this study:

1. The scope of this study was delimited to public schools in Northwestern New Jersey because the researcher could only access school districts within this region.
2. The study was delimited to the period of 2 months of data collection to ensure that the data collected in the study were applicable at the present time.

The following limitations were present in the study, “limitations are potential weaknesses or problems with the study identified by the researcher” (p. 199):

1. The study was limited to the constructs of grit and teacher self-efficacy measured using the survey instrument considered in the study. The survey used in this study included constructs of teacher grit, self-efficacy beliefs of teachers in classroom management and student engagement.
2. A limitation of this study was that participants might have different interpretations of survey questions, which might influence their responses. The data collected was self reported. “A basic issue concerning self-report measures is people’s ability to make accurate self-reports” (Whitley et al., 2013, p. 5246).
3. Another limitation of this study was using a convenience sampling of teachers. Convenience sampling increased the likelihood that the participants were not a representative sample of the population of PreK-8 teachers in the United States.
4. Using a quantitative nonexperimental correlational design was a limitation. Quantitative nonexperimental correlational researchers could not infer cause-and effect; causality could only be determined in experimental studies.

5. Due to size limitations of the convenience sample, the study had limited transferability and generalizability to a similar population. The research was analyzed with consideration to the implications of both transferability and generalizability of research. In order to make a generalizable claim, there will be a careful examination of the variables involved in the study. For this study, generalizability focused on making predictions based on a recurring experience. With sufficient data to support the research questions, the data was formulated to allow for generalization to similar circumstances.

Summary

The purpose of this quantitative, cross-sectional, and correlational design was to examine the impact of teacher grit on two teacher self-efficacy dimensions of classroom management and student engagement. The predictor variable, grit, was defined as an individual's perseverance and passion for a long-term goal, even in the face of adversity (Duckworth et al., 2007), while teacher self-efficacy was defined as the teacher's perception of their own abilities. Teacher self-efficacy was considered regarding two dimensions/outcomes: self-efficacy in classroom management and self-efficacy in student engagement (Tschannen-Moran & Hoy, 2001). Data collection in this study was conducted entirely through an online survey hosted by Microsoft Forms, which included administering the Grit-S and TSES to measure teacher grit and teachers' self-efficacy respectively.

Data analysis involved using the statistical program IBM SPSS, Version 29.02. Prior to analysis, the data were manually cleaned of responses that failed to complete all parts of the survey. In addition, any potential responses that were impossible due to data collection or survey errors were removed. Each of the interval variables was calculated from survey responses and collected into one data set for analysis. The study was guided by the following questions:

RQ1. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on classroom management?

RQ2. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on student engagement?

Chapter 4: Findings

The overall purpose of this study was to gain further knowledge about factors that influence teachers' abilities especially in elementary schools in the area of classroom management and student engagement. This study used a quantitative approach to explore the correlation between personal traits such as grit and their impact on self-efficacy beliefs of classroom management and student engagement among elementary teachers (PreK-8) in public schools. Through the analysis of data from this research study, additional exploration will seek to discover whether years of experience, educational level as well certification pathways in any way influence teacher abilities in managing classrooms and engaging students better. The conceptual framework of this study was derived from the gaps in the literature related to grit in elementary teachers and its correlation to self-efficacy beliefs about classroom management and student engagement. The theoretical framework of the study is grounded in Duckworth et al. 's (2007) theory of grit, Deci and Ryan's (1985) self-determination theory (SDT), and Bandura's (1977) self-efficacy theory.

The data for this study was collected through a Microsoft Forms anonymous online survey to collect demographic data in the areas of teacher educational qualifications, certification pathway, experience, and subjects taught. The survey also consisted of the Grit-S scale to measure grit and the TSES (short) scale to measure self-efficacy beliefs in classroom management and student engagement among teachers in grades PreK-8 in public schools in Northwestern New Jersey. This chapter will focus on the data analysis resulting from the data collected in the survey and discussion on the findings based on the two research questions:

RQ1. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on classroom management?

RQ2. Is there a relationship between PreK-8 teachers' level of grit and their self-efficacy beliefs on student engagement?

Descriptive Statistics

The sample size for the study consisted of 91 participants, 87 of the 91 respondents taught PreK-8. Of the 91 respondents to the survey, four participants were not able to complete the survey as they did not qualify the criteria that required them to teach grades PreK-8, one of the survey participants did not submit the survey. Eight-six valid responses were collected and $N = 86$ was considered for all data analysis.

The demographic data was self-reported and collected using a Microsoft Forms survey. It consisted of four questions related to years of teaching experience, teacher certification pathway, subject area taught, and the highest level of education. With regards to years of teaching experience, 17.4 % (15/86) reported up to 10 years of teaching experience. While 82.6% (71/86) of the survey takers reported 10–31 years of experience.

Further breakdown indicated that of those participants that had up to 10 years of experience, 5.81% (5/15) had between 0–5 years of experience while 11.6% (10/15) reported 6-10 years of teaching experience. Participants who reported above 10 years of experience, 19.8% (17/71) reported 11–15 years of experience, 23.3% (20/71) reported between 16–20 years of experience, 19.8% (17/71) between 21-25 years of experience, 12.8% (11/71) reported 26–30 years of experience, and 7% (6/71) reported 31 or more years of experience.

Table 6

Frequency and Percentage Summaries for Teacher Work Experience

Number of years	Frequency	Percent	Valid Percent	Cumulative Percent
0-5 years	5	5.8	5.8	5.8
6-10 years	10	11.6	11.6	100
11-15 years	17	19.8	19.8	25.6
16-20 years	20	23.3	23.3	48.8
21-25 years	17	19.8	19.8	68.6
26-30 years	11	12.8	12.8	81.4
31 +	6	7.0	7.0	88.4
Total	86	100.0	100.0	

In terms of certification pathways as shown in Table 2, 84.9% (75/86) of the survey participants reported having pursued a traditional teacher certification pathway and 15.1% (11/86) of the participants reported having taken an alternate route certification pathway to becoming a teacher.

Table 7

Descriptive Statistics for Certification Pathway

Certification Route	Frequency	Percent	Valid Percent	Cumulative Percent
Alternate route	13	15.1	15.1	15.1
Traditional program	73	84.9	84.9	100.0
Total	86	100.0	100.0	

With regards to subject areas that the participants taught as shown in Table 3, 84.9% (73/86) of the participants reported teaching all subjects which included ELA, math, social studies, and science. Furthermore, 15.1% (13/86) of participants taught special area subjects such as Spanish, music, art, gym, health, and technology.

Table 8

Percentage of Teachers Teaching All Subjects and Special Area Subjects

Subject	*N*	Percent	Valid Percent	Cumulative Percent
All Subjects	73	84.9	84.9	84.9
Special area	13	15.1	15.1	100.0
Total	86	100.0	100.0	

Table 9

Detailed Percentages of Subjects Taught

Subjects taught	Frequency	Percent	Valid %	Cumulative %
ELA	11	12.8	12.8	12.8
ELA; Math	2	2.3	2.3	15.1
ELA; Math; Science	1	1.2	1.2	16.3
ELA; Math; Science; Social Studies	17	19.8	19.8	36.0
ELA; Math; Science; Social Studies; Special Education	2	2.3	2.3	38.4
ELA; Math; Science; Social Studies; Special Education; Health	1	1.2	1.2	39.5
ELA; Math; Science; Social Studies; Technology	1	1.2	1.2	40.7
ELA; Math; Social Studies; Science	1	1.2	1.2	41.9
ELA; Math; Social Studies; Science; Special Education	1	1.2	1.2	43.0
ELA; Math; Special Education;	1	1.2	1.2	44.2
ELA; Social Studies	2	2.3	2.3	46.5

Table 9 (continued).

Subjects taught	Frequency	Percent	Valid %	Cumulative %
ELA; Special Education	1	1.2	1.2	47.7
Art	4	4.7	4.7	52.3
Foreign Languages;	2	2.3	2.3	54.7
Health; Physical Education;	1	1.2	1.2	55.8
Math	5	5.8	5.8	61.6
Math;ELA; Science	1	1.2	1.2	62.8
Math; ELA; Science; Social Studies	1	1.2	1.2	64.0
Math; ELA; Science; Social Studies; Special Education	1	1.2	1.2	65.1
Math; Science	4	4.7	4.7	69.8
Physical Education	1	1.2	1.2	70.9
Physical Education; Health	3	3.5	3.5	74.4
Science	1	1.2	1.2	75.6
Social Studies	2	2.3	2.3	77.9
Social Studies; Math; Special Education	1	1.2	1.2	79.1
Special Education	13	15.1	15.1	94.2
Special Education; ELA Math;Science; Social Studies	2	2.3	2.3	96.5
Technology	3	3.5	3.5	100.0
Total	86	100.0	100.0	

With regards to educational qualifications, as shown in Table 5, among the 86 teachers who participated in the survey, 30% (28/86) of the sample size reported having a Bachelor's degree as their highest qualification, 39% (36/86) participants reported a Master's degree as their highest qualification, 23% (21/86) of participants reported a Master's degree along with a specialist certification, 7% (6/86) of the survey participants reported holding a specialized degree as well as a specialized certification, and 1.2% (1/86) reported a doctoral degree as their highest qualification.

Table 10

Teacher Educational Qualification Levels

Degree	Frequency	Percent	Valid Percent	Cumulative Percent
Bachelor's Degree	29	33.7	33.7	33.7
Masters/Higher	57	66.3	66.3	100.0
Total	86	100.0	100.0	

The descriptive statistics for the quantitative variables of teacher grit, teacher self-efficacy in classroom management and student engagement that was analyzed using the data collected from 86 participants is presented in Table 6. The means of participants' responses for questionnaire items were calculated to determine the scores for each variable. For the teacher's grit, the range of score values among the 86 samples was from 2.37 to 4.62 with a mean score of 3.77 (*SD* = 0.47), indicating that the teachers had high grit levels. For the teacher's self-efficacy in classroom management, the range of score values among the 86 samples was from 5.25 to 9, with a mean score of 7.375 (*SD* = 0.88), indicating that the teachers had high efficacy in classroom management. For the teacher's self-efficacy in student engagement, the range of score values among the 86 samples was from 4.5 to 9.00, with a mean score of 6.87 (*SD* = 1.23), indicating high efficacy in student engagement.

Table 11

Descriptive Statistics of Study Variables-Grit, Self-efficacy Beliefs of Teachers' in Classroom Management and Student Engagement.

Variables	*N*	Minimum	Maximum	Mean	STDEV
Grit score	86	2.37	4.62	3.77	.477
Self-Efficacy in Classroom Management	86	5.25	9	7.37	.883
Self- Efficacy in Student Engagement	86	4.5	9	6.87	1.237

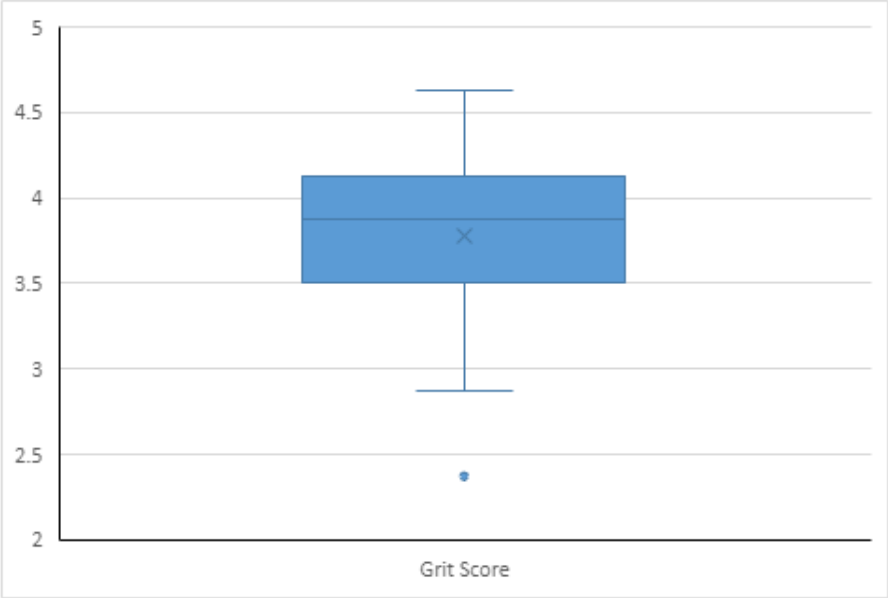
Quantitative Results

In this section, quantitative results will be discussed. Descriptive statistics were used in reporting and analyzing the results of the anonymous survey sent out to elementary teachers in grades PreK-8 in public schools in Northwestern New Jersey. Data collected from the survey were downloaded to the IBM Statistical Package for the Social Sciences (SPSS) version 29.02 for analysis. Data was analyzed utilizing the product moment correlation (Pearson *r*). Data for each research question was assessed through linearity assumptions utilizing scatter plots to determine the strength and direction of potential relationships, box and whisker plots were examined for extreme outliers, and bivariate normality assumptions were run.

An assumption of bivariate outliers was run to determine if any of the scores fell outside the region of the X, Y scatterplot. The extreme outliers should be identified and removed as they can unevenly impact the Pearson correlation (Warner, 2013). Box and Whisker plot graphs display variation within a set of data and are therefore useful in identifying outliers visually. Upon review of the graphs, some outliers were identified. Figures 5–7 display the Box and Whisker plot graphs for each variable. Figures 8 and 9 display scatterplots for the predictor variable (*x*) and the criterion variable (*y*).

Figure 1

Boxplot Graph for Grit Score



Note. The box plot shows outliers.

Figure 2

Boxplot Graph for Self-Efficacy Scores in Classroom Management

Figure 3

Boxplot Graph for Self-Efficacy Scores in Student Engagement

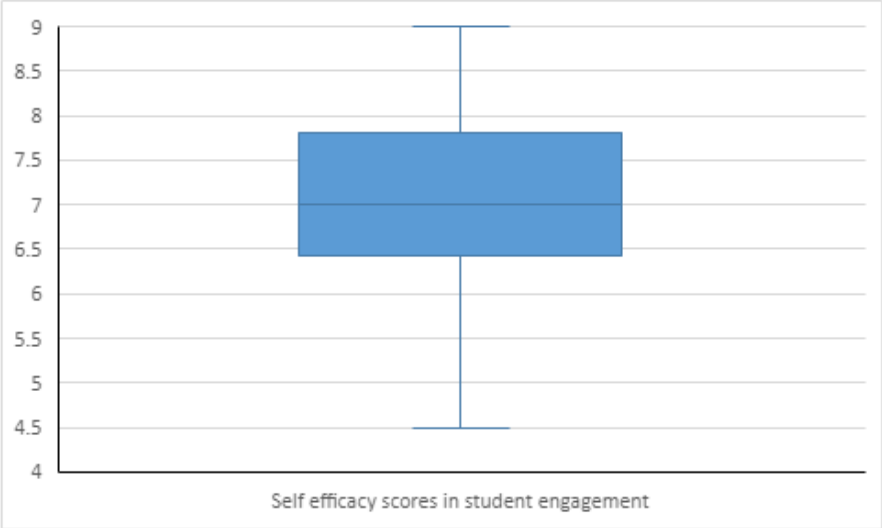
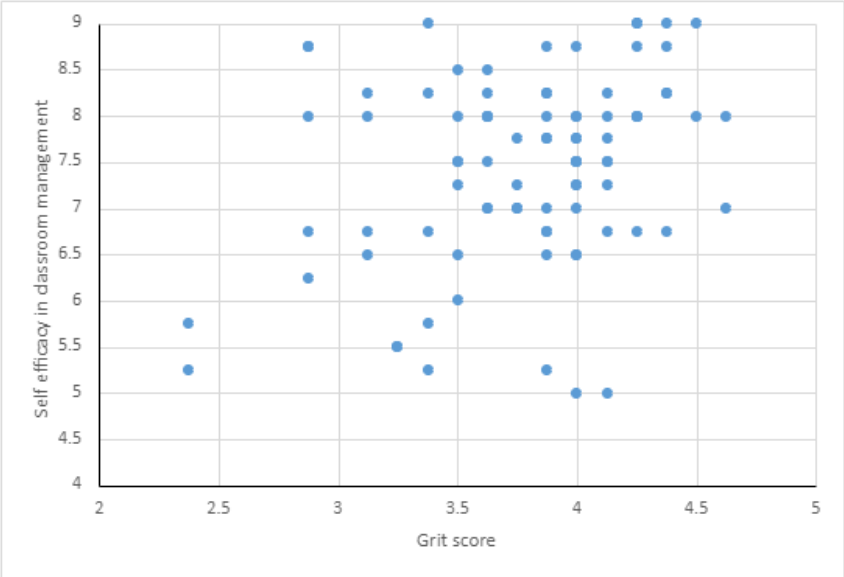


Figure 4

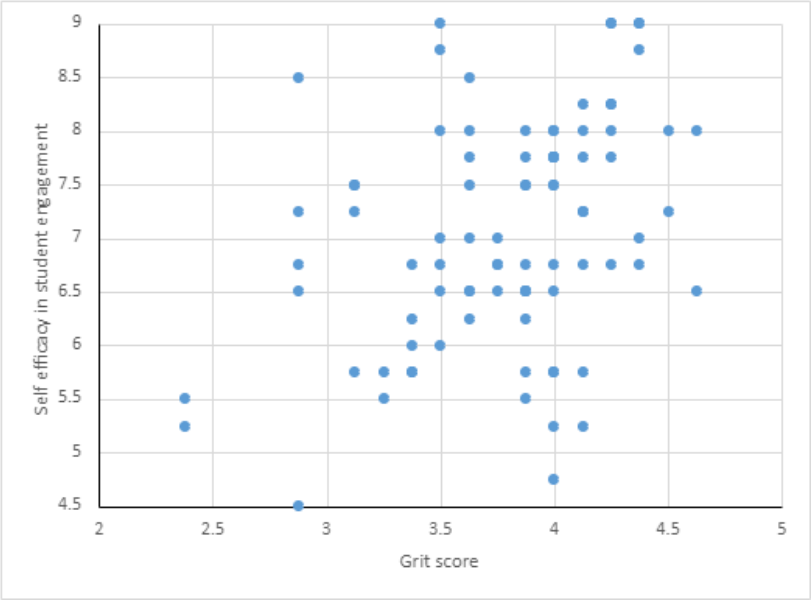
Assumption of Bivariate Outliers for the Relationship between Grit Score (x) and Efficacy in Classroom Management (y).



Note. The scatter plot shows outliers.

Figure 5

Assumption of Bivariate Outliers for the Relationship between Grit Score (x) and Efficacy in Student Engagement (y)



Note. The scatter plot shows outliers.

The box plots revealed some outliers. When analyzing the data, it is good practice to plot the standardized residuals against the predicted values to determine if the points are distributed fairly across all the values of independent variables. In order to determine an assumption of normality for all the variables histogram charts were plotted for the criterion and predictor variables. Normality tests were run in SPSS version 29.02. Table 7 shows data for the Kolmogorov-Smirnov and Shapiro-Wilk tests.

Table 12

Results of the Kolmogorov-Smirnov and Shapiro-Wilk Tests

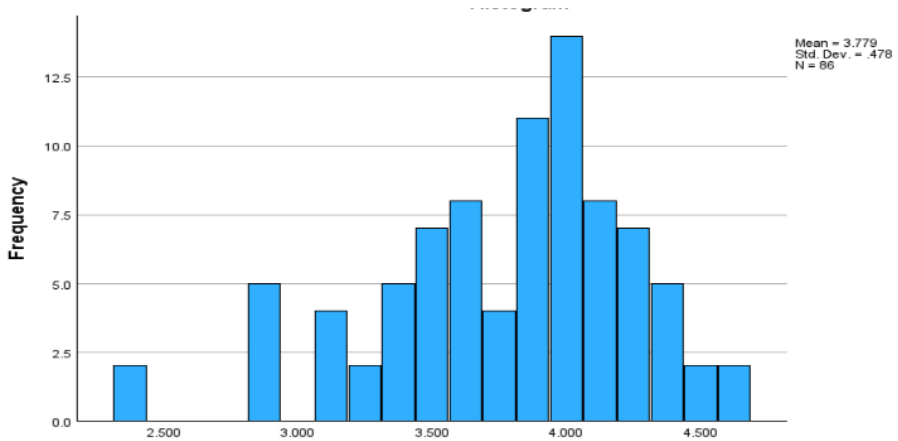
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	*df*	Sig.	Statistic	*df*	Sig.
Grit	.149	86	<.001	.952	86	.003
Classroom Management	.130	86	.001	.948	86	.002
Student Engagement	.085	86	.181	.977	86	.135

Note. a. Lilliefors Significance Correction.

Normality test data for Grit scores, Self-Efficacy in Classroom Management scores and Self-Efficacy in Student Engagement scores.

Figure 6

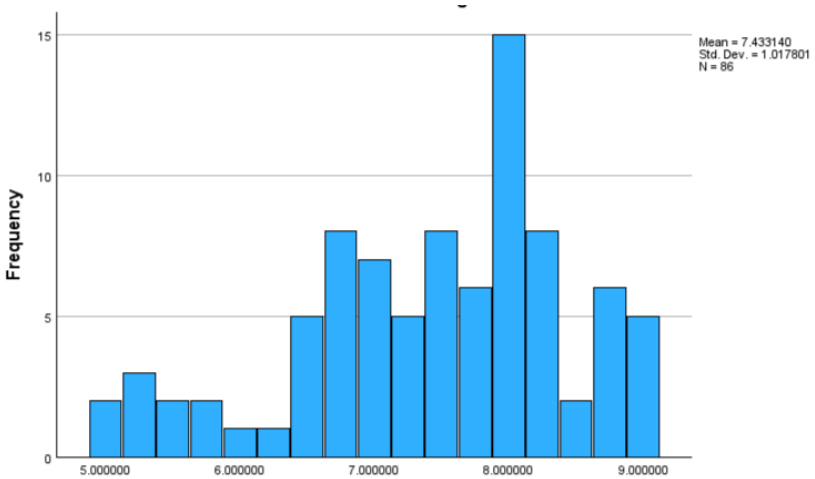
Histogram Showing Grit scores



Note. The data points are not distributed evenly across all the values of independent variables.

Figure 7

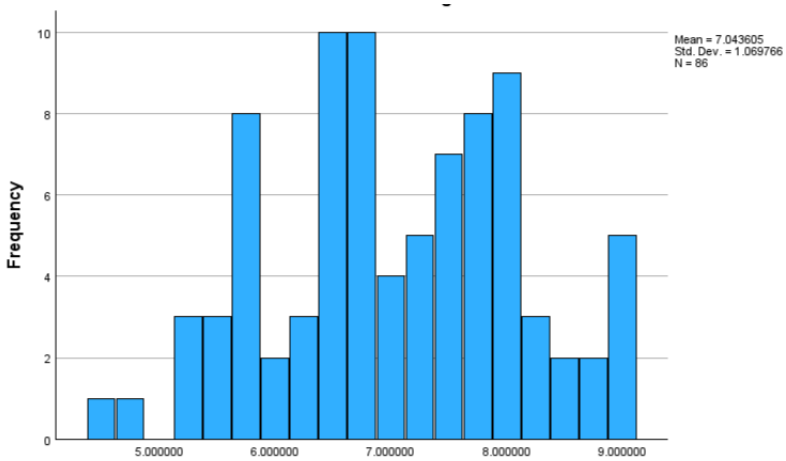
Histogram Showing Self-Efficacy Scores for Classroom Management



Note. The data points are not distributed evenly across all the values of independent variables.

Figure 8

Histogram Showing Self-Efficacy Scores for Student Engagement



Note. The data points are not distributed evenly across all the values of independent variables.

Figure 9

Normal Q-Q Plot for Grit scores

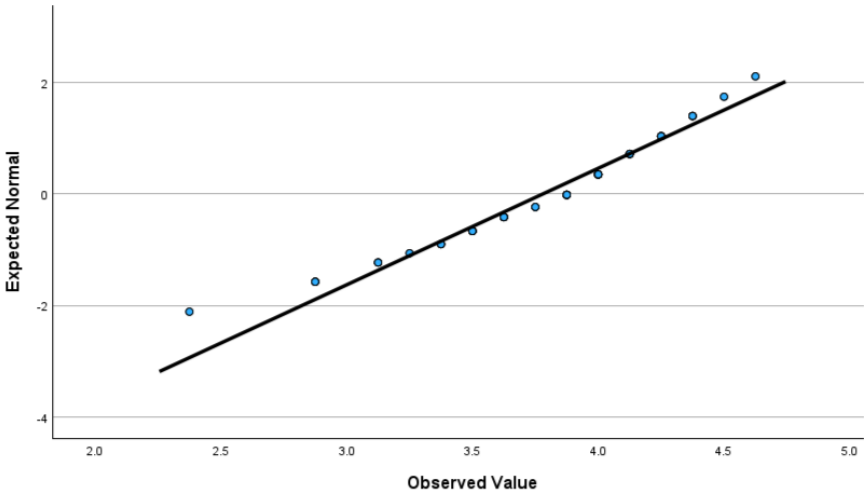


Figure 10

Detrended Q-Q Plot for Grit Scores

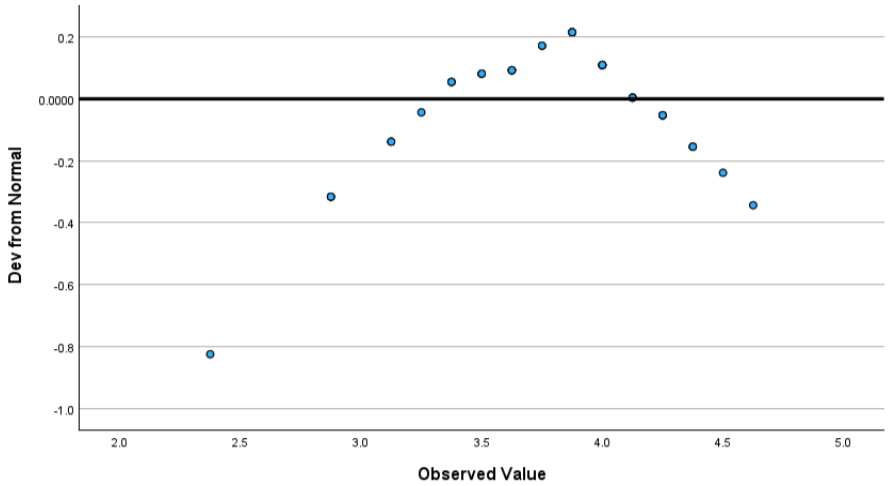


Figure 11

Normal Q-Q Plot for Self-Efficacy Beliefs in Classroom Management

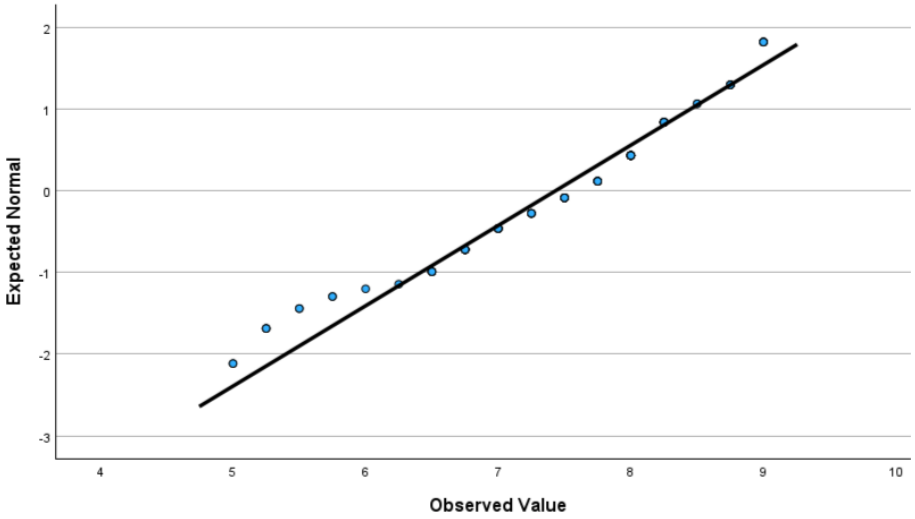


Figure 12

Detrended Q-Q Plot for Self-Efficacy Beliefs in Classroom Management

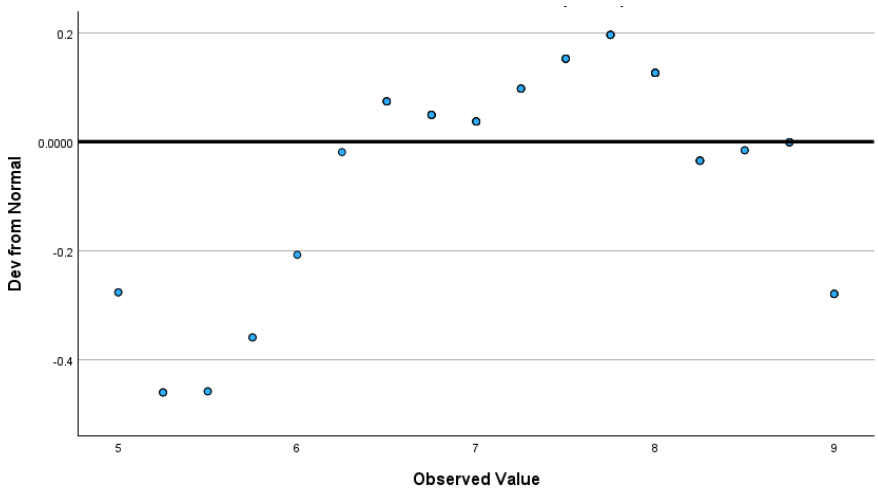


Figure 13

Normal Q-Q Plot for Self-Efficacy Beliefs in Student Engagement

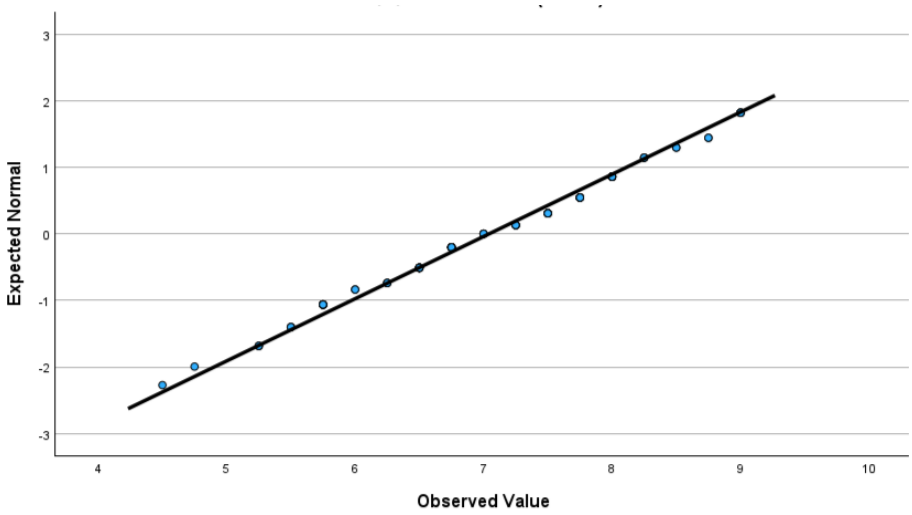
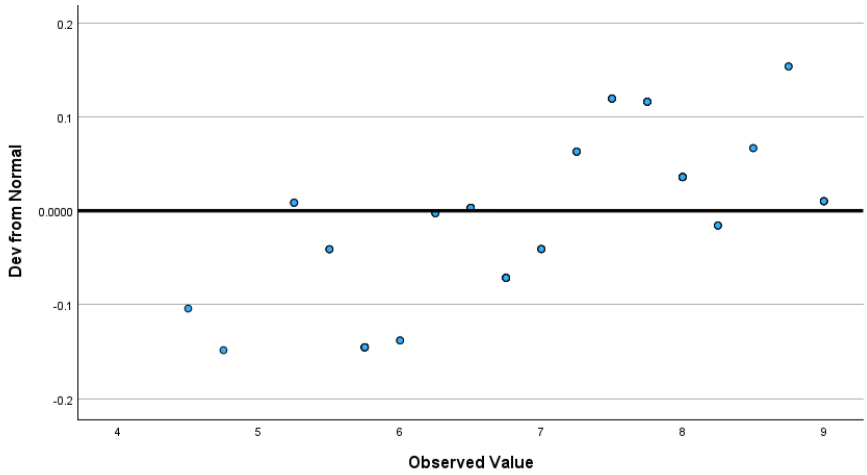


Figure 14

Detrended Q-Q Plot for Self-Efficacy Beliefs in Student Engagement



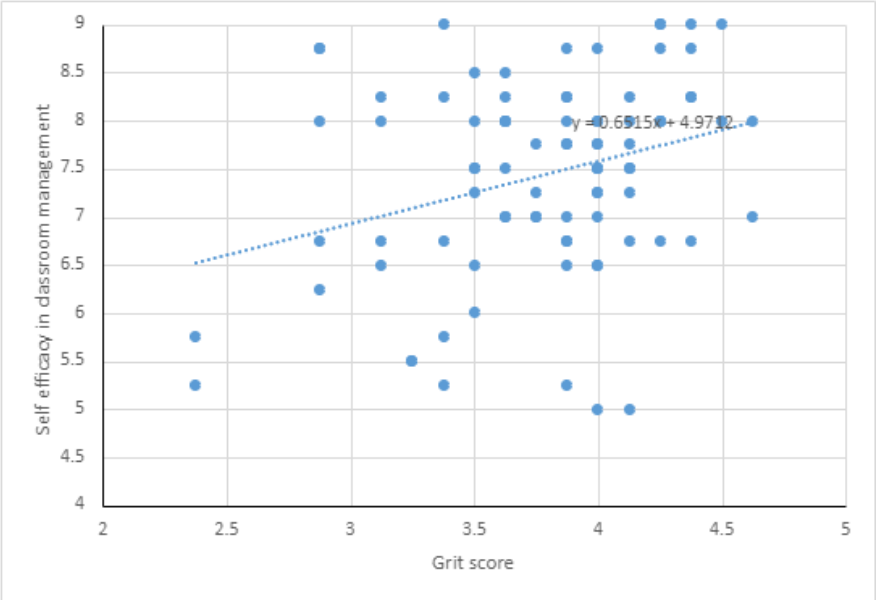
Linearity

In order to continue with the Pearson correlational data analysis, data must be linear (Warner, 2013), which essentially means a linear relationship between the dependent and independent variables. The best way to check for a linear relationship is to create scatter plots and then visually inspect the scatterplots for linearity. If the relationship displayed in the scatterplot is found to be not linear, then further analysis will be required for a non-linear regression or transformation of the data using statistical software, such as SPSS. If the

predictor variable which in this case is grit (*x*) is linearly related to the criterion variables classroom management and student engagement (*y*) the line of fit on the scatter plot should show the direction of the relation between the data (Warner, 2013). Review of the scatterplots with the line of fit shows that linearity was met for Grit and self-efficacy beliefs of teachers in classroom management as well as Grit and self-efficacy beliefs of teachers in student engagement. Figures 13 and 14 show the scatterplots with the line of fit.

Figure 15

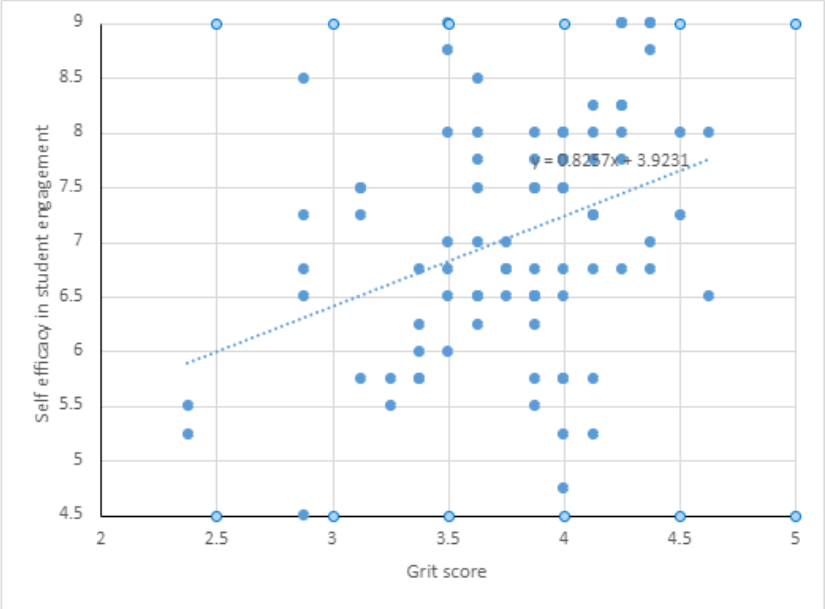
Scatterplot with Line of Fit for Grit Score (x) and Self-Efficacy Beliefs in Classroom Management



Note. The Line of fit shows that linearity was met for Grit and Self-Efficacy beliefs of teachers in Classroom Management.

Figure 16

Scatterplot with Line of Fit for Grit Score (x) and Self-efficacy Beliefs in Student Engagement



Note. The Line of fit shows that linearity was met for Grit and Self-Efficacy beliefs of teachers in Student Engagement

Research Question 1

To find out if there is a correlation between PreK-8 teachers' level of grit and its impact on self-efficacy beliefs of PreK-8 teachers about classroom management, the Pearson's correlations test was run. This was done to check if there is a statistically significant correlation between grit level measured by the Short Grit scale (Grit-S) and self-efficacy beliefs of classroom management measured by the Teacher self-efficacy Scale (TSES) of elementary school teachers. Data was collected through a Microsoft Forms anonymous survey that consisted of four questions related to participant demographics, the Grit-S scale, the TSES short scale for the quantitative variables of Grit (M = 3.77, SD = 0.47), and teachers' self-efficacy beliefs in classroom management (M = 7.37, SD = 0.88). An assumption of bivariate outliers was run to check for extreme outliers. There was just one outlier for grit.

The Kolmogrov-Smirnov test and Shapiro Wilk tests were run to check for normality of the data (N = 86). The normality for data on grit was found to be $p < .001$ while the normality for data on teachers' self-efficacy beliefs in classroom management was found to be $p = .001$. These values indicate that data for grit and classroom management did not meet the assumption of normality because the p-value or significance was found to be less than .05.

Data analysis with Pearson's correlation was continued and at a 0.05 confidence level, alpha level $p < 0.05$ where $r = 86$ and $p = .004$. The analysis shows a significant positive relationship between grit and self-efficacy beliefs of elementary teachers (PreK-8) in classroom management. Table 13 shows Pearson's Correlations results for Grit and teachers' self-efficacy beliefs in classroom management.

The results confirm the importance of grit and its impact on self-efficacy beliefs in classroom management. According to Madden (2015), grit is directly linked to teacher effectiveness. When teachers demonstrate higher grit, which is defined by their passion and perseverance for their work, they are more effective in classrooms. Perseverance in teachers is also related to being successful in using effective classroom management strategies which reinforces competence (Milner, 2002).

Overall, the results of RQ1 revealed that among elementary and middle school teachers (PreK-8), higher grit scores indicate higher self-efficacy beliefs in classroom management. The results were interesting as the majority of the teachers in the sample population were teachers who had been teaching for over ten years. Although this study did not take into consideration specific parameters of teacher age, years of experience, and qualification levels, it was interesting to see it reflected in the findings.

The findings support that grittier individuals persist and persevere despite challenges (Duckworth, 2016). These findings are consistent with Duckworth's (2016) theory of grit that

when an individual enjoys what they do which can be associated with passion, they tend to persevere even though they meet obstacles. In the context of classroom teachers, developing expertise with repeated practice of skills is a definition of continuous improvement that helps build confidence and effectiveness in classroom management.

Table 13

Results of the Pearson's Correlation Test between Grit and Self-Efficacy Beliefs of Teachers in Classroom Management and Student Engagement

		Classroom Management	Student Engagement
Grit	Pearson's Correlation	.306**	.369**
	Sig. (2-tailed)	.004	<.001
	N	86	86

Note. *. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Research Question 2

In order to find out if there is a correlation between PreK-8 teachers' level of grit and its impact on teachers' self-efficacy beliefs about student engagement, the Pearson's correlations test was run to see if there is a statistically significant correlation between grit level measured by the Grit-S scale and teachers self-efficacy beliefs of student engagement measured by the Teacher self-efficacy Scale (TSES) among elementary (PreK-8) teachers. Data was collected through a survey that contained four demographic questions, the Grit-S scale, and the TSES short scale for the quantitative variables of Grit (M = 3.77, SD = 0.477) and self-efficacy beliefs in student engagement (M = 6.87, SD = 1.23). An assumption of bivariate outliers was run to check for extreme outliers. The Kolmogrov-Smirnov test was run

to check for the normality of the data (N = 86). The normality for data grit was found to be $p < .001$ while the normality for data on teachers' self-efficacy beliefs in student engagement was found to be $p = .181$. Both these values indicate that the data for grit and engagement did not meet the assumption of normality because the p-value or significance was found to be less than .05.

Data analysis with Pearson's correlation was continued and at a 0.05 confidence level, alpha level $p < 0.05$ where $r = 86$ and $p < .001$. The analysis shows a significant positive relationship between grit and self-efficacy beliefs of elementary teachers (PreK-8) in student engagement. Table 13 shows Pearson's correlation results for Grit and teachers' self-efficacy beliefs in student engagement.

Teacher self-efficacy is the confidence teachers have in their ability to make an impact on student learning. Studies indicate that student engagement is associated with increased responsiveness, inquisitiveness, and attention among students (Junco et al., 2013). Grit is the intrinsic desire within an individual to accomplish or succeed at something (Duckworth, 2018; Duckworth et al., 2007). Teachers with higher grit consistently work towards making lessons and activities within classrooms interesting and engaging for students to ensure continued engagement.

Overall, the results of RQ2 revealed that among elementary and middle school teachers (PreK-8) higher grit scores indicate higher self-efficacy beliefs in student engagement. The results were interesting because the majority of the teachers in the sample population were teachers who had been teaching for over ten years. Although this study did not take into consideration specific covariates of teacher age, years of experience, and qualification levels, it was interesting to see it reflected in the findings.

Summary

The purpose of the current quantitative, correlational study was to examine if teacher grit had any significant impact on two of the dimensions of teacher self-efficacy beliefs, classroom management, and student engagement. The results of the Pearson's correlation test between grit scores obtained from the Grit-s scale and the self-efficacy beliefs of teachers obtained from the TSES short scale, show a significant positive relationship between these variables among elementary (PreK-8) teachers.

Chapter 5 concludes this study. Chapter 5 contains findings from the study. Chapter 5 also includes findings related to literature, implications for action, and recommendations for future research.

Chapter 5: Discussions, Conclusions & Recommendations

This chapter includes the purpose of the study, a discussion of the findings, implications for teacher retention strategies in schools, recommendations for further research, and conclusions. A summary and discussion of the findings is organized around both the research questions. Chapter 4 discussed the results of the study through descriptive statistics from the data collected using the sample population of teachers who participated in the survey. In this chapter, the findings are discussed with recommendations for future research. Each research question is discussed separately. The final section of the chapter includes implications and recommendations for future research with regard to strategies that teacher education programs and administrators in school districts can implement in order to retain teachers in the profession and prevent attrition.

Purpose of the Study

There has been increased interest among researchers in the study of the grit construct. Grit has been shown to be a strong indicator of academic success, Grit has also proven to be an important attribute that is being studied in the context of marriage, business, and the military (Duckworth, 2016; Eskreis-Winkler et al., 2014). Based on previous research, studying grit in correlation to performance can be a great predictor in varied fields. Traditionally achievement has been known to be reflected by IQ. However, according to Fitzgerald and Laurian-Fitzgerald (2016), grit has an impact on an individual's intellect. When individuals intentionally persevere with passion, it leads to persistence which in turn leads to increased intellectual capacity.

The purpose of this quantitative correlational study design was to determine if there was a correlation between grit and self-efficacy beliefs in classroom management and student engagement among elementary and middle school teachers (PreK-8) in public schools. There is extensive literature about self-efficacy. Despite the extensive body of work on teacher self-

efficacy, the examination of task-specific teacher self-efficacy beliefs remains lacking in the empirical literature (Zee & Koomen, 2016). Grit is a prominent topic in recent educational research. Understanding the potential of grit in predicting self-efficacy is important because researchers have linked teacher self-efficacy to important outcomes, such as instructional quality in classrooms (Künsting et al., 2016; Ryan et al., 2015). The relevance of grit in academic success (Datu et al., 2016) and how grit can benefit teachers (Jacobson & Kilpatrick, 2015) are key indicators that further studies are needed to study this construct.

The problem that launched the need for the study was a desire to gain an insight into the essential personality traits that help teachers stay in the profession. Education in public schools has seen a multitude of changes and with every change teachers have had to learn and adapt themselves on the go while also taking on additional responsibilities. This applies to those teachers who are in the early years of their careers and also veteran teachers. Furthermore, elementary and middle school teachers have the additional responsibility of teaching young minds how to think while introducing concepts that are often abstract. Teachers in PreK-8 classrooms also have to teach students multiple content areas while ensuring that classrooms are well managed and students are well engaged in activities planned for them.

According to studies, individuals who choose teaching as a career believe that they can make a difference and that it is their way of helping society (Farkas et al., 2000). Drawn to their career as a result of passion many teachers according to studies feel overwhelmed or face burnout. Burnout is common among individuals who do "people work" or work as human service professionals (Farber, 1991). Teaching involves work as a human service professional and there is often an emotional drain as a result of repetitive stress for people involved in such work which can lead to burnout (Maslach & Jackson, 1993). Burnout is a

syndrome characterized by a reduced sense of accomplishment, emotional incapacitation, and detachment Maslach and Jackson (1981).

This study explored the non-cognitive construct of grit in teachers. The study was narrowly focused on capturing the characteristics of teachers, specifically traits related to grit and its impact on self-efficacy beliefs in classroom management and student engagement among elementary and middle school teachers in public schools. Through this research, it is evident that there are strong correlations between grit and self-efficacy beliefs in classroom management and student engagement among elementary and middle school teachers.

Grit is an aspect of one’s mental toughness. Constant effort and hard work under challenging circumstances, maintenance of passion, interest, and perseverance of effort to achieve a target are the primary essence of this mental construct (Duckworth, 2006). Grit is a non-cognitive trait that helps one complete and achieve complex and difficult tasks even when they are tempted to quit. Research shows that grittier individuals perform better than people who are less gritty, and are hence, less likely to give up mid-way (Duckworth, 2006). Grit is related to the psychological concepts of conscientiousness, resilience, aspiration, hardiness, and perseverance, although each construct is distinct and unique (Alahdab et al., 2020). Individuals who demonstrate high levels of grit have higher motivation and do not abandon their objectives, despite the lack of positive feedback (Alahdab et al., 2020). Grit can predict an individual's success in challenging areas beyond just their talent (Alahdab et al., 2020).

As administrators and human resources personnel in school districts struggle to fill positions for teachers, identifying traits in potential candidates and creating professional development opportunities for teachers who are already in their districts will help improve the current teacher crisis. Developing a climate within school districts that promotes learning through professional learning communities, setting up better mentoring for new teachers and

establishing greater administrator support for all teachers so that they focus on engaging students in learning and expression would set up a framework for change in how teachers manage stress and persevere in their goals. This study focused on explicitly looking at the importance of grit among teachers and how it impacts their overall belief in their effectiveness in classrooms. The questions addressed in this study were to determine whether grit levels among teachers in the sample population had a correlation to their self-efficacy beliefs in classroom management and student engagement.

Discussion of the Research Results

A quantitative, non-experimental correlational design was chosen for the study. The predictor variable was grit. The criterion variable was teacher self-efficacy, as measured by two scales of teacher self-efficacy in classroom management and student engagement. Descriptive statistics were calculated, and hierarchical multiple regression analyses were conducted using the IBM Statistical Package for the Social Sciences (SPSS) version 29.02 to answer the research questions.

Based on this study, the following conclusions were drawn. The study concluded that grit has a significantly high correlation to self-efficacy beliefs of teachers in classroom management among elementary and middle school teachers. Secondly, grit also has a significantly high correlation to self-efficacy beliefs of teachers in student engagement among elementary and middle school teachers. The theoretical foundation of this study is grounded in Duckworth et al. 's (2007) theory of grit, Deci and Ryan’s (1985) self-determination theory (SDT), and Bandura’s (1977) self-efficacy theory.

Research Question 1

Research Question 1 asked, “Is there a relationship between PreK-8 teachers’ level of grit and their self-efficacy beliefs on classroom management?” The survey asked respondents four demographic questions related to years of experience, certification pathway, education,

and subject area taught. It also asked respondents to answer questions from the Grit-S scale Grit short scale that consisted of eight questions and the TSES (Teacher Self-Efficacy scale-short version that consisted of eleven questions).

Based on the survey results 82.6% (71/86) of the respondents reported 10–31 years of experience while 17.4 % (15/86) reported up to 10 years of teaching experience. For the teacher's grit, the range of score values among the 86 samples was from 2.37 to 4.62 with a mean score of 3.77 (SD = 0.47), indicating that the teachers had high grit levels. For the teacher's self-efficacy in classroom management, the range of score values among the 86 samples was from 5.25 to 9, with a mean score of 7.375 (SD = 0.88), indicating that the teachers had high efficacy in classroom management. Pearson's correlation results showed a significant and positive correlation between teacher grit scores and their self-efficacy beliefs in classroom management at a confidence level $p < 0.05$, and a statistical power of 0.07 at the medium effect where r (86) = .306, p = .004.

The results confirm the importance of grit and its impact on self-efficacy beliefs in classroom management. Teacher grit has been shown by various studies to be directly linked to teacher effectiveness (Madden, 2015). Teachers who demonstrate greater passion and perseverance for their work therefore are more effective in classrooms. Effectiveness in classrooms is most commonly demonstrated through classroom management strategies. When teachers feel competent in classroom management, they are determined to persevere even though they are new to the profession (Milner, 2002). Good classroom management strategies have been shown to have a direct impact on higher academic achievement among students along with an increase in student motivation (Dicke et al., 2015).

According to Aydin (2002), teaching is a difficult profession and teachers are affected by varying factors that can range from the school environments in which they practice, performance reviews, and the intensity that is expected when being in a profession that

requires so much in terms of people skills. The performance of a teacher has the ability to impact student learning in classrooms (Aydin, 2002). According to Sparks (1988), teachers with higher self-efficacy beliefs use effective instructional strategies in classrooms which enables better performance from students and therefore classrooms are well managed.

Overall, the results of RQ1 revealed that among elementary and middle school teachers (PreK-8) higher grit scores indicate higher self-efficacy beliefs in classroom management. The findings support that grittier individuals persist and persevere despite challenges (Duckworth, 2016). These findings are consistent with Duckworth's (2016) theory of grit that when an individual enjoys what they do which can be associated with passion, they tend to persevere even though they meet obstacles. In the context of classroom teachers developing expertise, with repeated practice of skills is a definition of continuous improvement that helps build confidence and effectiveness in classroom management.

Research Question 2

Research Question 2 asked, “Is there a relationship between PreK-8 teachers’ level of grit and their self-efficacy beliefs on classroom management?” The survey asked respondents four demographic questions related to years of experience, certification pathway, education, and subject area taught. It also asked respondents to answer questions from the Grit-S scale Grit short scale that consisted of eight questions and the TSES (Teacher Self-Efficacy scale-short version that consisted of eleven questions).

Based on the survey results, 82.6% (71/86) of the respondents reported 10–31 years of experience while 17.4 % (15/86) reported up to 10 years of teaching experience. For the teacher’s grit, the range of score values among the 86 samples was from 2.37 to 4.62 with a mean score of 3.77 (*SD* = 0.47), indicating that the teachers had high grit levels. For the teacher's self-efficacy in student engagement, the range of score values among the 86 samples was from 4.5 to 9.00, with a mean score of 6.87 (*SD* = 1.23), indicating high efficacy in

student engagement. Pearson's correlation results showed a significant and positive correlation between teacher grit scores and their self-efficacy beliefs in student engagement among elementary and middle school (PreK-8) teachers. The results showed a significant positive relationship does exist at a confidence level $p < 0.05$, and a statistical power of 0.07 at the medium effect where r (86) = .369, $p < .001$.

Teacher self-efficacy is the confidence teachers have in their ability to make an impact on student learning and outcomes. It is the sense from an educator that they can make a difference to students even if students appear to be unmotivated or seem challenging in the classroom (Tschannen-Moran & Hoy, 2001). Grit is the intrinsic desire within an individual to accomplish or succeed at something (Duckworth, 2018; Duckworth et al., 2007). Grit comprises perseverance and consistency of interest in the face of impediments and hurdles to succeed and achieve a set goal (Duckworth, 2018; Duckworth et al., 2007).

Self-efficacy beliefs in student engagement are the dimension that refers to teachers' perceptions of their abilities to engage their students in class content (Tschannen-Moran & Hoy, 2001). Studies indicated that student engagement is associated with increased responsiveness, inquisitiveness, and attention among students (Junco et al., 2013). Furthermore, research shows teacher competence, classroom environment, teacher-student relations, and student self-esteem to be associated with student engagement (Junco et al., 2013; Kahu, 2013; van Uden et al., 2014).

Overall, the results of RQ2 revealed that among elementary and middle school teachers (PreK-8) higher grit scores indicate higher self-efficacy beliefs in student engagement. Studies on teacher grit emphasize the importance of teacher grit in an occupational sense (Atapattu, 2015; Dobbins, 2016; Eskreis-Winkler et al., 2016; Madden, 2015; Robertson-Kraft & Duckworth, 2014; Silvia et al., 2013; West, 2015). The reason is that persistence and perseverance, allow individuals to follow long-term goals (Datu et al.,

2016). Teaching requires emotional labor, enthusiasm, passion, and persistence in achieving long-term goals. However, in the case of teachers, it is usually difficult to see immediate results especially when teaching at the elementary and middle school level.

The findings support the importance of the power to perceive that one's efforts will result in something worthy (Duckworth, 2016). In the context of classroom teachers, their sense of purpose can increase their level of grit and help them persevere despite challenges. Studies by Von Culin et al. (2014) show that "grittier individuals were more likely than less gritty individuals to see happiness through engagement" (p. 5). Being able to teach using creative methods of student engagement requires perseverance and continuous effort. Therefore, higher grit in teachers helps them achieve greater effectiveness in engaging students in classrooms.

According to Bandura (1997), mastery experiences are the most powerful because they show an individual's ability to take on a task. When an individual experiences success in doing a particular task repeatedly, their sense of self-efficacy increases when presented with a similar task in the future. Teacher performance is most commonly viewed from the perspective of student achievement. Although there is a relationship between the two, individual grit among teachers is a better indicator of the level of passion and perseverance as well as the ability to achieve set goals for student performance in classrooms. According to Duckworth (2016) using the lens of grit, individual achievements can be viewed in two parts. One is the talent that an individual possesses and the other is the effort one puts in. The combination of the two in a ratio where effort is double that of talent, leads to skill (Duckworth, 2016).

Implications

This study examined the correlation between grit and self-efficacy beliefs of teachers in the area of classroom management and student engagement. The study used the theoretical

foundations of Duckworth's (2007) theory of grit, Deci and Ryan's (1985) self-determination theory (SDT), and Bandura's (1977) self-efficacy theory. The results of the study indicate that grit has a positive and significant relationship to self-efficacy beliefs of teachers in elementary and middle school. Previous studies have found grit to be a significant predictor of self-efficacy (Rozell, 2019). Previous studies have also found both grit (Jang & Kim, 2019; Shin et al., 2018) and teacher self-efficacy (Afzal et al., 2019; Shin & Kim, 2017; Skaalvik & Skaalvik, 2016; Swanson & Huff, 2010) were negatively correlated to turnover intentions among teachers.

The knowledge of personality traits such as grit would benefit the advancement of scientific knowledge to determine the specific job stressors that most often contribute to turnover intention among elementary and middle school teachers. Teachers in elementary and middle schools have a difficult time teaching young learners and keeping them engaged in productive expression and learning. Besides teaching, teachers also have to ensure that the social-emotional needs of these learners are taken care of. Dealing with behavioral issues in classrooms while ensuring learning is not discontinued, managing classrooms as well the burden of other duties within the school building and preparing materials are some of the other daily workloads that teachers have to deal with. The overload causes stress and in some cases a feeling of failure if issues stay unresolved. The knowledge of traits such as grit would benefit administrators and human resource departments by providing valuable insight as to where to specifically concentrate resources to eliminate the reasons that cause job stress among teachers. Ensuring that adequate support systems are in place so that teachers feel supported as well as opportunities for professional development that are meaningful in enhancing teacher effectiveness in order to improve self-efficacy beliefs among teachers.

Studies show that "new teachers at risk of leaving the profession express strong dissatisfaction with their teaching assignments, and frustration with the politics of their

profession, the lack of adequate resources, and inadequate mentoring support" (Tait, 2008, p. 58). Furthermore, "teaching is one of the few professions in which beginners have as much responsibility as their experienced colleagues" (Tait, 2008, p. 58). New teachers carry the same workload and classroom schedules, as veteran teachers yet earn less due to lack of experience. New teachers also often feel unheard and that is why they have low self-efficacy levels. "It is essential that schools have in place a mentoring system that will support teachers' growth and retention" (Bell & Thomas, 2007, p. 2). Well-planned, mentoring programs that provide opportunities for mutual dialogue, observation, reflection, and establishment of shared goals (Bell & Thomas, 2007) can help new teachers feel a sense of community and belonging.

Numerous studies on employee retention show that when employees have a sense of psychological well-being at their place of work, they tend to be more successful and effective (Madden et al., 2006). Furthermore, when employees experience less anxiety and have a higher sense of self-esteem at work they contribute positively to institutional goals and take on changes as a means to improve themselves (Sezgin, 2012). Cases of burnout are also minimal in workplaces where anxiety and stress are less (Chan, 2003; Nartgün & Sahin, 2015). Thus, it is evident that there is a positive relationship between the perseverance of effort and the consistency of interest (Duckworth & Quinn, 2009a). This has a direct impact on creating a positive influence of the teachers in the development of their school environment. When individuals see that others are successful with their efforts, they also begin to believe that if they work hard, they can also succeed (Bandura, 2004).

Irrespective of teacher qualification, teacher motivation and performance are affected by the psychological state and grit of teachers. Higher grit levels ensure that the performance, health, and mood of teachers remain positive even in stressful environments (Maddi et al., 1998). Moreover, teachers are less affected by the negative results and their individual

failures (Maddi et al., 1998). Grit directly influences diligence, determination, and perseverance to reach long-term goals. Studies show that individuals with higher grit are more committed to their goals (Hill et al., 2014). Grit among teachers is reflected in their ability to be effective in the classroom (Yılmaz, 2007). Characteristics such as having a sense of trust, self-efficacy beliefs, optimism, extroversion, hope, and psychological endurance makes teachers gritty (Eryılmaz, 2013). These are often some of the requirements sought in teacher competencies. Grit influences teaching creativity, organizational commitment, and knowledge management (Widodo & Gunavan, 2021). The practical implications could help teachers work with building administrators to improve their grit sustainably by utilizing available school resources (Widodo & Gunavan, 2021).

Limitations

Several limitations to this study need to be considered. According to Creswell (2013), “Limitations are potential weaknesses or problems with the study identified by the researcher” (p. 199). The scope of this study was limited to public schools in Northwestern New Jersey because I only had access to schools in this region and was only able to send the survey to only those districts that had provided permission within this region. The study was limited to the period of 2 months of data collection to ensure that the data collected in the study were applicable at the present time.

The study was limited to the constructs of grit and teacher self-efficacy measured using the survey instruments considered in the study. Self-reporting measures are easy and inexpensive and easy to score; however, validity is a concern. Participants might have different interpretations of survey questions, which might influence their responses (Gilner et al., 2009, p. 181). The survey used in this study included constructs of teacher grit, self-efficacy beliefs of teachers in classroom management and student engagement.

A limitation of this study was the use of convenience sampling of teachers. Convenience sampling increased the likelihood that the participants were not a representative sample of the population of PreK-8 teachers in the United States. Using a quantitative nonexperimental correlational design was a limitation. Quantitative nonexperimental correlational research cannot infer cause-and effect; causality could only be determined in experimental studies. Lastly, due to the size limitations of the convenience sample, the study may have limited transferability and generalizability to a similar population.

Recommendations for Future Research

Upon review of the findings, results, and limitations of this study, there are a number of recommendations for future research to consider. The following recommendations should be considered to increase knowledge and formulate strategies for supporting novice and veteran teachers across all settings and creating quality and effective educational opportunities for all students in classrooms.

1. Replicate the study and include teachers at the elementary and middle school levels across the country.
2. Replicate the study within private and charter schools. Consider comparing and
3. contrasting the results from these educational settings with the public schools.
4. Supplement the quantitative data gathered with qualitative interviews to gather further
5. information and drill deeper into the reasoning for teacher responses.
6. Replicate the study to examine if there exists a difference in grit and self-efficacy among teachers in less affluent school districts and teachers in more affluent school districts.
7. Replicate the study to include covariates such as teacher qualification levels, age, and years of experience.

8. Replicate the study in rural and urban school districts to examine if there exists a difference in grit and self-efficacy among teachers based on demographics.
9. Replicate the study across elementary and middle schools in different states in America to examine if there exists a difference in grit and self-efficacy among teachers in different states.

Conclusion

Grit is a non-cognitive trait with potential benefits. The variables of classroom management and student engagement have been researched previously linking them to self-efficacy. However, there are no specific studies linking classroom management and student engagement to grit, among elementary teachers providing a gap for further investigation (Day et al., 2016; Dicke et al., 2015; van Uden et al., 2014). According to Jacobson and Kilpatrick (2015), existing literature is limited to the potential benefits of noncognitive traits, such as grit, in teachers.

The results of this study may be useful for school districts in creating layered support systems for experienced staff as well as new staff, and to craft professional development programs for teachers that can help them learn strategies that would make them successful in classrooms. Administrators can use strategies to create positive work environments in their schools by ensuring teachers feel a sense of belonging.

Administrators in school districts can improve teacher morale and decrease attrition rates by creating a collaborative and supported work environment. This will help promote creativity and optimism. Creativity has been shown to influence self-efficacy (Pachler et al., 2019; Regier & Savic, 2019; Vally et al., 2019) as well as innovative behavior (Zocche et al., 2018). Creativity among employees influences their organizational citizenship (Deng & Guan, 2017) as well as their performance (Rohmaniyah & Nurhayati, 2017) and professionalism (Kamenez et al., 2019). When individuals believe in a purpose, they are

focused on working towards it by putting in the required effort and engaging in continuous improvement.

Human resource departments in districts can design professional development opportunities that can help teachers develop a growth mindset. The mindset of an individual impacts their motivation, self-regulation, achievement, and interpersonal processes (Dweck, 2017). According to Dweck (2017), mindsets are the beliefs one has about their qualities such as intelligence, talents, and personality. Individuals with a fixed mindset view their abilities as static, they put in less effort to succeed and reject challenging learning opportunities because they fear that they might fail.

In contrast, individuals with a growth mindset believe that their abilities can change, they consider mistakes to be part of the learning process and they persevere when faced with challenges or setbacks (Dweck, 2007). Individuals may have one or the other kind of mindset at different times in their lives and for different tasks. Studies show that mindsets are influenced by the messages that an individual receives around them and mindsets can be changed (Dweck, 2007). Positive messaging within districts through constructive feedback, open conversations, creation of professional learning communities that help teachers share knowledge and strategies are key to promoting a growth mindset among teachers as evidence from studies show that mindset can be affected by failure, criticism, and praise (Cain & Dweck, 1995; Smiley & Dweck, 1994).

Lastly, teachers can benefit from professional development opportunities that can help them develop a growth mindset. Districts can benefit by promoting a positive work environment and improve teacher grit by putting in place supports that can help veteran teachers when they start to face burnout. School districts can set up structured mentorship programs for new teachers to increase self-efficacy beliefs and ensure that all teachers feel valued and have a sense of community within their buildings.

References

Afzal, S., Arshad, M., Saleem, S., & Farooq, O. (2019). The impact of perceived supervisor support on employees' turnover intention and task performance: Mediation of self-efficacy. *Journal of Management Development*, *38*(5), 369–382. https://doi.org/10.1108/JMD-03-2019-0076

Aldridge, J., & Fraser, B. (2016). Teachers' views of their school climate and its relationship with teacher self-efficacy and job satisfaction. *Learning Environments Research*, *19*, 291–307. https://doi.org/10.1007/s10984-015-9198-x

Al-Ghamdi, N. G. (2017). Role overload and job stress among the female university teachers-Saudi context. *European Online Journal of Natural and Social Sciences*, *6*(2), 288–295. https://european-science.com/eojnss/article/view/5117/pdf

Alahdab, F., Halvorsen, A. J., Mandrekar, J. N., Vaa, B. E., Montori, V. M., West, C. P., Murad, M. H., & Beckman, T. J. (2020). How do we assess resilience and grit among internal medicine residents at the Mayo Clinic? A longitudinal validity study including correlations with medical knowledge, professionalism and clinical performance. *BMJ Open*, *10*(12), e040699. https://doi.org/10.1136/bmjopen-2020-040699

Aloe, A. M., Amo, L. C., & Shanahan, M. E. (2014). Classroom management self-efficacy and burnout: A multivariate meta-analysis. *Educational Psychology Review*, *26*(1), 101–126. https://doi.org/10.1007/s10648-013-9244-0

Althauser, K. (2015). Job-embedded professional development: Its impact on teacher self-efficacy and student performance. *Teacher Development*, *19*(2), 210–225. https://doi.org/10.1080/13664530.2015.1011346

Atapattu, R. (2015). Grit and engagement as mediators of the relation between academic support and literacy achievement in ethnic minority elementary school students [Doctoral dissertation, University of Maryland]. University of Maryland Archive. http://drum.lib.umd.edu/bitstream/handle/1903/17400/Atapattu_umd_0117N_16826.pdf

Aydın, İ. P. (2002). *Stress in business life*. Pegem Publishing.

Baird, T. D., Kniola, D. J., Lewis, A. L., & Fowler, S. B. (2015). Pink time: Evidence of self-regulated learning and academic motivation among undergraduate students. *Journal of Geography*, *114*, 146–157. http://dx.doi.org/10.1080/00221341.2014.977334

Bandura, A. (1977). *Self-efficacy: Toward a unifying theory of behavioral change*. *Psychological Review*, *84*(2), 191–215. https://doi.org/10.1037/0033-295X.84.2.191

Bandura, A., Taylor, C. B., Williams, S. L., Mefford, I. N., Barchas, J. D. (1985). Catecholamine secretion as a function of perceived coping self-efficacy. *Journal of Consulting Clinical Psychology*, *53*(3), 406–414. https://doi.org/10.1037//0022-006x.53.3.406

Bandura, A. (1986). The explanatory and predictive scope of self-efficacy theory. *Journal of Social and Clinical Psychology*, *4*(3), 359–373. https://doi.org/10.1521/jscp.1986.4.3.359

Bandura, A. (1994). Self-efficacy. In V. S. Ramachaudran (Ed.), *Encyclopedia of human behavior* (Vol. 4, pp. 71–81). Elsevier.

Barańczuk, U. (2021). The five-factor model of personality and generalized self-efficacy: A meta-analysis. *Journal of Individual Difference*s. *42*(4). https://doi.org/10.1027/1614-0001/a000345

Bashant, J. (2014). Developing grit in our students: Why grit is such a desirable trait, and practical strategies for teachers and schools. *Journal for Leadership and Instruction*, *13*(2), 14–17. https://files.eric.ed.gov/fulltext/EJ1081394.pdf

Barni, D., Danioni, F., & Benevene, P. (2019). Teachers' self-efficacy: The role of personal values and motivations for teaching. *Frontiers in Psychology*, *10*, 1645. https://doi.org/10.3389/fpsyg.2019.01645

Barroga, E., Matanguihan, G. J., Furuta, A., Arima, M., Tsuchiya, S., Kawahara, C., Takamiya, Y., & Izumi, M. (2023). Conducting and writing quantitative and qualitative research. *Journal of Korean Medical Science*, *38*(37), e291. https://doi.org/10.3346/jkms.2023.38.e291

Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin*, *117*(3), 497–529. https://doi.org/10.1037/0033-2909.117.3.497

Bell, D., & Thomas, E. (2007) "A Mentoring Process to Support Teachers' Growth and Retention." *Academic Leadership: The Online Journal (2003-2012)*, 5(3). https://doi.org/10.58809/lovy7125

Bettini, E., Jones, N., Brownell, M., Conroy, M., Park, Y., Leite, W., Crockett, J., & Benedict, A. (2017). Workload manageability among novice special and general educators: Relationships with emotional exhaustion and career intentions. *Remedial and Special Education*, *38*(4), 246–256. https://doi.org/10.1177/0741932517708327

Billingsley, B., Bettini, E., (2019). Special education teacher attrition and retention: A review of the literature. *Review of Educational Research*, *89*(5), 697–744. https://doi.org/10.3102/0034654319862495

Bobek, B.L. (2002). Teacher resiliency: A key to career longevity. *The Clearing House*, *75*(4), 202–205. https://doi.org/10.1080/00098650209604932

Bowman, N. A., Hill, P. L., Denson, N., & Bronkema, R. (2015). Keep on truckin' or stay the course? Exploring grit dimensions as differential predictors of educational achievement, satisfaction, and intentions. *Social Psychological and Personality Science, 6*(6), 639–645. https://doi.org/10.1177/1948550615574300

Boyle, G. J., Borg, M. G., Falzon, J. M., & Baglioni, A. J. (1995). A structural model of the dimensions of teacher stress. *British Journal of Educational Psychology, 65*(1), 49–67. https://doi.org/10.1111/j.2044-8279.1995.tb01130.x

Bray-Clark, N., & Bates, R. (2003). Self-efficacy beliefs and teacher effectiveness: Implications for professional development. *Professional Educator, 26*(1), 13–22. https://files.eric.ed.gov/fulltext/EJ842387.pdf

Bruso, J., Stefaniak, J., & Bol, L. (2020). An examination of personality traits as a predictor of the use of self-regulated learning strategies and considerations for online instruction. *Educational Technology Research Development, 68*, 2659–2683. https://doi.org/10.1007/s11423-020-09797-y

Burton, L. (2021). *Grit as passion and perseverance and not sandpaper in retaining teachers: A correlational study* (Publication No. 28863179) [Doctoral dissertation, Northcentral University]. ProQuest Dissertations & Theses Global.

Cain, K. M., & Dweck, C. S. (1995). The relation between motivational patterns and achievement cognitions through the elementary school years. *Merrill-Palmer Quarterly, 41*(1), 25–52.

Caprara, G. V., Barbaranelli, C., Borgogni, L., & Steca, P. (2003). Efficacy beliefs as determinants of teachers' job satisfaction. *Journal of Educational Psychology, 95*(4), 821–832. https://doi.org/10.1037/0022-0663.95.4.821

Caprara, G. V., Barbaranelli, C., Steca, P., & Malone, P. S. (2006). Teachers' self-efficacy beliefs as determinants of job satisfaction and students' academic achievement: A study at the school level. *Journal of School Psychology, 44*(6), 473–490. https://doi.org/10.1016/j.jsp.2006.09.001

Carrillo, C., & Assunção Flores, M. (2018). Veteran teachers' identity: what does the research literature tell us? *Cambridge Journal of Education*, *48*(5), 639–656. https://doi.org/10.1080/0305764X.2017.1394982

Carver-Thomas, D., & Darling-Hammond, L. (2019). The trouble with teacher turnover: How teacher attrition affects students and schools. *Education Policy Analysis Archives*, *27*(36), 1–32. https://doi.org/10.14507/epaa.27.3699

Cedoline, A. J. (1982). *Job burnout in public education*. Teachers College Press.

Celep, C. (2000). Teachers' organizational commitment in educational organizations. *National FORUM of Teacher Education Journal*, *10*(3), 1999-2000.

Centers for Disease Control and Prevention. (n.d.). *Professional development 201: From basic to dynamic full course content*. https://www.cdc.gov/healthyschools/professional_ development/e-learning/pd201/_assets/PD201_Full_Course_Content.pdf

Chan, D. W. (2003). Hardiness and its role in the stress-burnout relationship among prospective Chinese teachers in Hong Kong. *Teaching and Teacher Education*, *19*, 381–395. https://doi.org/10.1016/S0742-051X(03)00023-4

Cherniss, C. (1980). *Professional burnout in human service organizations*. Praeger Publishers.

Clark, K. N., & Malecki, C. K. (2019). Academic grit scale: Psychometric properties and associations with achievement and life satisfaction. *Journal of School Psychology*, *72*, 49–66. https://doi.org/10.1016/j.jsp.2018.12.001

Cook, B., & Cook, L. (2008). Nonexperimental quantitative research and its role in guiding instruction. Intervention in school and clinic. *Intervention School Clinic*, *44*, 98–104. https://doi.org/10.1177/1053451208321565

Constantine, J.M., Player, D.W., Silva, T., Hallgren, K., Grider, M., & Deke, J. (2009). An Evaluation of Teachers Trained through Different Routes to Certification. Final Report. NCEE 2009-4043. *National Center for Education Evaluation and Regional Assistance*.

Corbell, K. & Osborne, J., & Reiman, A. (2010). Supporting and retaining beginning teachers: A validity study of the Perceptions of Success Inventory for Beginning Teachers. *Educational Research and Evaluation*, *16*, 75–96. https://doi.org/10.1080/13803611003722325.

Cormier, D. L., Dunn, J. G. H., & Causgrove Dunn, J. (2019). Examining the domain specificity of grit. Personality and individual differences, *139*, 349–354. https://doi.org/10.1016/j.paid.2018.11.026

Coronado, D. M. (2016). *Teacher's sense of self-efficacy and grit and its relationship to student achievement* (Publication No. 10152948) [Doctoral dissertation, University of Texas]. ProQuest Dissertation and Theses Global.

Costa, P. T., & McCrae, R. R. (1992). The five-factor model of personality and its relevance to personality disorders. *Journal of Personality Disorders*, *6*(4), 343–359. https://doi.org/10.1521/pedi.1992.6.4.343

Credé, M., Tynan, M. C., & Harms, P. D. (2016). Much ado about grit: A meta analytic synthesis of the grit literature. *Journal of Personality and Social Psychology*, *113*(3), 492–511. https://doi.org/10.1037/pspp0000102

Creswell, J. W. (2013). *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage.

Csikszentmihalyi, M., & Rathunde, K. (1993). The measurement of flow in everyday life: Toward a theory of emergent motivation. In J. E. Jacobs (Ed.), *Nebraska Symposium on Motivation, 1992: Developmental perspectives on motivation* (pp. 57–97). University of Nebraska Press.

Cullinane, C., & Montacute, R. (2020). *COVID-19 and social mobility impact brief #1: School Shutdown*. Research Brief.

Dale, G., Sampers, D., Loo, S., & Green, C.S. (2018). Individual differences in exploration and persistence: Grit and beliefs about ability and reward. *PLoS ONE*, *13*(9): e0203131. https://doi. org/10.1371/journal.pone.0203131

Datu, J. A., Valdez, J. P. M., & King, R. B. (2016). Perseverance counts but consistency does not! Validating the short grit scale in a collectivist setting. *Current Psychology*, *35*(1), 121–130.

Datu, J. A. D., Yuen, M., & Chen, G. (2017). Development and validation of the Triarchic Model of Grit Scale (TMGS): Evidence from Filipino undergraduate students. *Personality and Individual Differences, 114,* 198–205. https://doi.org/10.1016/j.paid.2017.04.012

Darling-Hammond, L., Chung, R., & Frelow, F. (2002). Variation in teacher preparation: How well do different pathways prepare teachers to teach? *Journal of Teacher Education*, *53*(4), 286–302.

Darling-Hammond, L. (2010). Teacher education and the American future. *Journal of Teacher Education*, *61*(1-2), 35–47.

Day, C., Gu, Q., & Sammons, P. (2016). The impact of leadership on student outcomes: How successful school leaders use transformational and instructional strategies to make a difference. *Educational Administration Quarterly*, *52*(2), 221–258. https://doi.org/10.1177/0013161X15616863

deCharms, R. (1992). Personal causation and the origin concept. In C. P. Smith, J. W. Atkinson, D. C. McClelland, & J. Veroff (Eds.), *Motivation and personality: Handbook of thematic content analysis* (pp. 325–333). Cambridge University Press. https://doi.org/10.1017/CBO9780511527937.024

Deci, E. L, & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. Springer.

Deci, E. L., & Ryan, R. M. (2008). Self-determination theory: A macrotheory of human motivation, development, and health. *Canadian Psychology/Psychologie Canadienne*, *49*(3), 182–185. https://doi.org/10.1037/a0012801

Demirdag, S. (2015). The relationship between critical thinking abilities and classroom management skills of high school teachers. *Educational Research and Reviews*, *10*(7), 850–855.

Deci, E. L. (1975). *Intrinsic motivation.* Plenum Press. https://doi.org/10.1007/978-1-4613-4446-9

Deng, M., Wang, S., Guan, W., & Wang, Y. (2017). The development and initial validation of a questionnaire of inclusive teachers' competency for meeting special educational needs in regular classrooms in China. *International Journal of Inclusive Education*, *21*(4), 416–427. https://doi.org/10.1080/13603116.2016.1197326

Dicke, T., Elling, J., Schmeck, A., & Leutner, D. (2015). Reducing reality shock: The effects of classroom management skills training on beginning teachers. *Teaching and Teacher Education*, *48*(1), 1–12. https://doi.org/10.1016/j.tate.2015.01.013

Djigić, G., Stojiljković, S., & Dosković, M. (2014). Basic personality dimensions and teachers' self-efficacy. *Procedia - Social and Behavioral Sciences*, *112*, 593–602.

Djonko-Moore, C. M. (2016). An exploration of teacher attrition and mobility in high poverty racially segregated schools. *Race Ethnicity and Education*, *19*(5), 1063–1087. https://doi.org/10.1080/13613324.2015.1013458

Dobbins, D. (2016). Teacher effectiveness: Examining the relationship between teacher grit and teacher self-efficacy [Doctoral dissertation, Oklahoma State University]. Open Research Oklahoma State University. https://openresearch.okstate.edu/server/api/core/bitstreams/ac6a936f-2078-4f46-aa47-77d92ce6813c/content

Doshi P. (2011). The elusive definition of pandemic influenza. *Bulletin of the World Health Organization*, *89*(7), 532–538. https://doi.org/10.2471/BLT.11.086173

Douglas, D., Obadan, A., Palius, M. F., & Walsh, S. (2024). *New Jersey teacher workforce landscape report*. Heldrich Center. https://heldrich.rutgers.edu/sites/default/files/2024-04/New_Jersey%E2%80%99s_Teacher_Workforce_Landscape_2024_Annual_Report.pdf

Duckworth, A. L. (2006). *Intelligence is not enough: Non -IQ predictors of achievement* (Publication No. 3211063) [Doctoral dissertation, University of Pennsylvania. ProQuest Dissertations & Theses Global.

Duckworth, A. (2016). *Grit: The power of passion and perseverance.* Scribner/Simon & Schuster.

Duckworth, A. (2018). *Grit: The power of passion and perseverance*. Scribner.

Duckworth, A., Peterson, C., Matthews, M. D., & Kelly, D. R. (2007). Grit: Perseverance and passion for long-term goals. *Journal of Personality and Social Psychology*, *92*(6), 1087–1101. https://doi.org/10.1037/0022-3514.92.6.1087

Duckworth, A., & Quinn, P. D. (2009). Development and validation of the Short Grit Scale (GRIT–S). *Journal of Personality Assessment*, *91*(2), 166–174. https://doi.org/10.1080/00223890802634290

Duckworth, A., Quinn, P., & Seligman, M. (2009). Positive predictors of teacher effectiveness. *The Journal of Positive Psychology*, *4*(6), 540–547. https://doi.org/10.1080/17439760903157232

Duckworth, A., & Quinn, P. (2012). *Development and validation of the Short Grit Scale (Grit–S)*. https://www.researchgate.net/publication/23996619_Development_and_validation_of_the_Short_Grit_Scale_GRIT-S

Duckworth, A. L., Weir, D., Tsukayama, E., & Kwok, D. (2012). Who does well in life? Conscientious adults excel in both objective and subjective success. *Frontiers in Psychology*, *3*, 356. https://doi.org/10.3389/fpsyg.2012.00356

Duffin, L. C., French, B. F., & Patrick, H. (2012). The teachers' sense of efficacy scale: Confirming the factor structure with beginning pre-service teachers, *Teaching and Teacher Education*, *28*(6), 827–834, https://doi.org/10.1016/j.tate.2012.03.004.

Dugan, R., Hochstein, B., Rouziou, M., & Britton, B. (2018). Gritting their teeth to close the sale: The positive effect of salesperson grit on job satisfaction and performance. *Journal of Personal Selling & Sales Management*, *39*(1), 81–101. https://doi.org/10.1080/08853134.2018.1489726

Dweck, C. S. (2006). *Mindset*. (Çev. U. Kaya). Yakamoz Yayıncılık.

Dweck, C. S. (2007). The perils and promises of praise. *Educational Leadership*, *65*(2), 34–39.

Dweck, C. S. (2008). Can personality be changed? The role of beliefs in personality and change. *Current Directions in Psychological Science*, *17*(6), 391–394.

Dweck C. (2014). Teachers' mindsets: "Every student has something to teach me" feeling overwhelmed? Where did your natural teaching talent go? Try pairing a growth mindset with reasonable goals, patience, and reflection instead. It's time to get gritty and be a better teacher. Education Horizons, *93*, 10–15. https://doi.org/10.1177/0013175X14561420

Dweck, C. S. (2016). *Mindset: The new psychology of success* (Updated ed.). Random House.

Dweck, C. S. (2017). The journey to children's mindsets and beyond. *Child Development Perspectives*, *11*(2), 139–144. https://doi.org/10.1111/cdep.12225

Eisenman, G., Edwards, S., & Cushman, A. (2015). Bringing reality to classroom management in teacher education. *The Professional Education*, *39*(1). https://files.eric.ed.gov/fulltext/EJ1062280.pdf

Eryılmaz, A. (2013). Using of positive psychology on psychological counseling and guidance context on character strength and preventive services. *The Journal of Happiness & Wellbeing*, *1*(1), 1–19.

Eskreis-Winkler, L., Shulman, E. P., Beal, S. A., & Duckworth, A. L. (2014). The grit effect: predicting retention in the military, the workplace, school and marriage. *Frontiers in Psychology*, *5*, 36. https://doi.org/10.3389/fpsyg.2014.00036

Farber, B. A. (1991). *Crisis in education*. Jossey-Bass Publishers.

Farkas, S., Thompson, J., & Foleno, T. (2000). *A sense of calling: Who teaches and why*. Public Agenda Foundation.

Fennick, R. (1992, March 19–21). *Combating new teacher burnout: Providing support networks for personal and professional growth* [Paper presentation]. Conference on College Composition and Communication, Cincinnati, OH, United States.

Ferguson, K., Frost, L., & Hall, D. (2012). Predicting teacher anxiety, depression, and job satisfaction. *Journal of Teaching and Learning*, *8*(1), 27–42, https://doi.org/10.22329/jtl.v8i1.2896

Field., A. (2013). *Discovering Statistics using IBM SPSS Statistics* (4th. ed.). Sage.

Finster, M. W. (2015). *Identifying, monitoring, and benchmarking teacher retention and turnover: Guidelines for TIF grantees*. Department of Education.

Fite, R. E., Lindeman, M. I., Rogers, A. P., Voyles, E., & Durik, A. M. (2017). Knowing oneself and long-term goal pursuit: Relations among self-concept clarity, conscientiousness, and grit. *Personality and Individual Differences*, *108*, 191–194. https://doi.org/10.1016/j.paid.2016.12.008

Fitzgerald, C. J., & Laurian-Fitzgerald, S. (2016). Helping students enhance their grit and growth mindsets. *Journal Plus Education / Educatia Plus*, 14, 52–67. https://www.uav.ro/jour/index.php/jpe/article/view/673/739

Fong, C. J., & Kim, Y. W. (2019). A clash of constructs? Re-examining grit in light of academic buoyancy and future time perspective, *Current Psychology*, *40*(4),1824–1837. https://doi.org/10.1007/s12144-018-0120-4

Fox, A. G., & Peters, M. L. (2013). First year teachers: Certification program and assigned subject on their self-efficacy. *Current Issues in Education*, *16*(1), 1–15. https://cie.asu.edu/ojs/index.php/cieatasu/article/view/1117

Gay, R. L., Mills, E. G., & Airasian, W. P. (2014). *Educational research competencies for analysis and application*. Pearson Education Limited

Gilner, J. A., Morgah, G. A., & Leech, N. L. (2009). *Research methods in applied settings: An integrated approach to design and analysis* (2nd ed.). Routledge

Gold, Y. (1984). Burnout: A major problem for the teaching profession. *Education*, *104*(3), 271–274.

Haderer, A. M. (2020). *Exploring the relationship between teacher efficacy and grit* (Publication No. 27964169) [Doctoral dissertation, Shenandoah University]. ProQuest Dissertations & Theses Global.

Hakanen, J. J., Bakker, A. B., & Schaufeli, W. B. (2006). Burnout and work engagement among teachers. *Journal of School Psychology*, *43*(6), 495–513. https://doi.org/10.1016/j.jsp.2005.11.001

Harris, J. A., Mihalas, S., Hirokawa, K. E., Whitesell, J. D., Choi, H., Bernard, A., Bohn, P., Caldejon, S., Casal, L., Cho, A., Feiner, A., Feng, D., Gaudreault, N., Gerfen, C. R., Graddis, N., Groblewski, P. A., Henry, A. M., Ho, A., Howard, R., Knox, J. E., … Zeng, H. (2019). Hierarchical organization of cortical and thalamic connectivity. *Nature*, *575*(7781), 195–202. https://doi.org/10.1038/s41586-019-1716-z

Harter, S. (1978). Effectance motivation reconsidered: Toward a developmental model. *Human Development*, *21*(1), 34–64. https://doi.org/10.1159/000271574

Henke, R. R., Chen, X. & Geis, S. (2015). *Progress through the teacher pipeline: 1992-93 college graduates and elementary/secondary school teaching as of 1997*. National Center of Education Statistics.

Herman, K. C., Hickmon-Rosa, J., & Reinke, W. M. (2018). Empirically derived profiles of teacher stress, burnout, self-efficacy, and coping and associated student outcomes. *Journal of Positive Behavior Interventions*, *20*(2), 90–100. https://doi.org/10.1177/1098300717732066

Hill, P. L., Burrow, A. L., & Bronk, K. C. (2016). Persevering with positivity and purpose: An examination of purpose commitment and positive affect as predictors of grit. *Journal of Happiness Studies: An Interdisciplinary Forum on Subjective Well-Being*, *17*(1), 257–269. https://doi.org/10.1007/s10902-014-9593-5

Hise, M. (2022). *Attrition, Mobility, and Retention Patterns of Public-School Teachers* (Publication No. 29061930) [Doctoral dissertation, Pepperdine University]. ProQuest Dissertation and Theses Global.

Hochanadel, A., & Finamore, D. (2015). Fixed and growth mindset in education and how grit helps students persist in the face of adversity. *Journal of International Education Research*, *11*(1), 47–50. https://files.eric.ed.gov/fulltext/EJ1051129.pdf

Holzberger, D., Philipp, A., & Kunter, M. (2014). Predicting teachers' instructional behaviors: The interplay between self-efficacy and intrinsic needs. *Contemporary Educational Psychology*, *39*(2), 100–111. https://doi.org/10.1016/ j.cedpsych.2014.02.001

Hoy, A. (2000). *Changes in teacher efficacy during the early years of teaching*. https://www.researchgate.net/publication/237218148_Changes_in_Teacher_Efficacy_During_the_Early_Years_of_Teaching

Huisman, S., Singer, N., & Catapano, S. (2010). Resiliency to success: Supporting novice urban teachers. *Teacher Development, 14*(4), 483–499.

Hull, C. L. (1943). *Principles of behavior: An introduction to behavior theory.* Appleton-Century.

Irwin, V., Wang, K., Tezil, T., Zhang, J., Filbey, A., Jung, J., Bullock Mann, F., Dilig, R., & Parker, S. (2023). Report on the condition of education 2023 (NCES 2023-144). U.S. Department of Education. https://nces.ed.gov/pubsearch/pubsinfo.asp?pubid=2023144.

Jacobson, E., & Kilpatrick, J. (2015). Understanding teacher affect, knowledge, and instruction over time: an agenda for research on productive disposition for teaching mathematics. *Journal of Mathematics Teacher Education. 18*. 401–406. 10.1007/s10857-015-9316-9.

Jachimowicz, J. M., Hauser, O. P., O'Brien, J. D., Sherman, E., & Galinsky, A. D. (2018). The critical role of second-order normative beliefs in predicting energy conservation. *Nature Human Behavior*, *2*(10), 757–764. https://doi.org/10.1038/s41562-018-0434-0

Jang, D. H., & Kim, S. L. (2019). The effects of early childhood teachers' job stress and grit on their turnover intention. *The Journal of the Convergence on Culture Technology*, *5*(1): 177–183.

Jennings, P. A., Brown, J. L., Frank, J. L., Doyle, S., Oh, Y., Davis, R., Rasheed, D., DeWeese, A., DeMauro, A. A., Cham, H., & Greenberg, M. T. (2017). Impacts of the CARE for teachers program on teachers' social and emotional competence and classroom interactions. *Journal of Educational Psychology*, *109*(7), 1010–1028. https://doi.org/10.1037/edu0000187

Johnson, S. M. (1990). *Teachers at work*. Basic Books.

Jordan, R. L. P., Garwood, J., & Trathen, W. (2019). Assessing general education and special education majors' self–efficacy for teaching reading. *Learning Disabilities Research & Practice*, *34*(4), 185–193. https://doi.org/10.1111/ldrp.12207

Judge, A., Locke, E., Durham, C., & Kluger, A. (1998). Dispositional effects on job and life satisfaction: The role of core evaluations. *Journal of Applied Psycholog*y, *83*(1): 17–34. https://doi.org/10.1037/0021-9010.83.1.17

Junco, R., Elavsky, C. M., & Heiberger, G. (2013). Putting Twitter to the test: Assessing outcomes for student collaboration, engagement and success. *British Journal of Educational Technology*, *44*(2), 273–287. https://doi.org/10.1111/j.1467-8535.2012.01284.x

Kahu, E. R. (2013). Framing student engagement in higher education. *Studies in Higher Education*, *38*(5), 758–773. https://doi.org/10.1080/03075079

Kalleberg, A. L. (1977). Labor markets and inequality in earnings. *American Sociological Review*, *42*(1), 124–143.

Kamenez, N.V. & Smirnova, Z.V. & Vaganova, O.I. & Bystrova, N.V. & Tsarapkina, J.M.. (2019). Development of instructing techniques in professional training. *International Journal of Mechanical Engineering and Technology*, *10*. 899–907.

Kane, T. J., Rockoff, J. E., & Staiger, Douglas. (2006, April). What does certification tell us about teacher effectiveness? Evidence from New York City. NBER Working Paper No. w12155. https://ssrn.com/abstract=896463

Keigher, A. (2010). Teacher attrition and mobility: Results from the 2008-09 Teacher Follow-Up Survey. First Look. NCES 2010-353. *National Center for Education Statistics*. https://nces.ed.gov/pubs2010/2010353.pdf

Kelchtermans, G. (2017). 'Should I stay or should I go': Unpacking attrition/retention as an educational issue. *Teachers and Teaching*, *23*(8), 961–977. https://doi.org/10.1080/13540602.2017.1379793

Khan, B. M., & Khan, A. M. (2017). Grit, happiness and life satisfaction among professionals: A correlational study. *Journal of Psychology and Cognition*, *2*(2). ttps://doi.org/10.35841/psychology-cognition.2.2.123-132

Kim, L. E., Jörg, V. & Klassen, R. M. A Meta-Analysis of the Effects of Teacher Personality on Teacher Effectiveness and Burnout. *Educational Psychology Review*, *31*(1), 163–195. https://doi.org/10.1007/s10648-018-9458-2

Klimek, S. G. (2019) Prestige, status, and esteem and the teacher shortage. *Journal of Education and Learning*, *8*(4), 185–213. https://doi.org/10.5539/jel.v8n4p185

Knaggs, C. M., & Sondergeld, T. A. (2015). Science as a learner and as a teacher: Measuring science self-efficacy of elementary preservice teachers. *School Science and Mathematics*, *115*(3), 117–128. https://doi.org/10.1111/ssm.12110

Korpershoek, H., Harms, T., de Boer, H., van Kuijk, M., & Doolaard, S. (2016). A meta-analysis of the effects of classroom management strategies and classroom management programs on students' academic, behavioral, emotional, and motivational outcomes. *Review of Educational Research*, *86*(3), 643–680. https://doi.org/10.3102/0034654315626799

Korte, D. & Simonsen, J. (2018). Influence of social support on teacher self-efficacy in novice agricultural education teachers. *Journal of Agricultural Education*, 59, 100–123. https://doi.org/10.5032/jae.2018.03100.

Kucukoglu, H. (2014). Ways to cope with teacher burnout factors in ELT classrooms. *Procedia - Social and Behavioral Sciences*, *116*(1), 2741–2746. https://doi.org/10.1016/j.sbspro.2014.01.647

Künsting, J., Neuber, V., & Lipowsky, F. (2016). Teacher self-efficacy as a long-term predictor of instructional quality in the classroom. *European Journal of Psychology of Education*, *31*(3), 299–322. https://doi.org/10.1007/s10212-015-0272-7

Lai Lam, K. K., & Zhou, M. (2020). A serial mediation model testing growth mindset, life satisfaction, and perceived distress as predictors of perseverance of effort, *Personality and Individual Differences*, 167, 110262. https://doi.org/10.1016/j.paid.2020.110262.

Lee, L. W. (2020). The relationship between secondary teachers' grit and self-efficacy beliefs on classroom management and student engagement [Doctoral dissertation, Liberty University], Scholars Crossing. https://digitalcommons.liberty.edu/doctoral/2751

Lent, R. W., Brown, S. D., & Hackett, G. (1994). Toward a unifying social cognitive theory of career and academic interest, choice, and performance. *Journal of Vocational Behavior*, *45*(1), 79–122. https://doi.org/10.1006/jvbe.1994.1027

Lentfer, V., & Franks, B. A. (2016). The redirect behavior model and the effects on pre-service teachers' self-efficacy. *Journal of Education and Practice*, *6*(35), 79–87. https://files.eric.ed.gov/fulltext/EJ1086329.pdf

Lowe, G., Gray, C., Prout, P., Jefferson, S., Shaw, T. (2019). Positive veteran teachers: Who are they and where are they to be found? *Issues in Educational Research 29*(3), 823–840. http://www.iier.org.au/iier/html

Madden, J. H. (2015). *Relationships of principal grit, teacher trust in the principal, and faculty organizational commitment in Alabama public high schools* [Doctoral dissertation, The University of Alabama Tuscaloosa]. University of Alabama Institution Repository.

Maddi, S., R., Kahn, S., & Maddi, K. L. (1998). The effectiveness of hardiness training. *Consulting Psychology Journal: Practice and Research*, *50*(2), 78–86. https://doi.org/10.1037/1061-4087.50.2.78

Marzano, R. J., Marzano, J. S., & Pickering, D. (2003). *Classroom management that works: Research-based strategies for every teacher*. ASCD.

Maslach, C., & Jackson, S.E. (1981). The Measurement of experienced burnout. *Journal of Organizational Behavior*, *2*, 99-113.

Maslach, C., & Jackson, S. E. (1993). *Maslach Burnout Inventory manual*. Consulting Psychologists Press.

McCarthy, C. J. (2019). Teacher stress: Balancing demands and resources. *Phi Delta Kappan*, *101*(3), 8–14. https://doi.org/10.1177/0031721719885909

McClendon, C., Massey, R., & Neugebauer, A. K., (2017). Grit, growth mindset and deliberate practice. *Journal of Instructional Research*, *6*, 8–17. https://files.eric.ed.gov/fulltext/EJ1153307.pdf

McFarland, J., Hussar, B., Wang, X., Zhang, J., Wang, K., Rathbun, A., Barmer, A., Forrest Cataldi, E., and Bullock Mann, F. (2018). *The condition of education 2018* (NCES 2018-144). National Center for Education Statistics. https://nces.ed.gov/pubsearch/pubsinfo. asp?pubid=2018144.

McHenry-Sorber, E., & Campbell, M. P. (2019). Teacher shortage as a local phenomenon: District leader sensemaking, responses, and implications for policy. *Education Policy Analysis Archives*, *27*, 87. https://doi.org/10.14507/epaa.27.4413

Meristo, M., & Eisenschmidt, E. (2014). Novice teachers' perceptions of school climate and self-efficacy. *International Journal of Educational Research*, *67*(1), 1–10. https://doi.org/10.1016/j.ijer.2014.04.003

Menon, D., & Sadler, T. (2017). Sources of science teaching self-efficacy for preservice elementary teachers in science content courses. *International Journal of Science and Mathematics Education*, *16*. https://doi.org/10.1007/s10763-017-9813-7.

Meyer, S. J., Espel, E. V., Weston-Sementelli, J. L., & Serdiouk, M. (2019). *Teacher retention, mobility, and attrition in Colorado, Missouri, Nebraska, and South Dakota*. Regional Educational Laboratory Central.

Miller, A. D., Ramirez, E. M., & Murdock, T. B. (2017). The influence of teachers' self-efficacy on perceptions: Perceived teacher competence and respect and student effort and achievement. *Teaching and Teacher Education*, *64*, 260–269. https://doi.org/10.1016/j.tate.2017.02.008

Milner, H. R. (2002). A case study of an experienced English teacher's self- efficacy and persistence through 'crisis' situations: theoretical and practical considerations. *High School Journal*, *86*(1), 28–35. https://www.jstor.org/stable/40364343

Mott, J. M. (2020). *Capturing Teacher Perseverance: A Study of Veteran Teachers Who Have Remained in the Classroom* (Publication No. 28990709) [Doctoral dissertation, Xavier University]. ProQuest Dissertations & Theses Global.

Muex, R. S. (2021). *Leaving education: A descriptive single-case study examining what factors drive teachers from the profession in Oklahoma* (Publication No. 28774454) [Doctoral dissertation, Baylor University].

Multon, K. D., Brown, Steven, D., & Lent, R. W. (1991). Relation of self-efficacy beliefs to academic outcomes: A meta-analytical investigation. *Journal of Counseling Psychology*, *38*(1), 30–38, https://doi.org/10.1037/0022-0167.38.1.30

Nartgun, Z., & Şahin, M. (2015). Psychometric properties of data gathering tools used in thesis. *Procedia - Social and Behavioral Sciences*, *174*, 2849–2855. https://doi.org/10.1016/j.sbspro.2015.01.978

National Center for Education Statistics. (2023). *National assessment of educational progress*. http://www.nces.ed.gov/nationsreportcard/

Nye, C., Su, R., Rounds, J., & Drasgow, F. (2012). Vocational interests and performance: A quantitative summary of over 60 years of research. *Perspectives on Psychological Science, 7*(4), 384–403. https://doi.org/10.1177/1745691612449021

Nyugen, T.D. (2020). Examining the teacher labor market in different rural contexts: Variations by urbanicity and rural states. *American Educational Research Association Open, 6*(4), 1–24. https://doi.org/10.1177/2332858420966336

Office of Human Subjects Research. (1979). *The Belmont Report: Ethical principles and guidelines for the protection of human subjects research*. https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html

Pachler, D., Kuonath, A., & Frey, D. (2019). How transformational lecturers promote students' engagement, creativity, and task performance: The mediating role of trust in lecturer and self-efficacy. *Learning and Individual Differences*, *69*, 162–172. https://doi.org/10.1016/j.lindif.2018.12.004

Park, E.-Y., & Shin, M. (2020). A Meta-Analysis of Special Education Teachers' Burnout. *Sage Open*, *10*(2). https://doi.org/10.1177/2158244020918297

Paunesku, D., Walton, G. M., Romero, C. L., Smith, E. N., Yeager, D. S., & Dweck, C. S. (2015). Mindset interventions are a scalable treatment for academic underachievement. *Psychological Science*, *26*(6), 784–793. https://doi.org/10.1177/0956797615571017

Perkins-Gough, D., & Duckworth, A. (2013). The significance of grit. *Educational Leadership*. *71*. 14–20.

Park, N., Peterson, C., & Seligman, M. E. P. (2004). Strengths of character and well-being. *Journal of Social and Clinical Psychology*, *23*(5), 603–619. https://doi.org/10.1521/jscp.23.5.603.50748

Pfitzner-Eden, F. (2016). I feel less confident so I quit? Do true changes in teacher self-efficacy predict changes in preservice teachers' intention to quit their teaching degree? *Teaching and Teacher Education*, *55*, 240–254.

Pogere, E. F., López-Sangil, M. C., García-Señorán, M. M., & González, A. (2019). Teachers' job stressors and coping strategies: Their structural relationships with emotional exhaustion and autonomy support. *Teaching and Teacher Education*, *85*, 269–280. https://doi.org/10.1016/j.tate.2019.07.001

Pines, A. M., & Aronson, E. (1988). *Career burnout*. The Free Press.

Qureshi, M. I., Iftikhar, M., Abbas, S. G., Hassan, U., Khan, K., & Zaman, K. (2013). Relationship between job stress, workload, environment and employees turnover intentions: What we know, what should we know. *World Applied Sciences Journal*, *23*(7), 764–770. https://doi.org/10.5829/idosi.wasj.2013.23.06.313

Rattan, A., Savani, K., Chugh, D., & Dweck, C. S. (2015). Leveraging mindsets to promote academic achievement: Policy recommendations. *Perspectives on Psychological Science*, *10*(6), 721–726. https://doi.org/10.1177/1745691615599383

Ratan, S., Anand, T., Ratan, J. (2019). Formulation of research question – Stepwise approach. *Indian Association Pediatric Surgery*, *24*(1), 15–20. https://doi.org/10.4103/jiaps.JIAPS_76_18

Reeve, J., & Deci, E. L. (1996). Elements of the competitive situation that affect intrinsic motivation. *Personality and Social Psychology Bulletin*, *22*(1), 24–33. https://doi.org/10.1177/0146167296221003

Regier, P., & Savic, M. (2019). How teaching to foster mathematical creativity may impact student self-efficacy for proving. *The Journal of Mathematical Behavior*, *57*, 100720. https://doi.org/10.1016/j.jmathb.2019.100720

Roberts, B. W., Jackson, J. J., Fayard, J. V., Edmonds, G., & Meints, J. (2009). Conscientiousness. In M. R. Leary & R. H. Hoyle (Eds.), *Handbook of individual differences in social behavior* (pp. 369–381). The Guilford Press.

Roberts, B. W., Lejuez, C., Krueger, R. F., Richards, J. M., & Hill, P. L. (2014). What is conscientiousness and how can it be assessed? *Developmental Psychology*, *50*(5), 1315–1330. https://doi.org/10.1037/a0031109

Robertson-Kraft, C., & Duckworth, A. L. (2014). True grit: Trait-level perseverance and passion for long-term goals predicts effectiveness and retention among novice teachers. *Teachers College Record*, *116*(3). http://www.tcrecord.org/Content.asp?ContentId=17352

Rohmaniyah, A. & Nurhayati, T. (2017). Improving teacher performance based on creative model. *Internation Journal of Islamic Business Ethics*, *2*(2). http://dx.doi.org/10.30659/ijibe.2.2.347-364

Riddle, R. A. (2018). *A predictive correlational study of the relationship between grit and self-efficacy beliefs among pre-service teachers* [Doctoral dissertation, Liberty University]. Scholars Crossing. https://digitalcommons.liberty.edu/doctoral/1723/

Rozell, Z. J. (2019). *A quantitative examination of grit and student engagement, instructional strategies, and classroom management teacher self-efficacy among alternatively certified educators* (Publication No. 13896825) [Doctoral dissertation, Grand Canyon University]. ProQuest Dissertations & Theses Global.

Rumrill, P. D., Jr (2004). Non-manipulation quantitative designs. *Work*, *22*(3), 255–260.

Ryan, A. M., Kuusinen, C. M., & Bedoya-Skoog, A. (2015). Managing peer relations: A dimension of teacher self-efficacy that varies between elementary and middle school teachers and is associated with observed classroom quality. *Contemporary Educational Psychology*, *41*, 147–156. https://doi.org/10.1016/j.cedpsych.2015.01.002

Ryan, S. V., von der Embse, N. P., Pendergast, L. L., Saeki, E., Segool, N., & Schwing, S. (2017). Leaving the teaching profession: The role of teacher stress and educational accountability policies on turnover intent. *Teaching and Teacher Education*, *66*, 1–11. https://doi.org/10.1016/j.tate.2017.03.016

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, *55*(1), 68–70. https://doi.org/10.1037/0003-066x.55.1.68

Saltman, K. J. (2014). The austerity school: Grit, character, and the privatization of public education. *Symploke*, *22*(1-2), 1–17. https://muse.jhu.edu/journal/190

Sanders, W., & Rivers, J. (1996). Cumulative and residual effects of teachers on future student academic achievement. University of Tennessee Value-Added Research and Assessment Center.

Shaeffer, K. (2022). *A dwindling number of new U.S. college graduates have a degree in education*. Pew Research Center. https://www.pewresearch.org/short-reads/2022/09/27/a-dwindling-number-of-new-u-s-college-graduates-havea-degree-in-education/

Schunk, D. H., & DiBenedetto, M. K. (2016). Self-efficacy theory in education. *Advances in Motivation Science*, *8*, 153–179. https://doi.org/10.1016/bs.adms.2020.10.001

Seligman, M. E. P. (1975). *Helplessness: On depression, development, and death.* Henry Holt & Co.

Seligman, M. E. P. (2011). *Flourish: A visionary new understanding of happiness and well-being.* Free Press.

Sezgin, F. (2012). Examination of psychological endurance levels of primary school teachers. *Kastamonu Education Magazine*, *20*(2), 489–502.

Shaw, J. L., & Newton, J. (2014). Teacher retention and satisfaction with a servant leader as principal. *Education*, *135*(1), 101–106.

Shaukat, S. & Iqbal, H. (2012). Teacher self-efficacy as a function of student engagement, instructional strategies and classroom management. *Pakistan Journal of Social and clinical Psychology*, *10*, 82–86.

Sheldon, K. M., Ryan, R., & Reis, H. T. (1996). What makes for a good day? Competence and autonomy in the day and in the person. *Personality and Social Psychology Bulletin*, *22*(12), 1270–1279. https://doi.org/10.1177/01461672962212007

Shin, K., & Kim, S. (2017). The moderating effect of the self-efficacy on the relationship between job satisfaction and turnover intention of nursery school's teachers. *Journal of Digital Convergence*, *15*(1), 403–412. https://doi.org/10.14400/jdc.2017.15.1.403

Shin, Y., Koo, M., & Park, D. (2018). The mediating effect of childcare teacher's job stress on the relationship between grit and turnover intention. *Korean Journal of Child Studies*, *39*(2), 15–26. https://doi.org/10.5723/kjcs.2018.39.2.15

Silvia, P. J., Eddington, K. M., Beaty, R. E., Nusbaum, E. C., & Kwapil, T. R. (2013). Gritty people try harder: Grit and effort-related cardiac autonomic activity during an active coping challenge. *International Journal of Psychophysiology*, *88*(2), 200–205

Singh, J., & Chopra, V.G. (2018). Workplace spirituality, grit and work engagement. *Asia-Pacific Journal of Management Research and Innovation*, *14*(1-2), 50–59. https://doi.org/10.1177/2319510X18811776

Skaalvik, E. M., & Skaalvik, S. (2016). Teacher stress and teacher self-efficacy as predictors of engagement, emotional exhaustion, and motivation to leave the teaching profession. Creative Education, 07(13), 1785-1799. https://doi.org/10.4236/ce.2016.713182

Skaalvik, E. M., & Skaalvik, S. (2017). Motivated for teaching? Associations with school goal structure, teacher self-efficacy, job satisfaction and emotional exhaustion. *Teaching and Teacher Education*, *67*, 152–160. https://doi.org/10.1016/j.tate.2017.06.006

Smiley, P. A., & Dweck, C. S. (1994). Individual differences in achievement goals among young children. *Child Development*, *65*(6), 1723–1743. https://doi.org/10.2307/1131290

Smith, T. G. (2023). *The impact of teacher grit and growth mindset on teacher self-efficacy: A phenomenological approach* (Publication No. 30566894) [Doctoral dissertation, Saint Peter's University]. ProQuest Dissertations & Theses Global.

Sorensen, L. C., & Ladd, H. F. (2020). The hidden costs of teacher turnover. *AERA Open*, *6*(1). https://doi.org/10.1177/2332858420905812

Sparks, G. M. (1988). Teachers' attitudes toward change and subsequent improvements in classroom teaching. *Journal of Educational Psychology*, *80*(1), 111–117. https://doi.org/10.1037/0022-0663.80.1.111

Starnaman, S. M., & Miller, K. I. (1992). A test of a causal model of communication and burnout in the teaching profession. *Communication Education*, *41*, 40–53. https://doi.org/10.1080/03634529209378869

Stoffel, J. M., & Cain, J. (2018). Review of grit and resilience literature within health professions education. *American Journal of Pharmaceutical Education*, *82*(2), 6150. https://doi.org/10.5688/ajpe6150

Strayhorn, T. L. (2014). What role does grit play in the academic success of black male collegians at predominantly white institutions? *Journal of African American Studies*, *18*(1), 1–10. http://www.jstor.org/stable/43525531

Stichler, J. F. (2016). Research, Research-informed design, evidence-based design: What is the difference and does it matter? *HERD: Health Environments Research & Design Journal*, *10*(1), 7–12. https://doi.org/10.1177/1937586716665031

Sutcher, L., Darling-Hammond, L., & Carver-Thomas, D. (2016). *A coming crisis in teaching? Teacher supply, demand, and shortages in the U.S*. https://learningpolicyinstitute.org/product/coming-crisis-teaching

Sutcher, L., Darling-Hammond, L., & Carver-Thomas, D. (2019). Understanding teacher shortages: An analysis of teacher supply and demand in the United States. *Education Policy Analysis Archives*, *27*(35). http://dx.doi.org/10.14507/epaa.27.3696

Swanson, P., & Huff, R. (2010). Georgia's rural foreign language teachers' sense of efficacy and how it relates to teacher attrition. *The Rural Educator*, *31*(3), 16–29

Tabachnick, B., & Fidell, L. (2013). *Using multivariate statistics*. (6th ed.). Pearson International. https://elibrary.pearson.de/book/99.150005/9781292034546

Tait, M. (2008). Resilience as a contributor to novice teacher success, commitment, and retention. *Teacher Education Quarterly*, *35*, 57–75.

Tomkins S. S. (1995). Frontmatter. In E. V. Demos (Ed.), *Exploring affect: The selected writings of Silvan S Tomkins*. Studies in Emotion and Social Interaction (pp. vii-xii). Cambridge University Press.

Tournaki, N., & Podell, D. M. (2005). The impact of student characteristics and teacher efficacy on teachers' predictions of student success. *Teaching and Teacher Education*, *21*, 299–314. https://doi.org/10.1016/j.tate.2005.01.003

Turner, C., & Cohen, N. (2023, March 23). *6 things to know about U.S. teacher shortages and how to solve them*. NPR. https://www.npr.org/2023/03/23/1164800932/teacher-shortages-schools explainer

Thompson, A. (1991). On teaching: A letter to Luke. In P. R. Kane (Ed.), *The first year of teaching: Real world stories from America's teachers* (pp. 103-107). Walker and Company.

Tschannen-Moran, M., Anita, W. H., & Hoy, W. K. (1998). Teacher efficacy: Its meaning and measure. *Review of Educational Research*, *68*(2), 202–248. https://doi.org/10.2307/1170754

Tschannen-Moran, M., & Hoy, A. W. (2001). Teacher efficacy: Capturing an elusive construct. *Teaching and Teacher Education*, *17*(7), 783–805. https://doi.org/10.1016/S0742- 051X(01)00036-1

Tschannen-Moran, M., & Hoy, A. W. (2007). The differential antecedents of self-efficacy beliefs of novice and experienced teachers. *Teaching and Teacher Education*, *23*(6), 944–956. https://doi.org/10.1016/j.tate.2006.05.003

U.S. Department of Education. (2014). Condition of education—Teacher turnover: Stayers, movers, and leavers. https://nces.ed.gov/programs /coe/indicator_slc.asp

Vallerand, R. J. (2012). From motivation to passion: In search of motivational processes involved in a meaningful life. *Canadian Psychology*, *53*(1), 42–52. https://doi.org/10.1037/a0026377

Vally, Z., & D'Souza, C. G. (2019). Abstinence from social media use, subjective well-being, stress, and loneliness. *Perspectives in Psychiatric Care*, *55*(4), 752–759. https://doi.org/10.1111/ppc.12431

Van Droogenbroeck, F., Spruyt, B., & Vanroelen, C. (2014). Burnout among senior teachers: Investigating the role of workload and interpersonal relationships at work. *Teaching and Teacher Education*, *43*(1), 99–109. https://doi.org/10.1016/j.tate.2014.07.005

van Uden, J. M., Ritzen, H., & Pieters, J. M. (2014). Engaging students: The role of teacher beliefs and interpersonal teacher behavior in fostering student engagement in vocational education. *Teaching and Teacher Education*, *37*(1), 21–32. https://doi.org/10.1016/j.tate.2013.08.005

Von Culin, K. R., Tsukayama, E., & Duckworth, A. L. (2014). Unpacking grit: Motivational correlates of perseverance and passion for long-term goals. *Journal of Positive Psychology*, *9*(4), 306-312. https://doi.org/10.1080/17439760.2014.898320

Viner, R. M., Russell, S. J., Croker, H., Packer, J., Ward, J., Stansfield, C., Mytton, O., Bonell, C., & Booy, R. (2020). School closure and management practices during coronavirus outbreaks including COVID-19: A rapid systematic review. *The Lancet. Child & adolescent Health*, *4*(5), 397–404. https://doi.org/10.1016/S2352-4642(20)30095-X

Warner, R. M. (2013). *Applied statistics: From bivariate through multivariate techniques* (2nd ed.). Sage Publications.

Wasylkiw, L., Hanson,S., Lynch, L.M., Vaillaincourt, & E.,Wilson, C. (2020). Predicting undergraduate student outcomes:Competing or complementary roles of self-esteem, self compassion, self-efficacy and mindsets? *Canadian Journal of Higher Education*, *50*(2), 1–14

Weber, M. (2020). New Jersey's shrinking pool of teacher candidates, New Jersey Policy Perspective, *Economic Justice.* https://www.njpp.org/wp-content/uploads/2020/05/NJPP-Report-New-Jerseys-Shrinking-Pool-of-Teacher-Candidates.pdf

Weinstein, C. S. (1989). Teacher Education Students' Preconceptions of Teaching. *Journal of Teacher Education*, *40*(2), 53–60. https://doi.org/10.1177/002248718904000210

West, C. P., Dyrbye, L. N.,Sloan, J. A., & Shanafelt,T. D. (2009). Single item measures of emotional exhaustion and depersonalization are useful for assessing burnout in medical professionals. *Journal of General Internal Medicine*, *24*(12), 1318–1321. https://doi.org/10.1007/s11606-009-1129-z PMID:19802645

West, M. R., Kraft, M. A., Finn, A. S., Martin, R. E., Duckworth, A. L., Gabrieli, C. F., & Gabrieli, J. D. (2015). Promise and paradox measuring students' non-cognitive skills and the impact of schooling. Educational Evaluation and Policy Analysis, 38(1), 148–170. https://doi.org/10.3102/0162373715597298

White, R. W. (1963). *Ego and reality in psychoanalytic theory*. International Universities Press.

Whitley, J., Smith, J. D., & Vaillancourt, T. (2013). Promoting mental health literacy among educators: Critical in school-based prevention and intervention. *Canadian Journal of School Psychology*, *28*(1), 56–70. https://doi.org/10.1177/0829573512468852

Widodo, W., Robertus, M., & Gunawan, B. (2021). Effect of grit on the teaching creativity of Indonesian teachers: The mediating role of organizational commitment and knowledge management. *Cogent Education*, *8*(1). https://doi.org/10.1080/2331186X.2021.2006111.

Wolters, C. A., & Daugherty, S. G. (2007). Goal structures and teachers' sense of efficacy: Their relation and association to teaching experience and academic level. *Journal of Educational Psychology*, *99*(1), 181–193. https://doi.org/10.1037/0022-0663.99.1.181

World Health Organization. (2019, May 28). *Burn-out an "occupational phenomenon": International Classification of Diseases*. https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases

Wright, D. S., Balgopal, M. M., Sample McMeeking, L. B., & Weinberg, A. E. (2019). Developing resilient K-12 STEM teachers. *Advances in Developing Human Resources*, *21*(1), 16–34. https://doi.org/10.1177/1523422318814483

Xie, Q. (2012). Relationship among physical exercise, physical self–esteem and general self–efficacy of undergraduates. *Journal of Guangzhou Sport University*, *32*(3), 95–99. https://doi.org/10.3390/ijerph191811596

Yeager, D. S., & Dweck, C. S. (2012). Mindsets that promote resilience: When students believe that personal characteristics can be developed. *Educational Psychologist*, *47*(4), 302–314. https://doi.org/10.1080/00461520.2012.722805

Yost, D. (2006). Reflection and self-efficacy: Enhancing the retention of qualified teachers from a teacher education perspective. *Teacher Education Quarterly*, *33*(4) 59–76. https://files.eric.ed.gov/fulltext/EJ795226.pdf

Yılmaz, I. T. (2007, June 21–24). *A review of grit*. Psychology Graduate Students Congress. İzmir Economy University, Bildiri Metinleri Kitabı.

Zhang, G., & Zeller, N. (2016). A longitudinal investigation of the relationship between teacher preparation and teacher retention. *Teacher Education Quarterly*, *43*(2), 73–92. https://files.eric.ed.gov/fulltext/EJ1100322.pdf

Zee, M., & Koomen, H. (2016). Teacher self-efficacy and its effects on classroom processes, student academic adjustment and teacher well-being: A synthesis of 40 years of research. *Review of Educational Research*, *86*(4), 981–1015. https://doi.org/10.3102/0034654315626801

Zocche, L., de Paula, I. C., Kunrath, S. E., Martins, V. L. M., & Lermen, F. H. (2018). Variables that influence creativity in perception of professionals: A case study in innovative Brazilian companies. *Thinking Skills and Creativity*, *29*, 170–184. https://doi.org/10.1016/j.tsc.2018.07.002

ProQuest Number: 31768526

INFORMATION TO ALL USERS
The quality and completeness of this reproduction is dependent on the quality
and completeness of the copy made available to ProQuest.

Distributed by
ProQuest LLC a part of Clarivate (2025).
Copyright of the Dissertation is held by the Author unless otherwise noted.

This work is protected against unauthorized copying under Title 17,
United States Code and other applicable copyright laws.

This work may be used in accordance with the terms of the Creative Commons license
or other rights statement, as indicated in the copyright statement or in the metadata
associated with this work. Unless otherwise specified in the copyright statement
or the metadata, all rights are reserved by the copyright holder.

ProQuest LLC
789 East Eisenhower Parkway
Ann Arbor, MI 48108 USA